Stochastic Analysis on Manifolds

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Chapter 1

SDE and Diffusion

Fix probability space $(\Omega, \mathcal{F}, \mathbb{P})$ equipped with a filtration $\mathbb{F} = (\mathcal{F}_t)_{t\geq 0}$ of σ-fields contained in $\mathcal{F}$ with assumption that $\mathcal{F}_\infty = \mathcal{F}$. Also, $\mathbb{F}$ satisfies the usual condition, i.e., complete and right-continuous. Without specification, all semimartingales are assumed continuous. We use Einstein summation convention for convenience.

1.1 SDE on Euclidean Space

Solution of SDE: Let $Z = (Z_t)_{t\geq 0}$ be a continuous $\mathbb{R}^\ell$-valued $\mathbb{F}$-semimartingale, i.e.,

$$Z = M + A,$$

where M is a local martingale and A is an adapted process of local bounded variation such that $A_0 = 0$. Let $\sigma\colon \mathbb{R}^N \to \mathbb{R}^{N\times\ell}$ be smooth and locally Lipschitz, i.e. for all $R > 0$ there exists $R > 0$ and $C(R) > 0$ such that

$$\|\sigma(x) - \sigma(y)\|_{\mathrm{F}} \leq C(R)\,\|x - y\|\,, \quad \forall\, x, y \in B(R).$$

Let $X_0 \in \mathbb{R}^N$ be $\mathcal{F}_0$ measurable. For a stopping time τ, consider a semimartingale $X = (X_{t\wedge\tau})_{t\geq 0}$ of the form

$$X_t = X_0 + \int_0^t \sigma(X_s)dZ_s, \quad 0 \leq t \leq \tau.$$

We say X is a solution of a SDE driving by the semimartingale Z, and such equation denoted as $\mathrm{SDE}(\sigma, Z, X_0)$. Moreover, by Itô formula, for $f \in C^2(\mathbb{R}^N)$,

$$\begin{aligned}
f(X_t) &= f\left(X_0\right) + \int_0^t f_{x_i}\left(X_s\right)\sigma_\alpha^i\left(X_s\right)dZ_s^\alpha \\
&\quad + \frac{1}{2}\int_0^t f_{x_ix_j}\left(X_s\right)\sigma_\alpha^i\left(X_s\right)\sigma_\beta^j\left(X_s\right)d\left\langle Z^\alpha, Z^\beta\right\rangle_s \\
&= f\left(X_0\right) + \int_0^t f_{x_i}\left(X_s\right)\sigma_\alpha^i\left(X_s\right)dM_s^\alpha + \int_0^t f_{x_i}\left(X_s\right)\sigma_\alpha^i\left(X_s\right)dA_s^\alpha \\
&\quad + \frac{1}{2}\int_0^t f_{x_ix_j}\left(X_s\right)\sigma_\alpha^i\left(X_s\right)\sigma_\beta^j\left(X_s\right)d\left\langle M^\alpha, M^\beta\right\rangle_s
\end{aligned}$$

Remark 1.1.1. Note that if $X_0 \in L^2$ and σ is globally Lipschitz continuous, then we don't need the stopping time τ and $\mathrm{SDE}(\sigma, Z, X_0)$ has a unique solution $X = (X_t)_{t\geq 0}$.

The problem why we need to consider a stopping time is because of the local Lipschitz property of σ. The local Lipschitz would make the usual solution exploded. To consider it rigorously, we first need the definition of explosion time.

Let M be a locally compact metric space and $\widehat{M} = M \cup \{\partial_M\}$ be its one-point compactification. A continuous map $x\colon [0,\infty) \to \widehat{M}$ has an explosion time $e = e(x) > 0$ if $x_t \in M$ for $0 \le t < e$ and $x_t = \partial M$ for $t \ge e$ when $e < \infty$, i.e.

$$e\colon W(M) \to [0,\infty]$$

where $W(M) \subset C([0,\infty), \widehat{M})$ such that e can be defined. Moreover, $W(M)$ can be equipped with the canonical filtration $\mathbb{F}_M = (\mathcal{F}_{M,t})_{t\ge 0}$, i.e., the natural filtration generated by coordinate processes. Then e is a $\mathbb{F}_M$-stopping time.

Definition 1.1.2. Let τ be a stopping time. A continuous $X = (X_{t\wedge\tau})_{t\ge 0}$ is called a semimartingale up to τ if there exists a sequence of stopping times $\tau_n \uparrow \tau$ such that $X^{\tau_n} = (X_{t\wedge\tau_n})$ is a semimartingale.

Definition 1.1.3. A semimartingale X up to a stopping time τ is a solution of $\mathrm{SDE}(\sigma, Z, X_0)$ if there exists a sequence of stopping times $\tau_n \uparrow \tau$ such that $X^{\tau_n} = (X_{t\wedge\tau_n})$ is a semimartingale and

$$X_{t\wedge\tau_n} = X_0 + \int_0^{t\wedge\tau_n} \sigma\left(X_s\right) dZ_s, \quad t \ge 0.$$

Theorem 1.1.4. *If σ is locally Lipschitz continuous, then there exists a unique $W(\mathbb{R}^N)$-valued random variable X which is a solution of $\mathrm{SDE}(\sigma, Z, X_0)$ up to its explosion time $e(X)$. Moreover, if X, Y are two solutions up to τ and η respectively, then $X_{t\wedge\tau\wedge\eta} = Y_{t\wedge\tau\wedge\eta}$. In particular, if X is a solution up to $e(X)$, then $\eta \le e(X)$ and $X_{t\wedge\tau} = Y_{t\wedge\tau}$.*

Theorem 1.1.5. *Suppose σ and b are locally Lipschitz. Then the weak uniqueness holds for*

$$X_t = X_0 + \int_0^t \sigma(X_s) dW_s + \int_0^t b(X_s) ds,$$

where W is a Brownian motion.

Stratonovich integral. For two continuous semimartingales H, Z, the Stratonovich integral is defined as

$$\int_0^t H_s \circ dZ_s = \int_0^t H_s dZ_s + \frac{1}{2}\left\langle H, Z\right\rangle_t.$$

In such case, the Itô formula becomes

$$df(X_t) = \frac{\partial f}{\partial x^\alpha}(X_t) \circ dX_t^\alpha.$$

Suppose that V_α, $\alpha = 1, 2, \cdots, \ell$ are smooth vector fields in $\mathbb{R}^N$, that is,

$$V_\alpha = (V_\alpha^1, \cdots, V_\alpha^N)\colon \mathbb{R}^N \to \mathbb{R}^N$$

and vector fields means that for any $f \in C^1(\mathbb{R}^N)$,

$$V_\alpha f = \sum_{i=1}^N V_\alpha^i \frac{\partial f}{\partial x^i}.$$

More generally, if

$$X_t = X_0 + \int_0^t V_\alpha\left(X_s\right) \circ dZ_s^\alpha,$$

then

$$X_t = X_0 + \int_0^t V_\alpha\left(X_s\right) dZ_s^\alpha + \frac{1}{2}\int_0^t \nabla_{V_\beta} V_\alpha\left(X_s\right) d\left\langle Z^\alpha, Z^\beta\right\rangle_s.$$

Proof. By definition,

$$dX_t = V_\alpha(X_t) \circ dZ_t^\alpha = V_\alpha(X_t) dZ_t^\alpha + \frac{1}{2} d \left\langle V_\alpha(X_t), Z_t^\alpha \right\rangle .$$

or

$$dX_t^i = \sum_{\alpha=1}^{\ell} V_\alpha^i(X_t) dZ_t^\alpha + \frac{1}{2} \sum_{\alpha=1}^{\ell} d \left\langle V_\alpha^i(X_t), Z_t^\alpha \right\rangle .$$

For $d \langle V_\alpha^i(X_t), Z_t^\alpha \rangle$, first because $\sum_\alpha V_\alpha^i(X_t) dZ_t^\alpha$ is the local martingale part of dX_t^i,

$$d \left\langle X^i, Z^\alpha \right\rangle_t = \sum_{\beta=1}^{\ell} V_\beta^i(X_t) d \left\langle Z^\beta, Z^\alpha \right\rangle_t .$$

Then by Itô formula,

$$dV_\alpha^i(X_t) = \sum_{j=1}^{n} \frac{\partial V_\alpha^i}{\partial x^j} dX_t^j + \text{ bounded variation part } .$$

So

$$\begin{aligned} d \left\langle V_\alpha^i(X_t), Z_t^\alpha \right\rangle &= \sum_{j=1}^{n} \frac{\partial V_\alpha^i}{\partial x^j} d \left\langle X^j, Z^\alpha \right\rangle_t \\ &= \sum_{\beta=1}^{\ell} \sum_{j=1}^{n} V_\beta^j \frac{\partial V_\alpha^i}{\partial x^j} d \left\langle Z^\beta, Z^\alpha \right\rangle_t \\ &= \nabla_{V_\beta} V_\alpha^i d \left\langle Z^\beta, Z^\alpha \right\rangle_t . \end{aligned}$$

Therefore,

$$dX_t = V_\alpha(X_t) dZ_t^\alpha + \frac{1}{2} \nabla_{V_\beta} V_\alpha(X_t) d \left\langle Z^\beta, Z^\alpha \right\rangle_t . \qquad \square$$

Remark 1.1.6. Let X, Y, Z, Y, H, K are be $\mathbb{R}$-semimartingales.

(1) Note that, if $dX = H \circ dW$ and $dY = K \circ dZ$, because

$$H \circ dW = HdW + \frac{1}{2} \langle H, W \rangle , \quad K \circ dZ = KdZ + \frac{1}{2} \langle K, Z \rangle$$

and $\langle H, W \rangle$ and $\langle K, Z \rangle$ are of bounded variation, then

$$d \langle X, Y \rangle = HKd \langle W, Z \rangle .$$

(2) In fact, we have

$$H \circ d \left(\int X \circ dW \right) = (HX) \circ dW,$$

because

$$d \langle HX, W \rangle = Hd \langle X, W \rangle + Xd \langle H, W \rangle ,$$

which is because of Itô formula, $d(HX) = HdX + XdH + d \langle H, X \rangle$.

In the sense of Stratonovich integral, if $X_t = X_0 + \int_0^t V_\alpha(X_s) \circ dZ_s^\alpha$ and $f \in C^2(\mathbb{R}^N)$, then

$$f(X_t) = f(X_0) + \int_0^t (V_\alpha f)(X_s) \circ dZ_s^\alpha .$$

For a Itô SDE,

$$dX_t = b(X_t)dt + \sigma_n(X_t)dW_t^n,$$

where $W = (W^1, \cdots, W^\ell)$ is a ℓ-dimensional Brownian motion. Let

$$Z_t^1 = t, \quad Z^2 = W^1 \ , \ \cdots \ , \ Z_{\ell+1} = W^\ell.$$

Clearly, they are semimartingales. Then by above, we can see if let

$$V_1(x) = b(x) - \frac{1}{2}\sigma_n^i \frac{\partial \sigma_n}{\partial x^i}, \quad V_2(x) = \sigma_1(x), \ \cdots \ , V_{\ell+1}(x) = \sigma_\ell(x),$$

the Itô SDE is equivalent to

$$dX_t = V_\alpha(X_t) \circ dZ_t^\alpha = V_1(X_t)dt + V_\beta(X_t) \circ dZ_t^\beta.$$

1.2 SDE on Manifolds

Definition 1.2.1. Let M be a smooth manifold. A continuous M-valued process X is called a M-valued semi-martingale if $f(X)$ is a real-valued semi-martingale for all $f \in C^\infty(M)$.

Remark 1.2.2. We can also define it up to a stopping time τ as before. By Itô formula, it is obviously that $M = \mathbb{R}^N$ gives the usual definition of semimartingales. Moreover, the test function space can be chosen as $C_c^\infty(M)$.

Let $V_1, \cdots, V_\ell \in \Gamma(TM)$, Z be an $\mathbb{R}^\ell$-valued semimartingale, and an M-valued random variable $X_0 \in \mathcal{F}_0$. Consider an equation symbolically written as

$$dX_t = V_\alpha \circ dZ_t^\alpha,$$

and denoted it as $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$.

Definition 1.2.3. An M-valued semimartingale X defined up to a stopping time τ is a solution of $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$ up to τ if for all $f \in C^\infty(M)$,

$$f(X_t) = f(X_0) + \int_0^t (V_\alpha f)(X_s) \circ dZ_s^\alpha.$$

Proposition 1.2.4. *Suppose* $\Phi\colon M \to N$ *is a diffeomorphism and* X *a solution of* $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$. *Then* $\Phi(X)$ *is a solution of* $\mathrm{SDE}(\Phi_* V_1, \cdots, \Phi_* V_\ell; Z, \Phi(X_0))$ *on* N.

Proof. Let $Y = \Phi(X)$. For any $f \in C^\infty(N)$, because $f \circ \Phi \in C^\infty(M)$ and X is a solution of $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$,

$$f \circ \Phi(X_t) = f \circ \Phi(X_0) + \int_0^t V_\alpha(f \circ \Phi)(X_s) \circ dZ_s^\alpha.$$

Note that $\Phi_*\colon \Gamma(TM) \to \Gamma(TN)$ such that for and $f \in C^\infty(N)$,

$$\Phi_* V(f)(\Phi(x)) = V(f \circ \Phi)(x).$$

It follows that

$$f(Y_t) = f(Y_0) + \int_0^t \Phi_* V_\alpha(f)(Y_s) \circ dZ_s^\alpha,$$

which means that Y is a solution of $\mathrm{SDE}(\Phi_* V_1, \cdots, \Phi_* V_\ell; Z, \Phi(X_0))$. □

Note that by Whitney's embedding theorem, any smooth manifold M can be properly embedded in a Euclidean space, i.e., $i\colon M \hookrightarrow \mathbb{R}^N$, where i is a smooth proper map. Therefore, $i(M)$ is a closed submanifold in $\mathbb{R}^N$. Note that here closed submanifold means that a submanifold that is a closed subset. Therefore, we can always view $M \subset \mathbb{R}^N$ as an embedded submanifold that is closed. Moreover, we usually assume M without boundary. When M is non-compact, we can consider its one-point compactification $\widehat{M} = M \cup \{\partial_M\}$. So $\{x_n\}_n \subset M$ such that $x_n \to \partial_M$ in $\widehat{M}$ if and only if $\|x_n\|_{\mathbb{R}^N} \to \infty$.
In the following we always assumed that $M \subset \mathbb{R}^N$ is a submanifold that is closed and without boundary. Moreover, let

$$i = (f^1, \cdots, f^N)\colon M \hookrightarrow \mathbb{R}^N$$

and f^i are coordinate functions that $f^i(x) = x^i$.

Proposition 1.2.5. *Suppose $M \subset \mathbb{R}^N$ is a submanifold that is closed. Let $f^1, \cdots, f^N$ be coordinate functions. Let X be an M-valued continuous process.*

(1) X is a semimartingale on M if and only if it is an $\mathbb{R}^N$-valued semimartingale, i.e. $f^i(X)$ is a $\mathbb{R}$-valued semimartingale for each $i = 1, 2, \cdots, N$.

(2) X is a solution of $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$ up to a stopping time σ if and only if for each $i = 1, 2, \cdots, N$,

$$f^i(X_t) = f^i(X_0) + \int_0^t (V_\alpha f^i)(X_s) \circ dZ_s^\alpha, \quad 0 \le t < \sigma.$$

Proof. (1) $\Rightarrow$: It is by the definition.

$\Leftarrow$: For any $f \in C^\infty(M)$, since M is closed $\mathbb{R}^N$, it can be extended on $\tilde{f} \in C^\infty(\mathbb{R}^N)$ such that $f(X) = \tilde{f}(X)$. Because X is $\mathbb{R}$-valued semimartingale, $f(X) = \tilde{f}(X)$ is also a semimartingale.

(2) $\Rightarrow$: It is by the definition.

$\Leftarrow$: For any $f \in C^\infty(M)$, let $\tilde{f} \in C^\infty(\mathbb{R}^N)$ be an extension. Then

$$f(X_t) = \tilde{f}\left(f^1(X_t), \cdots, f^N(X_t)\right).$$

Therefore,

$$\begin{aligned} df(X_t) &= \tilde{f}\left(f^1(X_t), \cdots, f^N(X_t)\right) \circ d(f^i(X_t)) \\ &= \tilde{f}\left(f^1(X_t), \cdots, f^N(X_t)\right) \circ V_\alpha f^i(X_t) \circ dZ_t^\alpha \\ &= \Big(\tilde{f}\left(f^1(X_t), \cdots, f^N(X_t)\right) V_\alpha f^i(X_t)\Big) \circ dZ_t^\alpha \\ &= V_\alpha \tilde{f}(X_t) \circ dZ_t^\alpha = V_\alpha f(X_t) \circ dZ_t^\alpha, \end{aligned}$$

where the final equality is because $V_\alpha \in \Gamma TM$ and $\tilde{f}$ is an extension of f. □

Remark 1.2.6. Note that for $f \in \mathbb{R}^k \to \mathbb{R}$ and $g\colon \mathbb{R}^n \to \mathbb{R}^k$

$$\begin{aligned} d(f \circ g(M)) &= f_{x_i}(g(M)) \circ dg^i(M) \\ &= f_{x_i}(g(M)) \circ g^i_{x_j}(M) \circ dM^j \\ &= \nabla f(g(M))^\top Dg(M) \circ dM \end{aligned}$$

under Stratonovich integral.

Remark 1.2.7. If X is a solution of $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$ up to its explosion time $e(X)$ and $X_0 \in M$, then $X_t \in M$ for $0 \le t < e(X)$. Moreover, $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$ is unique up its explosion time.

1.3 Diffusion Process

Let L be a smooth second order elliptic, but not necessarily non degenerate, differential operator on a smooth manifold M, that is,

$$L\colon C^\infty(M) \to C^\infty(M)$$

such that on every chart $(U; x^1, \cdots, x^d)$

$$Lf(x) = a^{ij}(x)\frac{\partial^2}{\partial x^i \partial x^j} f(x) + b^i(x)\frac{\partial}{\partial x^i} f(x)$$

for some $a^{ij}, b^i \in C^\infty(U)$ and $\sum_{i,j} a^{ij}(x)\xi_i\xi_j \geq 0$.

Remark 1.3.1. If M is equipped with a torsion-free connection ∇, like a Riemannian manifold, then L can be written as a coordinate-free formulation, that is,

$$Lf := \left\langle A, \nabla^2 f \right\rangle + \langle b, \nabla f\rangle ,$$

for $b \in \Gamma(TM)$, and A, a symmetric $(2,0)$-tensor, where $\langle \cdot, \cdot \rangle$ is the natural dual operation.

For $f \in C^2(M)$ and $\omega \in W(M)$, let

$$M^f(\omega)_t = f(\omega_t) - f(\omega_0) - \int_0^t Lf(\omega_s)ds.$$

Definition 1.3.2. An $\mathbb{F}$-adapted process X defined on a filtered $(\Omega, \mathbb{F}, \mathbb{P})$ and that is valued in M is called a diffusion process generated by L if X is a M-valued semimartingale w.s.t. $\mathbb{F}$ up to $e(X)$ and

$$M^f(X)_t = f(X_t) - f(X_0) - \int_0^t Lf(X_s)ds, \quad 0 \leq t < e(X),$$

is a $\mathbb{F}$-local martingale for all $f \in C^\infty(M)$.

A probability measure μ defined on $(W(M), \mathcal{F}_{M,\infty})$ is called a diffusion measure generated by L if

$$M^f(\omega)_t = f\left(\omega_t\right) - f\left(\omega_0\right) - \int_0^t Lf\left(\omega_s\right) ds, \quad 0 \leq t < e(\omega),$$

is a local $\mathbb{F}_M$ local martingale for all $f \in C^\infty(M)$.

Remark 1.3.3. If X is an L-diffusion, then $\mu^X = X_\# \mathbb{P}$ is an L-diffusion measure. Conversely, if μ is an L-diffusion measure, then the coordinate process is an L-diffusion process.

For given L, assume locally

$$L = \frac{1}{2}\sum_{i,j=1}^d a^{ij}(x)\frac{\partial^2}{\partial x^i \partial x^j} + \sum_{i=1}^d b^i(x)\frac{\partial}{\partial x^i},$$

where $(a^{ij}(x))$ is positive semi-definite. Define

$$\Gamma(f,g) = L(fg) - fLg - gLf, \quad f,g \in C^\infty(M).$$

So locally

$$\Gamma(f,g) = a^{ij}\frac{\partial f}{\partial x^i}\frac{\partial f}{\partial x^j}.$$

And the semi-elliptic means that for any $f \in C^\infty(M)$, $\Gamma(f,f) \geq 0$. Combining this with bilinearity, it implies that if $g^1, \cdots, g^n$, the matrix $(\Gamma(g^i, g^j))$ is positive semimartingale.
Here is another explanation for Γ. For any $f, g \in C^\infty(M)$, let $h = fg$. Let $Y = f(X)$ and $Z = g(X)$, so $h(X) = YZ$. First,

$$Y_t Z_t = h(X)_t = Y_0 Z_0 + M_t^h + \int_0^t L(fg)(X_s)ds.$$

On the other hand, because

$$\begin{aligned} Y_t &= Y_0 + M_t^f + \int_0^t Lf(X_s)ds \\ Z_t &= Z_0 + M_t^g + \int_0^t Lf(X_s)ds. \end{aligned}$$

and M^f, M^g are local martingales, by Itô formula,

$$\begin{aligned} Y_t Z_t = Y_0 Z_0 &+ \int_0^t f(X_s)dM_s^g + \int_0^t g(X_s)dM_s^f \\ &+ \int_0^t \left(fLg + gLf\right)(X_s)ds \\ &+ \left\langle M^f, M^g \right\rangle_t. \end{aligned}$$

Therefore, we have

$$\int_0^t \Gamma\left(f,g\right)(X_s)ds - \left\langle M^f, M^g \right\rangle_t = \int_0^t f(X_s)dM_s^g + \int_0^t g(X_s)dM_s^f - M_t^h.$$

Note that the LHS is a process with bounded variation and the RHS is a continuous local martingale, which implies that the RHS is constant, i.e., 0. Then we have

$$M_t^h = \int_0^t f(X_s)dM_s^g + \int_0^t g(X_s)dM_s^f.$$

Moreover,

$$\left\langle Y, Z \right\rangle_t = \left\langle M^f, M^g \right\rangle_t = \int_0^t \Gamma\left(f,g\right)(X_s)ds.$$

Existence of Diffusion Process. Assume M is a submanifold of $\mathbb{R}^N$, which is closed. Let $z = (z^1, \cdots, z^N)$ be coordinates in $\mathbb{R}^N$ and f^α $(\alpha = 1, 2, \cdots, N)$ are coordinate maps $M \to \mathbb{R}^N$ by $f^\alpha(z) = z^\alpha$. Let

$$\tilde{a}^{\alpha\beta} = \Gamma(f^\alpha, f^\beta), \quad \tilde{b}^\alpha = Lf^\alpha.$$

Then they are in $C^\infty(M)$ and $(\tilde{a}^{\alpha\beta})$ is positive semi-definite. By the closedness of M, $\tilde{a}, \tilde{b}$ can be extended to $\mathbb{R}^N$ and let

$$\tilde{L} = \frac{1}{2}\sum_{\alpha,\beta=1}^N \tilde{a}^{\alpha\beta}\frac{\partial^2}{\partial z^\alpha \partial z^\beta} + \sum_{\alpha=1}^N \tilde{b}^\alpha \frac{\partial}{\partial z^\alpha}$$

be defined on $\mathbb{R}^N$. It is a true extension of L.

Lemma 1.3.4. *Suppose $f \in C^\infty(M)$. Then for any $\tilde{f} \in C^\infty(\mathbb{R}^N)$ which extends f from M to $\mathbb{R}^N$, $\tilde{L}\tilde{f} = Lf$ on M.*

Proof. Let $(x^1, \cdots, x^d)$ be a local coordinate on M. Note that

$$f(x) = \tilde{f}(f^1(x), \cdots, f^N(x)).$$

Then

$$\begin{aligned} Lf(x) &= L\tilde{f}(f^1(x), \cdots, f^N(x)) \\ &= \frac{1}{2}a^{ij}(x)\frac{\partial^2 \tilde{f}(f^1(x), \cdots, f^N(x))}{\partial x^i \partial x^j} + b^i(x)\frac{\partial \tilde{f}(f^1(x), \cdots, f^N(x))}{\partial x^i} \\ &= \tilde{L}\tilde{f}(z). \end{aligned}$$

□

For $\tilde{L}$ on $\mathbb{R}^n$, if $\tilde{a} = \tilde{\sigma}\tilde{\sigma}^\top$, it is the generator of diffusion process that satisfies

$$X_t = X_0 + \int_0^t \tilde{\sigma}(X_s)dW_s + \int_0^t \tilde{b}(X_s)ds.$$

We can clearly choose $\tilde{\sigma} = \tilde{a}^{\frac{1}{2}}$ and W a standard Brownian motion in $\mathbb{R}^N$ independent of X_0.

Remark 1.3.5. The existence of solution up to a stopping time needs the local Lipschitz of $\tilde{\sigma}$, which can be guaranteed by the local Lipschitz of $\tilde{a}$.

Moreover, for such diffusion process X on $\mathbb{R}^N$, $\mu^X = X_{\#}\mathbb{P}$, the distribution on $W(\mathbb{R}^N)$, is concentrated on $W(M)$, i.e., X is in fact a L-diffusion on M. In fact, we have the following theorem.

Theorem 1.3.6. *If L is a smooth second order semi-elliptic operator a C^∞-manifold M and μ_0 is a probability measure on M, then there exists a unique L-diffusion measure with initial distribution μ_0.*

Remark 1.3.7. The existence and uniqueness of such $\mathbb{P}_{\mu_0}$ guarantee that the L martingale problem is well-posed. So if X is a L-diffusion process, then it is a strong Markov process w.s.t. $\mathbb{F}^X$ and its transition group $P_t f = \mathbb{E}^x[X_t]$ has the generator extending L. In this case, in fact, X is a strong Markov process w.s.t. $\mathbb{F}$.

Driving by Brownian Motion. Let M be a Riemannian manifold and consider all solutions of SDE

$$dX_t = V_\alpha(X_t) \circ dW_t^\alpha + V_0(X_t)dt,$$

where $V_\alpha, V_0 \in \Gamma(TM)$ and $W = (W^\alpha)$ is an ℓ-dimensional Brownian motion. First, by Itô formula,

$$dV_\alpha f(X_t) = V_0(V_\alpha f(X_t))dt + V_\beta(V_\alpha f(X_t)) \circ dW_t^\beta,$$

which implies that

$$d\left\langle V_\alpha f(X_t), W_t^\alpha \right\rangle = V_\beta(V_\alpha f(X_t))d\left\langle W^\beta, W^\alpha \right\rangle_t = \sum_\alpha V_\alpha(V_\alpha f(X_t))dt$$

Therefore, we have

$$\begin{aligned} f(X_t) - f(X_0) &= \int_0^t V_0 f(X_t)dt + \int_0^t V_\alpha f(X_t) \circ dW_t^\alpha \\ &= \int_0^t V_0 f(X_t)dt + \int_0^t V_\alpha f(X_t)dW_t^\alpha + \frac{1}{2}\left\langle V_\alpha f(X_t), W^\alpha \right\rangle_t \end{aligned}$$

$$= \int_0^t V_\alpha f(X_t) dW_t^\alpha + \int_0 \left(V_0 f(X_t) + \frac{1}{2} \sum_\alpha V_\alpha(V_\alpha f(X_t)) \right) dt$$

Denote $V_\alpha(V_\alpha f(x)) = V_\alpha^2 f(x)$ and let

$$L = \frac{1}{2} \sum_\alpha V_\alpha^2 + V_0,$$

which is obviously a semi-elliptic second order differential operator. From above,

$$f(X_t) - f(X_0) - \int_0^t Lf(X_t)dt = \int_0^t V_\alpha f(X_t) dW_t^\alpha.$$

So a solution to $dX_t = V_\alpha(X_t) \circ dW_t^\alpha + V_0(X_t)dt$ is a diffusion process generated by L. Moreover, assume that V_α, V_0 satisfy

$$\sup_{t \in [0,T]} \sup_{x \in U} \mathbb{E}\left[|D^\alpha f(X_t)| \mid X_0 = x \right] < \infty$$

for all $f \in C_c^\infty(M)$, $T > 0$, coordinate chart U with $\overline{U}$ compact, and index α, where D^α is the usual derivative.

Remark 1.3.8. In fact, assume $M \subset \mathbb{R}^n$ and V_α, V_0 are the restriction of $\mathbb{R}^n$-vector fields $\tilde{V}_\alpha, \tilde{V}_0$ on M. If $\tilde{V}_\alpha = (a_\alpha^j), \tilde{V}_0 = (b^j)$ satisfy the condition of the existence and uniqueness of solution of SDE, then it satisfies the above condition.

Then define

$$u(t,x) = \mathbb{E}[f(X_t) \mid X_0 = x], \quad f \in C_c^\infty(M).$$

Because $f \in C_c^\infty(M)$,

$$M_t^f = f(X_t) - f(X_0) - \int_0^t Lf(X_s)ds$$

is a martingale, which implies that $\mathbb{E}[M_t^f \mid X_0] = M_0^f = 0$. Therefore,

$$u(t,x) = \mathbb{E}[f(X_t) \mid X_0 = x] = f(x) + \int_0^t \mathbb{E}\left[Lf(X_s) \mid X_0 = x \right] ds,$$

called the Dynkin's formula. Moreover, for $u_t = u(t, \cdot)$,

$$\begin{cases} \dfrac{\partial u_t}{\partial t} = Lu_t, \\ u(0,x) = f(x). \end{cases}$$

This result can be extended to so-called Feynman-Kac formula. Let

$$v(t,x) = \mathbb{E}\left[\exp\left(\int_0^t g(X_s)ds \right) f(X_t) \mid X_0 = x \right].$$

Then $v_t = v(t, \cdot)$ satisfies

$$\begin{cases} \dfrac{\partial v_t(x)}{\partial t} = Lv_t(x) + g(x)v_t(x), \\ v(0,x) = f(x). \end{cases}$$

Chapter 2

Stochastic Differential Geometry

2.1 Horizontal Lift

Let M be d-dimensional smooth manifold equipped with an affine connection ∇. Let Γ_{ij}^k be the corresponding Christoffel symbol, i.e,

$$\nabla_{X_i} X_j = \Gamma_{ij}^k X_k,$$

where $X_i = \frac{\partial}{\partial x^i}$. ∇ can induce concepts, such as, parallel moving ($\nabla_{\dot{c}(t)} V = 0$) and geodesic ($\nabla_{\dot{c}(t)} \dot{c}(t) = 0$).
For $x \in M$, a frame at x is a $\mathbb{R}$-linear isomorphism $u \colon \mathbb{R}^d \to T_x M$. Let $F(M)_x$ be the set of all frameworks at x. $GL(d, \mathbb{R})$ acts on $F(M)_x$ as $u \mapsto ug = u \circ g$. Let $F(M) = \bigcup_{x \in M} F(M)_x$, called the frame bundle of dimension $d + d^2$, with the canonical projection $\pi \colon F(M) \to M$. Note that each $F(M)_x \cong GL(d, \mathbb{R})$. In fact, $(F(M), M, GL(d, \mathbb{R}))$ is a principal bundle $M \cong F(M)/GL(d, \mathbb{R})$. Let

$$F(M) \times_{GL(d,\mathbb{R})} \mathbb{R}^d := F(M) \times \mathbb{R}^d / \sim,$$

where $(ug, v) \sim (u, gv)$. Then define

$$\Phi \colon F(M) \times_{GL(d,\mathbb{R})} \mathbb{R}^d \to TM$$

by

$$\Phi([u, v]) = u(v) \in T_{\pi(u)} M.$$

Φ is a linear isomorphism and so $F(M) \times_{GL(d,\mathbb{R})} \mathbb{R}^d \cong TM$.
Consider the canonical projection $\pi \colon F(M) \to M$, because it is a smooth submersion and $\pi^{-1}(x) = F(M)_x \cong GL(d, \mathbb{R})$, for

$$d\pi_u \colon T_u F(M) \to T_{\pi(u)} M,$$

the vertical space

$$V_u F(M) := \ker(d\pi_u) = T_u F(M)_{\pi(u)} \cong \mathfrak{g}(d, \mathbb{R}).$$

Let M be equipped with an affine connection ∇. Let $c(t)$ be a smooth curve on M and $u(t)$ be a lift of $c(t)$ on $F(M)$, i.e., $\pi(u(t)) = c(t)$. Because $u(t) \in F(M)_{c(t)}$, for any $e \in \mathbb{R}^d$, $u_e(t) = u(t)e \in T_{c(t)} M$, which means $u_e \in \Gamma(TM)$ is a vector field along $c(t)$. We say $u(t)$ is a horizontal lift if for any $e \in \mathbb{R}^d$, u_e is parallel along c, i.e.,

$$\nabla_{\dot{c}(t)} u_e(t) \equiv 0.$$

Lemma 2.1.1. *For any smooth curve $c\colon (-\varepsilon, \varepsilon) \to M$, let $u_0 \in F(M)_{c(0)}$. Then there exists a unique $u\colon (-\varepsilon, \varepsilon) \to F(M)$ such that*

$$\pi \circ u = c, \quad u(0) = u_0, \quad \nabla_{\dot{c}(t)} \dot{u}_a(t) = 0, \ \forall\ a \in \mathbb{R}^d.$$

Moreover, u dependents smoothly on (c, u_0), and if $u_1 = u_0 g$, the $t \mapsto u(t)g$ is the unique horizontal lift starting from u_1.

Proof. WTLG, assume $c((-\varepsilon, \varepsilon))$ is contained in a chart $(U; x^1, \cdots, x^d)$ and let Γ^k_{ij} be the corresponding Christoffel symbol for ∇. Let $e_1, \cdots, e_d$ be the canonical basis of $\mathbb{R}^d$
Then a vector field $V(t) = V^i(t) \frac{\partial}{\partial x^i}\big|_{c(t)}$ along $c(t)$ is parallel to $c(t)$ if and only if $V(t)$ satisfies

$$\dot{V}^i(t) + A^i_k(t) V^k(t) = 0, \quad \forall\ i,$$

where $A^i_k(t) = \Gamma^i_{jk}(c(t))\dot{c}^j(t)$. For each e_j, above equation has a unique solution

$$V_{e_j}(t) = V^i_j(t) \left.\frac{\partial}{\partial x^i}\right|_{c(t)}$$

with initial value $V_{e_j}(0) = u_0 e_j = u^i_{0,j} \frac{\partial}{\partial x^i}\big|_{c(0)}$. So the matrix $U(t) = (V^i_j(t))$ is the unique solution of ODE

$$\dot{U}(t) + A(t)U(t) = 0, \quad U(0) = U_0 := (u^i_{0,j}).$$

Moreover, because

$$\frac{d}{dt} \det U(t) = \operatorname{tr}\left(U(t)^{-1}\dot{U}(t)\right) \det U(t) = -\operatorname{tr}(A(t)) \det U(t),$$

and $u_0 \in F(M)_{c(0)}$ (i.e., $\det u_0 \neq 0$),

$$\det U(t) = \det U_0 \cdot \exp\left(-\int_{t_0}^t \operatorname{tr}(A(s))ds\right) \neq 0,$$

it means $U(t) \in GL(d, \mathbb{R})$ for all t. Let

$$u(t)a := V^i_j(t) a^j \left.\frac{\partial}{\partial x^i}\right|_{c(t)}$$

for any $a = (a^j) \in \mathbb{R}^d$. Then $u(t) \in F(M)_{c(t)}$ and satisfies $u(0) = u_0$ and the parallel condition. The uniqueness of $u(t)$ is directly obtained by the uniqueness of solution of the ODE, which also implies that $t \mapsto u(t)g$ is the unique lift starting from u_1. Also, the smoothness on initial values is also by the theory of ODE. □

Remark 2.1.2. Moreover, give a curve $c(t)$ on M, let $u(t), v(t)$ be its horizontal lifts starting from u_0, v_0 respectively, because $u_0 = v_0(v_0^{-1}u_0)$, by above

$$u(t) = v(t)(v_0^{-1}u_0).$$

Moreover, for any t_0, t_1, it is obvious that

$$\mathcal{P}_{c,t_0,t_1} = u_{t_1} u_{t_0}^{-1} \colon T_{c(t_0)}M \to T_{c(t_1)}M$$

is the parallel moving along $c(t)$ by considering $V(t) = u_t(u_{t_0}^{-1}(X))$ and the uniqueness of parallel moving.

Remark 2.1.3. Moreover, if $\dot{c}(0) = X = X^i \left.\frac{\partial}{\partial x^i}\right|_{c(0)}$, then we can see

$$\dot{u}(0) = -\Gamma^i_{\ell k}(c(0))X^\ell u^k_{0,j} \left.\frac{\partial}{\partial v^i_j}\right|_{u_0} + X^i \left.\frac{\partial}{\partial x^i}\right|_{u_0}$$

which implies $c(0)$, $\dot{c}(0)$, and u_0 uniquely determined $\dot{u}(0) \in T_{u_0}F(M)$.
For a given $X \in T_xM$, let c be a smooth curve in M with $c(0) = x$ and $\dot{c}(0) = X$. Let $u \in F(M)_x$. Let $u(t)$ be the unique horizontal lift of c starting from u. Then define

$$X^H_u = \dot{u}(0) \in T_uF(M),$$

which is well-defined because it is independent of the choice of c, called a horizontal vector. Let

$$H_uF(M) := \left\{X^H_u : X \in T_xM\right\} \subset T_uF(M),$$

called the horizontal space.

Lemma 2.1.4. *For the canonical projection $\pi\colon F(M) \to M$ and any $u \in F(M)$ with $x = \pi(u)$,*

$$d\pi_u \colon H_uF(M) \to T_xM$$

is an isomorphism. Therefore,

$$T_uF(M) = V_uF(M) \oplus H_uF(M).$$

Proof. For any $X^H_u \in H_uF(M)$, $X^H_u = \dot{u}(0)$ for some $u(t)$ which is a horizontal lift of $c(t)$ with $c(0) = x$ and $\dot{c}(0) = X$ and starting from u with $\pi(u) = x$. Then

$$d\pi_u(X^H_u) = \left.\frac{d}{dt}\right|_{t=0} \pi(u(t)) = \left.\frac{d}{dt}\right|_{t=0} c(t) = X.$$

And from the construction, we have seen $X \neq 0$ implies that $X^H_u \neq 0$. $\square$

For any $e \in \mathbb{R}^d$ and $u \in F(M)$, $ue \in T_{\pi(u)}M$ and so we can consider

$$H_e(u) := (ue)^H_u = \dot{u}(0) \in H_uF(M),$$

where $u(t)$ is the horizontal lift of $c(t)$ for $c(0) = \pi(u)$ and $\dot{c}(0) = ue$ starting from u. By above, we know

$$dg_uH_e(u) = H_{g^{-1}e}(ug)$$

where $g\colon F(M) \to F(M)$ is $g(u) = u \circ g = ug$, because

$$dg_uH_e(u) = \left.\frac{d}{dt}\right|_{t=0} u(t)g.$$

Remark 2.1.5. Note that because

$$(d\pi_u)^{-1} \colon X \mapsto X^H_u$$

is $\mathbb{R}$-linear,

$$H_{\lambda e + \mu e'}(u) = \lambda H_e(u) + \mu H_{e'}(u), \quad \forall\ e, e' \in \mathbb{R}^d,\ u \in F(M),\ \lambda, \mu \in \mathbb{R}.$$

Let $e_1, \cdots, e_d$ be the canonical basis of $\mathbb{R}^d$. Then

$$H_j(u) = H_{e_j}(u) \in H_uF(M)$$

forms a basis of $H_uF(M)$ for $j = 1, \cdots, d$, because for any $v = (v^i) \in \mathbb{R}^d$, i.e., $v = v^ie_i$, then

$$H_v(u) = \sum_{i=1}^d v^iH_i(u).$$

$\{H_i\}_{i=1}^d$ are called the fundamental horizontal vector fields.

Remark 2.1.6. For any $X \in \Gamma(TM)$, its unique horizontal lift vector field $X^H \in \Gamma(TF(M))$ is defined as

$$X^H(u) := (X(\pi(u)))^H_u \in T_u F(M), \quad \forall\, u \in F(M).$$

Moreover, by definition, $d\pi_u(X^H(u)) = X(\pi(u))$, i.e,

$$\pi_* X^H = X.$$

This is the reason why X^H is called a lift.

Let $c(t)$ be a smooth curve on M and $u(t)$ be the horizontal lift of it starting from u_0. Then we can consider a curve $w(t)$ on $\mathbb{R}^d$, called the anti-development of $c(t)$ w.s.t. u_0, defined as

$$w_{u_0}(t) = \int_0^t u(s)^{-1}\dot{c}(s)ds,$$

which is because $\dot{c}(t) \in T_{c(t)}M$ and $u(t)^{-1} \colon T_{c(t)}M \to \mathbb{R}^d$. But note that $w_{u_0}(t)$ dependents on the choice of u_0. Let $v(t)$ be the horizontal lift of $c(t)$ starting from v_0. If $u_0 = v_0 g$, i.e., $v_0 = u_0 g^{-1}$, then $v(t) = u(t)g^{-1}$ and thus

$$w_{v_0}(t) = \int_0^t v(s)^{-1}\dot{c}(s)ds = g\int_0^t u(s)^{-1}\dot{c}(s)ds = g w_{u_0}(t).$$

Fix u_0 and let $w(t) = w_{u_0}(t)$. Because $\dot{w}(t) = u(t)^{-1}\dot{c}(t)$, i.e, $u(t)\dot{w}(t) = \dot{c}(t)$,

$$H_{\dot{w}(t)}(u(t)) = (u(t)\dot{w}(t))^H_{u(t)} = (\dot{c}(t))^H_{u(t)} = \dot{u}(t),$$

i.e.

$$\dot{u}(t) = \dot{w}^i(t)H_i(u(t)).$$

Note that from the definition of $H_i(u(t))$ this is linear in $u(t)$, when given $w(t)$ and u_0, this ODE has a unique solution $u(t)$ called the development of $w(t)$ in $F(M)$ and its projection $c(t) = \pi(u(t))$ is called the development of $w(t)$ in M.

Orthogonal Frame. When considering (M, g) be a Riemannian manifold with the Levi-Civita connection ∇. Instead of considering $F(M)$, we consider the orthogonal frame $O(M)$, that is, $O(M)_x$ consists of all $u \colon \mathbb{R}^d \to T_x M$ orthogonal map. The problem is if we can consider the horizontal lift on $O(M)$. To do that, let

$$u(t)a = V^i_j(t)a^j \frac{\partial}{\partial x^i}$$

be the parallel vector field by solving the ODE

$$\dot{U}(t) + A(t)U(t) = 0.$$

The problem is whether $u(t) \in O(M)_{c(t)}$, that is, whether

$$\langle u(t)e_i, u(t)e_j\rangle = \left\langle V^\alpha_i \frac{\partial}{\partial x^\alpha}, V^\beta_j \frac{\partial}{\partial x^\beta} \right\rangle = V^\alpha_i g_{\alpha\beta} V^\beta_j = (U(t)^\top G(t)U(t))_{ij} = \delta_{ij},$$

i.e., $U(t)^\top G(t)U(t) \equiv I$, where $U(t) = (V^i_j(t))$ and $G(t) = (g_{ij}(c(t)))$, the local expression of metric. Note that, by the consistency of metric

$$\dot{g}_{ij}(c(t)) = \dot{c}(t)\left\langle \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^i} \right\rangle$$

$$\begin{aligned}
&= \left\langle \nabla_{\dot c(t)} \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle + \left\langle \frac{\partial}{\partial x^i}, \nabla_{\dot c(t)} \frac{\partial}{\partial x^j} \right\rangle \\
&= \dot c^\alpha \Gamma^k_{\alpha i} g_{kj} + \dot c^\alpha \Gamma^k_{\alpha j} g_{ik} \\
&= A^\top G + GA,
\end{aligned}$$

where $A = (A^i_k)$ for $A^i_k(t) = \Gamma^i_{jk}(c(t))\dot c^j(t)$. Moreover, because

$$\dot U(t) + A(t)U(t) = 0,$$

we have

$$\begin{aligned}
\frac{d}{dt} U^\top G U &= \dot U^\top G U + U^\top \dot G U + U^\top G \dot U \\
&= -U^\top A^\top G U + U^\top (A^\top G + GA) U - U^\top G(AU) = 0.
\end{aligned}$$

Therefore,

$$U \top G U \equiv U^\top(0) G(0) U(0) = I,$$

when $u(0) \in O(M)_{c(0)}$. Therefore, the horizontal lift lies on $O(M)$. For Riemannian manifolds, the following $F(M)$ can be replaced by $O(M)$.

2.2 Tensor Fields Scalarization

Let θ be a (r, s)-tensor field on M. For any $u \in F(M)$, let

$$X_i = u e_i \in T_{\pi(u)} M$$

for $i = 1, \cdot, d$, where $\{e_i\}$ is the canonical basis of $\mathbb{R}^d$. Then $\{X_i\}$ is a basis of $T_{\pi(u)}M$. Choose $\{X^i\}$ be the its dual basis of $T^*_{\pi(u)}M$. Then under these basis, we have

$$\theta(\pi(u)) = \theta^{i_1 \cdots i_r}_{j_1 \cdots j_s} X_{i_1} \otimes \cdots \otimes X_{i_r} \otimes X^{j_1} \otimes \cdots \otimes X^{j_s}.$$

and we define the scalarization

$$\widetilde{\theta} \colon F(M) \to (\mathbb{R}^d)^{\otimes r} \otimes (\mathbb{R}^d)^{* \otimes s} =: \otimes^{(r,s)} \mathbb{R}^d,$$

as

$$\widetilde{\theta}(u) = \theta^{i_1 \cdots i_r}_{j_1 \cdots j_s} e_{i_1} \otimes \cdots \otimes e_{i_r} \otimes e^{j_1} \otimes \cdots \otimes e^{j_s},$$

where $\{e^i\}$ is the dual basis of $\{e_i\}$ in $(\mathbb{R}^d)^*$. Note that $\widetilde{\theta}$ is $GL(d, \mathbb{R})$-equivariant, i.e.,

$$\widetilde{\theta}(ug) = g\tilde{\theta}(u).$$

If θ is $(0,0)$-tensor, i.e., $\theta = f \colon M \to \mathbb{R}$,

$$\widetilde{f} = f \circ \pi \colon F(M) \to \mathbb{R}.$$

Therefore, $\widetilde{\theta}$ can be considering as a lift of θ on $F(M)$.

Remark 2.2.1. For a finite dimensional vector space V, let $\{v_i\}$ be a basis of V and $\{v_i^*\}$ be its dual basis in V^*. If

$$w_j = g^i_j v_i$$

and $\{w_j\}$ is another basis, i.e., $G = (g^i_j)_{i \times j} \in GL$, then for $w^*_\ell = \sum_k h^\ell_k (v_k)^*$, we have the matrix

$$H = (h^\ell_k)_{\ell \times k} = G^{-1}.$$

Moreover, for $u \in F(M)$, $u^{-1}\colon T_{\pi(u)}M \to \mathbb{R}^d$ and so we can consider $u^*\colon T^*_{\pi(u)}M \to (\mathbb{R}^d)^*$, which follows that we can defined $u^{-1}\colon \otimes^{(r,s)} T_{\pi(u)}M \to \otimes^{(r,s)}\mathbb{R}^d$ as

$$u^{-1} = \underbrace{u^{-1} \otimes \cdots \otimes u^{-1}}_{r} \times \underbrace{u^* \otimes \cdots \otimes u^*}_{s}.$$

Then it is obvious that for $\theta \in \Gamma(\otimes^{(r,s)}TM)$

$$\widetilde{\theta}(u) = u^{-1}\theta(\pi(u)).$$

On the other hand, for $u\colon \mathbb{R}^d \to T_xM$, consider $(u^{-1})^*\colon (\mathbb{R}^d)^* \to T^*_{\pi(u)}M$. Then it can extend u to be defined on $\otimes^{(r,s)}(\mathbb{R}^d)^* \to \otimes^{(r,s)}T^*_{\pi(u)}M$ as

$$u = \underbrace{(u^{-1})^* \otimes \cdots \otimes (u^{-1})^*}_{r} \otimes \underbrace{u \otimes \cdots \otimes u}_{s},$$

where $\otimes^{(r,s)}(\mathbb{R}^d)^* = \otimes^r(\mathbb{R}^d)^* \otimes \otimes^s\mathbb{R}^d$ and $\otimes^{(r,s)}T^*_{\pi(u)}M = \otimes^r T^*_{\pi(u)}M \otimes \otimes^s T_{\pi(u)}M$. Then for any $\theta \in \Gamma(\otimes^{(r,s)}TM)$,

$$\theta_{\pi(u)}\colon\ \otimes^{(r,s)}\, T^*_{\pi(u)}M \to \mathbb{R}, \quad \widetilde{\theta}_u\colon\ \otimes^{(r,s)}\, (\mathbb{R}^d)^* \to \mathbb{R},$$

they have

$$\theta_{\pi(u)} \circ u = \widetilde{\theta}(u),$$

because $\{(u^{-1})^*e^j\}$ is the dual basis of $\{ue_i\}$ in $T^*_{\pi(u)}M$. For $f \in C^\infty(M)$, clearly $f_{\pi(u)} = \widetilde{f}(u)$. And so

$$\nabla_X f(\pi(u)) = \widetilde{\nabla_X f}(u).$$

Proposition 2.2.2. *Let $X \in \Gamma(TM)$ and $\theta \in \Gamma(\otimes^{(r,s)}TM)$. Then*

$$\widetilde{\nabla_X\theta} = X^H\widetilde{\theta},$$

i.e., $\widetilde{\nabla_X\theta}(u) = X^H_u\widetilde{\theta}$ for all $u \in F(M)$, where $X^H_u\widetilde{\theta}$ acts on each component since $X^H_u \in T_uF(M)$.

Proof. Fix $u \in F(M)$. Let $c(t)$ be a curve in M with $c(0) = \pi(u)$ and $\dot{c}(0) = X_{\pi(u)}$. Let $u(t)$ be the unique horizontal lift of $c(t)$ starting from $u(0) = u$. Then as we know, $\tau_t = u(t)u^{-1}$ is the parallel moving along $c(t)$. So we have

$$\left.\frac{d}{dt}\right|_{t=0} \tau_t^{-1}\theta(c(t)) = \nabla_X\theta(\pi(u)).$$

On the other hand, because $\pi(u(t)) = c(t)$, we have

$$\widetilde{\theta}(u(t)) = u(t)^{-1}\theta(c(t)).$$

Moreover, because $X^H_u = \dot{u}(0)$,

$$\begin{aligned} X^H_u\widetilde{\theta} &= \left.\frac{d}{dt}\right|_{t=0} \widetilde{\theta}(u(t)) \\ &= \left.\frac{d}{dt}\right|_{t=0} u(t)^{-1}\theta(c(t)) \\ &= u^{-1} \left.\frac{d}{dt}\right|_{t=0} \tau_t^{-1}\theta(c(t)) \\ &= u^{-1}(\nabla_X\theta)(\pi(u)) = \widetilde{\nabla_X\theta}(u), \end{aligned}$$

where the first equality is because $\widetilde{\theta}$ is a tensor field on $\mathbb{R}^d$, i.e. the coordinate does not change along time. $\square$

Remark 2.2.3. By the definition of $X_u^H\widetilde{\theta}$, because it is a tensor field on $\mathbb{R}^d$,

$$(X_u^H\widetilde{\theta})(v^1, \cdots, v^r, w_1, \cdots, w_s) = X_u^H\left(\widetilde{\theta}(u)(v^1, \cdots, v^r, w_1, \cdots, w_s)\right)$$

for any $v^i \in (\mathbb{R}^d)^*$ and $w_j \in \mathbb{R}^d$.

Remark 2.2.4. Above is also true for $f \in C^\infty(M)$.

$$\begin{aligned}
X_u^H\widetilde{f} &= \left.\frac{d}{dt}\right|_{t=0} \widetilde{f}(u(t)) \\
&= \left.\frac{d}{dt}\right|_{t=0} f(\pi(u(t))) = \left.\frac{d}{dt}\right|_{t=0} f(c(t)) \\
&= \nabla_X f(c(0)) = \nabla_X f(\pi(u)) = \widetilde{\nabla_X f}(u).
\end{aligned}$$

Example 2.2.5 (Hessian)**.** For any $f \in C^\infty(M)$, consider the Hessian $\nabla^2 f \in \Gamma(\otimes^{0,2}TM)$,

$$\begin{aligned}
\nabla^2 f_{\pi(u)}(ue_i, ue_j) &= (\nabla_{ue_j}\nabla f)_{\pi(u)}(ue_i) = (\widetilde{\nabla_{ue_j}\nabla f})(u)(e_i) \\
&= \left((ue_j)_u^H\widetilde{\nabla f}\right)(e_i) = H_j\left(\widetilde{\nabla f}(u)(e_i)\right) \\
&= H_j\left(\nabla f_{\pi(u)}(ue_i)\right) = H_j\left((\nabla_{ue_i} f)_{\pi(u)}\right) \\
&= H_j\left(\widetilde{\nabla_{ue_i} f}(u)\right) = H_jH_i\widetilde{f}
\end{aligned}$$

where $\widetilde{f} = f \circ \pi$.

2.3 Stochastic Horizontal Lift

Given a SDE on $F(M)$:

$$dU_t = H(U_t) \circ dW_t = \sum_{i=1}^{d} H_i(U_t) \circ dW_t^i, \tag{2.1}$$

where W is an $\mathbb{R}^d$-valued semimartingale and $\{H_i\}_{i=1}^d$ are the fundamental horizontal vector fields on $F(M)$.

Definition 2.3.1. (1) An $F(M)$-valued semimartingale is said to be horizontal if there exists an $\mathbb{R}^d$-valued semimartingale W with $W_0 = 0$ such that SDE (2.1) holds. Then W is called the anti-development of U (or of its projection $X = \pi(U)$).

(2) Let W be an $\mathbb{R}^d$-valued semimartingale with $W_0 = 0$ and U_0 is an $F(M)$-valued, $\mathcal{F}_0$-measurable random variable. The solution of SDE (2.1) with initial condition U_0 is called a stochastic development of W in $F(M)$, so is its projection $X = \pi(U)$.

(3) Let X be an M-valued semimartingale. An $F(M)$-valued horizontal semimartingale U such that $\pi(U) = X$ is called a stochastic horizontal lift of X.

Note that $W \mapsto U$ is by considering the solution of SDE (2.1), and $U \mapsto X$ is just by projection, but we need $X \mapsto U$ and $U \mapsto W$.

Assume $M \subset \mathbb{R}^N$ is a submanifold that is closed. Let $f = (f^\alpha)\colon M \to \mathbb{R}^N$ be the coordinate function and let lift it to

$$\widetilde{f}\colon F(M) \to M \subset \mathbb{R}^N$$

as $\widetilde{f} = f \circ \pi$, i.e. $\widetilde{f}(u) = f(\pi(u)) = \pi(u)$ as in $\mathbb{R}^N$.
For any M-valued semimartingale, we regard $X = (X^\alpha)_{\alpha=1}^N = (f^\alpha(X))_{\alpha=1}^N \in \mathbb{R}^N$ as an $\mathbb{R}^N$-valued semimartingale.
For any $x \in M$, let $P(x)\colon \mathbb{R}^N \to T_xM$ be the orthogonal projection by viewing $T_xM \subset \mathbb{R}^N$. Let $P_\alpha(x) = P(x)e_\alpha \in T_xM$ and so $P_\alpha \in \Gamma(TM)$.

Remark 2.3.2. For the smoothness of P_α, first we can equip M with the induced metric. Then on each local chart, there exists a smooth orthonormal frames E_i on TM, which implies that

$$P_\alpha = \sum_i \langle e_\alpha, E_i\rangle E_i.$$

So P_α is smooth.
Define $P_\alpha^H \in \Gamma(TF(M))$ as

$$P_\alpha^H(u) = (P_\alpha(\pi(u)))_u^H \in T_uF(M), \quad \forall\, u \in F(M).$$

By using the fundamental horizontal lifts,

$$P_\alpha^H(u) = \big(u(u^{-1}P_\alpha(\pi(u)))\big)_u^H = (u^{-1}P_\alpha(\pi(u)))^i H_i(u).$$

Lemma 2.3.3. *For notations as above, when viewing $T_xM \subset \mathbb{R}^N$, we have*

$$P_\alpha^H(u)\widetilde{f} = P_\alpha(\pi(u)), \quad P_\alpha(\pi(u))H_i(u)\widetilde{f^\alpha} = ue_i.$$

Proof. Let $u(t)$ be the horizontal lift from $u(0) = u$ of a curve $c(t)$ on M with $c(0) = \pi(u)$ and $\dot{c}(0) = \widetilde{f}(u) = P_\alpha(\pi(u))$. It follows that $P_\alpha^H(u) = \dot{u}(0)$. Because $\widetilde{f}\colon F(M) \to \mathbb{R}^N$ and $P_\alpha^H(u) \in T_uF(M)$,

$$P_\alpha^H(u)\widetilde{f} = \frac{d}{dt}\bigg|_{t=0} \widetilde{f}(u(t)) = \frac{d}{dt}\bigg|_{t=0} \pi(u(t)) = \frac{d}{dt}\bigg|_{t=0} c(t) = P_\alpha(\pi(u)).$$

Next, let $v(t)$ be the horizontal lift start from $v(0) = u$ of curve $x(t)$ with $x(0) = \pi(u)$ and $\dot{x}(0) = ue_i$. So

$$H_i(u)\widetilde{f} = \frac{d}{dt}\bigg|_{t=0} \widetilde{f}(v(t)) = \frac{d}{dt}\bigg|_{t=0} \pi(v(t)) = \frac{d}{dt}\bigg|_{t=0} x(t) = ue_i,$$

which implies that $H_i(u)\widetilde{f} \in T_{\pi(u)}M$ and so

$$P(\pi(u))H_i(u)\widetilde{f} = H_i(u)\widetilde{f} \;\Rightarrow\; P_\alpha(\pi(u))H_i(u)\widetilde{f^\alpha} = H_i(u)\widetilde{f} = ue_i. \qquad \square$$

Remark 2.3.4. For a vector field $H \in \Gamma(TF(M))$, we usually denote $Hg(u) := H(u)g$ for any $u \in F(M)$ and $g\colon F(M) \to \mathbb{R}^n$. Under this notation,

$$P_\alpha^H\widetilde{f}(u) = P_\alpha(\pi(u)), \quad P_\alpha(\pi(u))H_i\widetilde{f^\alpha}(u) = ue_i.$$

Remark 2.3.5. Note that above two identities view $P_\alpha(\pi(u))$ as a vector in $\mathbb{R}^N$. More precisely, for any $g \in C^\infty(M)$, $g \circ \pi \in C^\infty(F(M))$. We similarly have

$$H_i(u)(g \circ \pi) = \frac{d}{dt}\bigg|_{t=0} g(c(t)) = (ue_i)g.$$

and

$$P_\alpha^H(u)(g \circ \pi) = P_\alpha(\pi(u))g$$

or in other words,

$$d\pi_u(P_\alpha^H(u)) = P_\alpha(\pi(u)),$$

which is the definition of horizontal lift.

Lemma 2.3.6. *Assume $M \subset \mathbb{R}^N$ is a submanifold that is closed. If X is an M-valued semimartingale, then*

$$X_t = X_0 + \int_0^t P(X_s) \circ dX_s = X_0 + \int_0^t P_\alpha(X_s) \circ dX_s^\alpha.$$

Proof. Let $Q_\alpha(x) = e_\alpha - P_\alpha(x)$, i.e., $Q_\alpha(x)$ normal to T_xM and $Q_\alpha \in \Gamma(NM)$. Let

$$Y_t = X_0 + \int_0^t P_\alpha(X_s) \circ dX_t^\alpha$$

By the closedness of M, there exists $f \in C^\infty(\mathbb{R}^n)$ such that $f(x) \geq 0$ and $f^{-1}(0) = M$. Then by Itô formula,

$$f(Y_t) = f(X_0) + \int_0^t P_\alpha f(X_t) \circ dX_t^\alpha.$$

Because $P_\alpha \in \Gamma(TM)$ and $f|_M = 0$, $P_\alpha f(x) = 0$ for all $x \in M$. Since X is M-valued, $f(Y_t) = 0$, i.e., $Y \in f^{-1}(0) = M$, and Y is an M-valued semimartingale.
Next, consider a tubular neighborhood U of M and defined the natural projection $\tilde{h}\colon U \to M$, which can be extended a smooth $h\colon \mathbb{R}^N \to M$. Because $Q_\alpha \in \Gamma(NM)$, by choosing Fermi coordinates, we have

$$Q_\alpha h(x) = 0 \;\Rightarrow\; P_\alpha h(x) = \frac{\partial h}{\partial x^\alpha}.$$

So

$$\begin{aligned} Y_t = h(Y_t) &= h(X_0) + \int_0^t P_\alpha h(X_s) \circ dX_s^\alpha \\ &= X_0 + \int_0^t \frac{\partial h}{\partial x^\alpha} \circ dX_s^\alpha \\ &= h(X_t) = X_t. \end{aligned}$$

□

Theorem 2.3.7. *A horizontal semimartingale U on $F(M)$ has a unique anti-development W. In fact,*

$$W_t = \int_0^t U_s^{-1} P_\alpha(X_s) \circ dX_s^\alpha,$$

for $X = \pi(U)$.

Proof. By definition, W should be

$$dU_t = H_i(U_t) \circ dW_t^i.$$

Let $\widetilde{f}$ be defined as above. Then

$$\widetilde{f}(U_t) = f(\pi(U_t)) = f(X_t) = X_t \in M \subset \mathbb{R}^N.$$

Then X_t should satisfy

$$dX_t^\alpha = H_i \widetilde{f^\alpha}(U_t) \circ dW_t^i.$$

Therefore, by above formulas, we have

$$U_t^{-1} P_\alpha(X_t) \circ dX_t^\alpha = U_t^{-1} P_\alpha(X_t) H_i \widetilde{f}^\alpha(U_t) \circ dW_t^i = e_i dW_t^i = dW_t$$

By the uniqueness of SDE, W_t uniquely exists and

$$W_t = \int_0^t U_s^{-1} P_\alpha(X_s) \circ dX_s^\alpha.$$

□

Remark 2.3.8. Note that if U_t^1 and U_t^2 are two horizontal lifts of X starting from U_0^1 and $U_0^2 = U_0^1 g$, then by above, we obviously have

$$W_t^1 = g^{-1} W_t^2.$$

Remark 2.3.9. On the other hand,

$$\begin{aligned} P_\alpha(X_t) \circ dX_t^\alpha &= P_\alpha(X_t) H_i \widetilde{f^\alpha}(U_t) \circ dW_t^i \\ &= U_t e_i \circ dW_t^i. \end{aligned}$$

Then by above,

$$X_t = X_0 + \int_0^s U_s e_i \circ dW_s^i.$$

In formally, we just denote

$$X_t = X_0 + \int_0^s U_s \circ dW_s.$$

Similarly, because $dX_t = P_\alpha(X_t) \circ dX_t^\alpha$, we can informally write

$$W_t = \int_0^t U_s^{-1} \circ dX_s.$$

Theorem 2.3.10. *Suppose X is an M-valued semimartingale (up to a stopping time τ), and U_0 is an $F(M)$-valued $\mathcal{F}_0$-random variable such that $\pi(U_0) = X_0$. Then there exists a unique horizontal lift U (up to τ) of X starting from U_0.*

Proof. Consider a SDE defined as

$$dU_t = P_\alpha^H(U_t) \circ dX_t^\alpha.$$

Let U_t be its unique solution, so U_t is clearly a horizontal lift since $P_\alpha^H \in \Gamma F(M)$. It suffices to show $\pi(U_t) = X_t$. Let

$$Y_t = \widetilde{f}(U_t) = f(\pi(U_t)) = \pi(U_t) \in M \subset \mathbb{R}^N.$$

By above, we have

$$dY_t = P_\alpha^H \widetilde{f}(U_t) \circ dX_t^\alpha = P_\alpha(Y_t) \circ dX_t^\alpha.$$

By above lemma, we know $dX_t = P_\alpha(X_t) \circ dX_t^\alpha$. By the uniqueness of SDE, $X = Y$.
Next, assume there exists another horizontal lift Π_t of X_t. Then there exists a semimartingale W such that

$$d\Pi_t = H_i(\Pi_t) \circ dW_t^i.$$

and by above theorem

$$dW_t = \Pi_t^{-1} P_\alpha(X_t) \circ dX_t^\alpha.$$

So

$$\begin{aligned} d\Pi_t &= \left(\Pi_t^{-1} P_\alpha(X_t)\right)^i H_i(\Pi_t) \circ dX_t^\alpha \\ &= P_\alpha^H(\Pi_t) \circ dX_t^\alpha. \end{aligned}$$

So $\Pi = U$ by the uniqueness of solution of SDE. ☐

Remark 2.3.11. When considering explosion time, there is a fact, if X on M is a semimartingale up to τ, then its horizontal lift H on $F(M)$ is also defined up to τ.

Similarly as the deterministic case, given a semimartingale X on M, the horizontal lift U_t provide the stochastic parallel moving along X_t,

$$\tau^X_{t_1t_2} = U_{t_2}U_{t_1}^{-1} \colon T_{X_{t_1}}M \to T_{X_{t_2}}M.$$

Proposition 2.3.12. *Let a semimartingale X on a manifold M be the solution of* $\mathrm{SDE}(V_1, \cdots, V_N; Z, X_0)$ *and V_α^H be the horizontal lift of V_α. Then the horizontal lift U of X starting from U_0 satisfy* $\mathrm{SDE}(V_1^H, \cdots, V_N^H; Z, U_0)$*, and the anti-development of X is given by*

$$W_t = \int_0^t U_s^{-1}V_\alpha(X_s) \circ dZ_s^\alpha.$$

Proof. Assume $M \subset \mathbb{R}^N$ that is closed and use above notations. First, we similar have

$$V_\alpha^H\widetilde{f}(u) = V_\alpha(\pi(u)).$$

Let Π be the solution of $\mathrm{SDE}(V_1^H, \cdots, V_N^H; Z, U_0)$ and $Y_t = \widetilde{f}(\Pi_t) = \pi(\Pi_t)$.

$$dY_t = V_\alpha^H\widetilde{f}(\Pi_t) \circ dZ_t^\alpha = V_\alpha(Y_t) \circ dZ_t^\alpha,$$

which implies that $Y = X$ by the uniqueness of solution of $\mathrm{SDE}(V_1, \cdots, V_N; Z, X_0)$. And by the uniqueness of horizontal lift, $U = \Pi$.
Next, the anti-development of X is given by

$$dW_t = U_t^{-1}P_\beta(X_t) \circ dX_t^\beta.$$

Because $dX_t = V_\alpha(X_t) \circ dZ_t^\alpha$, i.e., $dX_t^\beta = V_\alpha^\beta(X_t) \circ dZ_t^\alpha$,

$$dW_t = U_t^{-1}P_\beta(X_t)V_\alpha^\beta(X_t) \circ dZ_t^\alpha.$$

Because $V_\alpha(X_t) \in T_{X_t}M$,

$$P_\beta(X_t)V_\alpha^\beta(X_t) = P(X_t)V_\alpha(X_t) = V_\alpha(X_t).$$

So

$$dW_t = U_t^{-1}V_\alpha(X_t) \circ dZ_t^\alpha.$$

□

2.4 Stochastic Line Integral

Line Integral. For any 1-form $\omega = f(t)dt$ on $[a, b]$ of $\mathbb{R}$, define

$$\int_{[a,b]} \omega = \int_a^b f(t)dt.$$

Let M be a smooth manifold and $\omega \in \Gamma(T^*M)$. Let $\gamma \colon [a, b] \to M$ smooth. Define the line integral of ω along γ as

$$\int_\gamma \omega = \int_{[a,b]} \gamma^*\omega.$$

Clearly, it satisfies the linearity in ω and additivity w.s.t. γ. Moreover, for any smooth $F \colon M \to N$ and $\eta \in \Gamma(T^*N)$ and γ in M,

$$\int_\gamma F^*\eta = \int_{F\circ\gamma} \eta.$$

Proposition 2.4.1. *Let $\omega \in \Gamma(T^*M)$ and $\gamma\colon [a,b] \to M$ smooth.*

$$\int_\gamma \omega = \int_a^b \omega_{\gamma(t)}(\dot\gamma(t))dt$$

In particular, if $\omega = df$ for some $f \in C^\infty(M)$,

$$\int_\gamma df = \int_a^b (f\circ\gamma)'(t)dt = f(\gamma(b)) - f(\gamma(a)).$$

Proof. Let $\gamma(t) = (\gamma^1(t), \cdots, \gamma^d(t))$ and $\omega = \omega_i dx^i$ be the coordinate expression. Then

$$(\gamma^*\omega)_t = \omega_i(\gamma(t))dx^i(\gamma(t)) = \omega_i(\gamma(t))d\gamma^i(t) = \omega_i(\gamma(t))(\gamma^i)'(t)dt$$

On the other hand,

$$\omega_{\gamma(t)}(\dot\gamma(t)) = \omega_i(\gamma(t))dx^i\left((\gamma^j)'(t)\frac{\partial}{\partial x^j}\right) = \omega_i(\gamma(t))(\gamma^i)'(t).$$

Therefore, we get the desired result. $\square$

Stochastic Line Integral. Let $x\colon [0,t] \to M$ and $u(t)$ be its horizontal lift on $F(M)$ starting from u_0. Let $w(t)$ be the anti-development of $x(t)$, i.e., $\dot w_t = u_t^{-1}\dot x_t$ and $w_0 = 0$. Let $\{e_i\}$ be the canonical basis of $\mathbb{R}^d$ and so $\{u_t e_i\}$ is a basis of $T_{x_t}M$. Let

$$w_t = w_t^i e_i \;\Rightarrow\; \dot x_t = \dot w_t^i\,(u_t e_i)\,.$$

Let $\theta \in \Gamma(t^*M)$. Then

$$\int_x \theta = \int_0^t \theta_{x_s}(\dot x_s)ds = \int_0^t \dot w_s^i \theta_{x_s}(u_s e_i)ds.$$

Definition 2.4.2. Let $\theta \in \Gamma(t^*M)$ and X be an M-valued semimartingale. Let U be a horizontal lift of X and W be its anti-development. The stochastic line integral of θ along X is defined by

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta_{X_s}(U_s e_i)\circ dW_s^i.$$

Remark 2.4.3. For such θ, let $\widetilde\theta$ be the scalarization of 1-form θ. Then by the definition, $\widetilde\theta\colon F(M) \to \mathbb{R}^d$ by viewing $(\mathbb{R}^d)^* = \mathbb{R}^d$ and

$$\widetilde\theta(u) = \left(\theta_{\pi(u)}(ue_1), \cdots, \theta_{\pi(u)}(ue_d)\right)^\top \in \mathbb{R}^d.$$

So

$$\int_{X_{[0,t]}} \theta = \int_0^t \widetilde\theta(U_s)_i \circ dW_s^i = \int_0^t \widetilde\theta(U_s)^\top \circ dW_s.$$

Note that $\int_{X_{[0,t]}} \theta$ seems depend on the choice of horizontal lift, and so the connection on M. Actually, it is independent of the above choices by the following proposition and the fact that any semimartingale on M is a solution of some SDE shown in above section.

Proposition 2.4.4. *Let $\theta \in \Gamma(T^*M)$ and X be the solution of $dX_t = V_\alpha(X_t)\circ dZ_t^\alpha$. Then*

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta(V_\alpha)(X_s)\circ dZ_s^\alpha.$$

Proof. Choose a horizontal lift U and corresponding anti-development W of X. Then

$$dW_t = U_t^{-1} V_\alpha(X_t) \circ dZ_t^\alpha.$$

Therefore,

$$\begin{aligned}
\widetilde{\theta}(U_s)_i \circ dW_s^i &= \widetilde{\theta}(U_s)_i (U_t^{-1} V_\alpha(X_s))^i \circ dZ_s^\alpha \\
&= \theta_{X_s}(U_s e_i)(U_t^{-1} V_\alpha(X_s))^i \circ dZ_s^\alpha \\
&= \theta_{X_s}\left(U_s\left(e_i(U_t^{-1} V_\alpha(X_s))^i\right)\right) \circ dZ_s^\alpha \\
&= \theta_{X_s}(V_\alpha(X_s)) \circ dZ_s^\alpha.
\end{aligned}$$

Therefore, we have

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta(V_\alpha)(X_s) \circ dZ_s^\alpha. \qquad \square$$

Remark 2.4.5. Note that the stochastic line integral of 1-form θ along a given M-valued semimartingale X is independent of the choice of horizontal lift and also the connection. Symbolically,

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta \circ dX_t$$

Example 2.4.6. (1) For all $f \in C^\infty(M)$, assume $dX_t = V_\alpha(X_t) \circ dZ_t^\alpha$.

$$\int_{X_{[0,t]}} df = \int_0^t V_\alpha f(X_s) \circ dZ_s^\alpha = f(X_t) - f(X_0)$$

(2) Let $(U, x^1, \cdots, x^d)$ be a local chart of M. Let $\theta = \theta_i dx^i$. For any X a semimartingale on M, let $X^i = x^i(X)$ that is a semimartingale on $\mathbb{R}$.

$$dX_t = V_i(X_t) \circ dX_t^i, \quad V_i = \frac{\partial}{\partial x^i},$$

because $df(X_t) = \frac{\partial f}{\partial x^i}(X_t) \circ dX_t^i$. Then

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta(V_i(X_s)) \circ dX_s^i = \int_0^t \theta_i(X_s) \circ dX_s^i,$$

which is the local expression of stochastic line integral.

Definition 2.4.7. Let $\theta \colon \Gamma(TF(M)) \to C^\infty(F(M), \mathbb{R}^d)$, i.e., a $\mathbb{R}^d$-valued 1-form on $F(M)$. Define

$$\theta(Z)(u) = \theta_u(Z_u) := u^{-1}(d\pi_u(Z_u)),$$

for any $Z \in \Gamma(TF(M))$ and $u \in F(M)$, where $\pi \colon F(M) \to M$ is the natural projection. Then θ is called the solder form.

Proposition 2.4.8. *Let U be a horizontal semimartingale on $F(M)$. Then its the corresponding anti-development is given by*

$$W_t = \int_{U_{[0,t]}} \theta,$$

where θ is the solder form.

Proof. Let $X_t = \pi(U_t)$. Then we have

$$dU_t = P^H_\alpha(U_t) \circ dX^\alpha_t.$$

Note that $\pi_* P^H_\alpha = P_\alpha$. So we get

$$\begin{aligned}\int_{U_{[0,t]}} \theta &= \int_0^t \theta(P^H_\alpha)(U_s) \circ dX^\alpha_s \\ &= \int_0^t U_s^{-1} d\pi_{U_s}(P^H_\alpha(U_s)) \circ dX^\alpha_s \\ &= \int_0^t U_s^{-1} P_\alpha(\pi(U_s)) \circ dX^\alpha_s = W_t.\end{aligned}$$

□

Quadratic Variation. Let $h \in \Gamma(\otimes^{(0,2)}TM)$ and $\widetilde{h}$ be its scalarization. So

$$\widetilde{h}(u)(e, e') = h_{\pi(u)}(ue, ue'), \quad \forall\ e, e' \in \mathbb{R}^d,\ u \in F(M).$$

Let

$$h^{\text{sym}}(e, e') = \frac{h(e, e') + h(e', e)}{2}.$$

Definition 2.4.9. Let $h \in \Gamma(\otimes^{(0,2)}TM)$ and X be an M-valued semimartingale. Let U be a horizontal lift of X and W be its anti-development. Note that

$$dX_t = U_t e_i \circ dW^i_t,$$

Then the h-quadratic variation of X is defined as

$$\begin{aligned}\int_0^t h(dX_s, dX_s) &:= \int_0^t h_{X_s}(U_s e_i, U_s e_j) d\left\langle W^i, W^j \right\rangle_s \\ &= \int_0^t \widetilde{h}(U_s)(e_i, e_j) d\left\langle W^i, W^j \right\rangle_s.\end{aligned}$$

By viewing $dW^i_s dW^j_s = d\langle W^i, W^j\rangle_s$, then

$$\int_0^t h(dX_s, dX_s) = \int_0^t \widetilde{h}(U_s)(dW_s, dW_s).$$

Obviously,

$$\int_0^t h(dX_s, dX_s) = \int_0^t h^{\text{sym}}(dX_s, dX_s)$$

and if h is anti-symmetric, $\int_0^t h(dX_s, dX_s) = 0$.

Remark 2.4.10. Note that this definition is independent the choice of U, i.e., the choice of U_0, because the anti-development of $U_t g$ is $g^{-1}W_t$ and

$$\begin{aligned}h(U_t g e_i, U_t g e_j) d\left\langle (g^{-1}W)^i, (g^{-1}W)^j \right\rangle_t &= \widetilde{h}(U_t)(g e_i, g e_j) d\left\langle (g^{-1}W)^i, (g^{-1}W)^j \right\rangle_t \\ &= g^\alpha_i g^\beta_j \widetilde{h}(U_t)(e_\alpha, e_\beta) \tilde{g}^i_k \tilde{g}^j_\ell d\left\langle W^k, W^\ell \right\rangle_t \\ &= \widetilde{h}(U_t)(e_k, e_\ell) d\left\langle W^k, W^\ell \right\rangle_t,\end{aligned}$$

where $g e_i = g^j_i$ and $(\tilde{g}^j_i) = (g^j_i)^{-1}$. So h-quadratic variation is well-defined.

Proposition 2.4.11. *Let $h \in \Gamma(\otimes^{(0,2)}TM)$ and X be the solution of* $\mathrm{SDE}(V_1, \cdots, V_N; Z, X_0)$. *Then*

$$\int_0^t h(dX_s, dX_s) = \int_0^t h(V_\alpha, V_\beta)(X_s) d\left\langle Z^\alpha, Z^\beta \right\rangle_s.$$

Proof. Note that

$$dX_t = V_\alpha(X_t) \circ dZ_t^\alpha$$

Let U be a horizontal lift of X and W be its anti-development. Then

$$dW_t = U_t^{-1} V_\alpha(X_t) \circ dZ_t^\alpha.$$

and so

$$d\left\langle W^i, W^j \right\rangle_t = \left(U_t^{-1} V_\alpha(X_t)\right)^i \left(U_t^{-1} V_\beta(X_t)\right)^j d\left\langle Z^\alpha, Z^\beta \right\rangle_t.$$

It implies that

$$\begin{aligned} h(U_t e_i, U_t e_j) d\left\langle W^i, W^j \right\rangle_t &= h(U_t e_i, U_t e_j) \left(U_t^{-1} V_\alpha(X_t)\right)^i \left(U_t^{-1} V_\beta(X_t)\right)^j d\left\langle Z^\alpha, Z^\beta \right\rangle_t \\ &= h(V_\alpha, V_\beta)(X_t) d\left\langle Z^\alpha, Z^\beta \right\rangle_t. \end{aligned}$$

□

Example 2.4.12. (1) Let $(U, x^1, \cdots, x^d)$ be a coordinate and $h \in \Gamma(\otimes^{(0,2)}TM)$ locally written as

$$h = h_{ij} dx^i \otimes dx^j.$$

For $X = (X^i)$ on M,

$$\int_0^t h(dX_s, dX_s) = \int_0^t h_{ij}(X_s) d\left\langle X^i, X^j \right\rangle_s,$$

by $dX_t = \frac{\partial}{\partial x^i}(X_t) \circ dX_t^i$.

(2) For $f \in C^\infty(M)$, $\nabla^2 f \in \Gamma(\otimes^{(0,2)}TM)$ and so

$$\begin{aligned} \int_0^t \nabla^2 f(dX_s, dX_s) &= \int_0^t \nabla^2 f_{X_s}(U_s e_i, U_s e_j) d\left\langle W^i, W^j \right\rangle_s \\ &= \int_0^t H_j H_i \widetilde{f}(U_s) d\left\langle W^i, W^j \right\rangle_s. \end{aligned}$$

(3) If $f, g \in C^\infty(M)$,

$$\int_0^t (df \otimes dg)(dX_s, dX_s) = \langle f(X), g(X) \rangle_t.$$

Proof. For X an M-valued semimartingale, WTLG assume $dX_t = V_\alpha(X_t) \circ dZ_t^\alpha$. Then

$$\int_0^t (df \otimes dg)(dX_s, dX_s) = \int_0^t V_\alpha f(X_s) V_\beta f(X_s) d\left\langle Z^\alpha, Z^\beta \right\rangle_s.$$

On the other hand, because

$$df(X_t) = V_\alpha f(X_t) \circ dZ_t^\alpha, \quad dg(X_t) = V_\beta g(X_t) \circ dZ_t^\beta,$$

we have

$$\langle f(X), g(X) \rangle_t = \int_0^t V_\alpha f(X_s) V_\beta f(X_s) d\left\langle Z^\alpha, Z^\beta \right\rangle_s.$$

□

Proposition 2.4.13. *If $\theta \in \Gamma(\otimes^{(0,2)}TM)$ is positive semi-definite, then the θ-quadratic variation of a semimartingale X is nondecreasing.*

Proof. First,

$$\int_0^t h\left(dX_s, dX_s\right) = \int_0^t h\left(U_s e_i, U_s e_j\right) d\left\langle W^i, W^j\right\rangle_s .$$

Because h is positive semi-positive, the matrix $\left(h\left(U_s e_i, U_s e_j\right)\right)$ is positive semi-definite and let $(m_i^k(s))$ be its squared root matrix. Let

$$J_t^k = \int_0^t m_i^k(s) dW_s^i.$$

Therefore,

$$\int_0^t h\left(dX_s, dX_s\right) = \sum_k \int_0^t m_i^k(s) m_j^k(s) d\left\langle W^i, W^j\right\rangle_s = \sum_k \left\langle J^k, J^k\right\rangle_t . \qquad \square$$

2.5 Martingale on Manifold

Definition 2.5.1. Suppose M is a C^∞ manifold with a connection ∇. An M-valued semimartingale X is called a ∇-martingale if its anti-development W w.s.t ∇ is an $\mathbb{R}^d$-valued local martingale.

Remark 2.5.2. It is well defined because if W and W' are two anti-development of X, then there exists a $g \in GL(d, \mathbb{R})$ such that $W' = gW$.

Remark 2.5.3. Note that when $M = \mathbb{R}^N$ is equipped with the canonical connection. Then for any $c(t)$ on M, its horizontal lift starting from u_0 is

$$u(t)a = (u_0)_j^i a^j \left.\frac{\partial}{\partial x^i}\right|_{c(t)} .$$

When viewing $T_x M \subset \mathbb{R}^N$, $u(t)a \equiv u_0 a$. Define $f\colon F(M) \to \mathbb{R}^N$ as $f(u) = ue_i$. Then

$$P_\alpha^H(u) f = \left.\frac{d}{dt}\right|_{t=0} f(u_t) = \left.\frac{d}{dt}\right|_{t=0} u_t e_i = 0.$$

Let X be an M-valued semimartingale. Because $dU_t = P_\alpha^H(U_t) \circ dX_t^\alpha$,

$$df(U_t) = P_\alpha^H f(U_t) \circ dX_t^\alpha = 0 \;\Rightarrow\; f(U_t) = U_t e_i = U_0 e_i.$$

Therefore,

$$\begin{aligned} X_t &= X_0 + \int_0^t U_s e_i \circ dW_s^i \\ &= X_0 + \int_0^t U_0 e_i dW_s^i \\ &= X_0 + U_0 W_t. \end{aligned}$$

It follows that when $M = \mathbb{R}^N$, if X is an M-martingale, then it is a local martingale in the usual sense.

Proposition 2.5.4. *An M-valued semimartingale X is a ∇-martingale if an only if*

$$N^f(X)_t := f(X_t) - f(X_0) - \frac{1}{2}\int_0^t \nabla^2 f(dX_s, dX_s),$$

is an $\mathbb{R}$-valued local martingale for every $f \in C^\infty(M)$.

Remark 2.5.5. For Euclidean case, a continuous semimartingale X is a local martingale if and only if $N^f(X)$ is a local martingale for all $f \in C^\infty$ (or $f \in C_c^\infty$) by Itô formula.

Proof. $\Rightarrow$: Let U be a horizontal lift of X and W be its anti-development. Then

$$dU_t = H_i(U_t) \circ dW_t^i.$$

For $f \in C^\infty(M)$, let $\widetilde{f} = f \circ \pi \in C^\infty(F(M))$. So

$$\begin{aligned}
f(X_t) - f(X_0) &= \widetilde{f}(U_t) - \widetilde{f}(U_0) \\
&= \int_0^t H_i\widetilde{f}(U_s) \circ dW_s^i \\
&= \int_0^t H_i\widetilde{f}(U_s) dW_s^i + \frac{1}{2}\left\langle H_i\widetilde{f}(U), W^i \right\rangle_t.
\end{aligned}$$

Note that

$$dH_i\widetilde{f}(U_t) = H_j H_i \widetilde{f}(U_t) \circ dW_t^j,$$

which implies that

$$\begin{aligned}
\left\langle H_i\widetilde{f}(U), W^i \right\rangle_t &= \int_0^t H_j H_i \widetilde{f}(U_s) d\left\langle W^i, W^j \right\rangle_s \\
&= \int_0^t \nabla^2 f(dX_s, dX_s).
\end{aligned}$$

Therefore,

$$N^f(X)_t = \int_0^t H_i\widetilde{f}(U_t) dW_t^i.$$

When X is an M-local semimartingale, i.e., W is an $\mathbb{R}^d$-local martingale, then $N^f(X)$ is obviously a local martingale.
$\Leftarrow$: Assume $M \subset \mathbb{R}^N$ properly embedded. Let $f\colon M \to \mathbb{R}^N$ be the coordinate function $f(x) = x$ and $\widetilde{f} = f \circ \pi$ be its lift on $F(M)$. When viewing $T_xM \subset \mathbb{R}^N$, we already have $H_i\widetilde{f}(u) = ue_i$. So similarly as above calculation,

$$\begin{aligned}
N^f(X)_t &:= N^{f^\alpha}(X)_t e_\alpha \\
&= e_\alpha \int_0^t H_i\widetilde{f^\alpha}(U_s) dW_s^i \\
&= e_\alpha \int_0^t (U_s e_i)^\alpha dW_t^i \\
&= \int_0^t U_s e_i dW_t^i = \int_0^t U_s dW_s,
\end{aligned}$$

by viewing $U_s\colon \mathbb{R}^d \to \mathbb{R}^N$ where $U_s\colon \mathbb{R}^d \to T_xM$ is an isomorphism. Define $V_s\colon \mathbb{R}^N = T_xM \oplus N_xM \to \mathbb{R}^d$ as

$$V_s\xi = \begin{cases} U_s^{-1}\xi, & \xi \in T_{X_s}M, \\ 0, & \xi \perp T_{X_s}M \end{cases}$$

Then $V_sU_se = e$ for all $\mathbb{R}^d$. Therefore,

$$\int_0^t V_s dN^f(X)_s = \int_0^t V_sU_s dW_s = W_t.$$

Because $N^f(X)$ is a local martingale, W_t is a local martingale. It follows that X is an M-martingale. ☐

Remark 2.5.6. From the proof, we can see that when viewing $M \subset \mathbb{R}^N$ and let $f = (f^1, \cdots, f^\alpha)\colon M \to \mathbb{R}^N$ be the coordinate function, if N^{f^α} for $\alpha = 1, \cdots, N$ are local martingales, then X is an M-local martingale.

Proposition 2.5.7. *Suppose $(U, x^1, \cdots, x^d)$ is a local chart on M and $X = (X^i)$, i.e. $X^i = x^i(X)$, a semimartingale on M. Then X is an M-martingale if and only if*

$$N_t^i = X_t^i - X_0^i + \frac{1}{2}\int_0^t \Gamma_{jk}^i(X_s) d\left\langle X^j, X^k\right\rangle_s$$

is a local martingale.

Proof. $\Rightarrow$: Locally, we know

$$\nabla^2 x^i = -\Gamma_{jk}^i dx^j \otimes dx^k,$$

which implies that

$$\int_0^t \nabla^2 x^i(dX_s, dX_s) = -\int_0^t \Gamma_{jk}^i(X_s) d\left\langle X^j, X^k\right\rangle_s.$$

Therefore, this direction can be directly obtained by applying above theorem to each $f = x^i$.
$\Leftarrow$: Choose a local coordinate for $F(M)$ as $(\tilde{x}^i, e_j^i)$, where

$$\tilde{x}^i(u) = x^i(\pi(u)), \quad e_j^i(u) \text{ for } ue_j = e_j^i(u) \left.\frac{\partial}{\partial x^j}\right|_{\pi(u)}.$$

Let U be a horizontal lift with the anti-development W of X. Then

$$dU_t = H_j(U_t) \circ dW_t^j.$$

Because $X_t = \pi(U_t)$ and $U_te_j = e_j^i(t) \left.\frac{\partial}{\partial x^j}\right|_{X_t}$, where $e_j^i(t) = e_j^i(U_t)$, we have

$$\begin{aligned} dX_t^i = d\tilde{x}^i(U_t) &= H_j\tilde{x}^i(U_t) \circ dW_t^j \\ &= (U_te_j)(x^i) \circ dW_t^j \\ &= e_j^i(t) \circ dW_t^j. \end{aligned}$$

Moreover, we also know

$$dU_t = P_k^H(U_t) \circ dX_t^k.$$

Note that in the local chart, M can be viewed as $\mathbb{R}^d$ and so $P(x)\colon \mathbb{R}^d \to T_xM$ the orthogonal projection is the identity, which implies that

$$P_k(X_t) = P(X_t)e_k = e_k = \delta_k^\ell \left.\frac{\partial}{\partial x^\ell}\right|_{X_t}.$$

Let $v(s)$ be the horizontal lift starting from U_t of a curve $c(s)$ with $c(0) = U_t$ and $\dot{c}(0) = P_k(X_t) = \delta_k^m \frac{\partial}{\partial x^m}\big|_{X_t}$. Then by using the coordinates, we have

$$\dot{v}(0) = P_k^H(U_t) = -\Gamma^i_{m\ell}(X_t)\delta_k^m e_j^\ell(t) \left.\frac{\partial}{\partial e_j^i}\right|_{U_t} + \delta_k^m \left.\frac{\partial}{\partial \tilde{x}^m}\right|_{U_t}.$$

So

$$P_k^H e_j^i(U_t) = -\Gamma^i_{k\ell}(X_t)e_j^\ell(t).$$

It follows that

$$de_j^i(t) = -\Gamma^i_{k\ell}(X_t)e_j^\ell(t) \circ dX_t^k.$$

Let $(f_j^i) = (e_j^j)^{-1}$. Then we get

$$dW_t^j = f_k^j(t) \circ dX_t^k.$$

So

$$\begin{aligned} dX_t^i &= e_j^i(t)dW_t^i + \frac{1}{2}d\left\langle e_j^i, W^j\right\rangle_t \\ &= e_j^i(t)dW_t^i - \frac{1}{2}\Gamma^i_{k\ell}\left(X_t\right)e_j^\ell(t)f_m^j(t)d\left\langle X^k, X^m\right\rangle_t \\ &= e_j^i(t)dW_t^i - \frac{1}{2}\Gamma^i_{k\ell}\left(X_t\right)d\left\langle X^k, X^\ell\right\rangle, \end{aligned}$$

which implies that

$$dN_t^i = e_j^i(t)dW_t^i \;\Rightarrow\; dW_t^i = f_j^i(t)dN_t^i.$$

Therefore, if N_t^i are all local martingales, so is W_t and X is an M-valued martingale. □

2.6 Martingales on Submanifolds

Let M, N be two manifolds equipped with connection ∇^M, ∇^N respectively, and $M \subset N$ be a submanifold. Consider the second fundamental form $\mathrm{II}\colon \Gamma(TM) \times \Gamma(TM) \to \Gamma(TN|_M)$ defined as

$$\mathrm{II}(V, W) = \nabla_V^N W - \nabla_V^M W,$$

which is obviously bilinear over $C^\infty(M)$, so II is a $TN|_M$-valued $(0,2)$-tensor field on M.

Remark 2.6.1. A direct result of this is, $\mathrm{II}(V, W)(p)$ is fully determined by V_p and W_p

Moreover, for a connection ∇, let

$$T(V, W) = \nabla_V W - \nabla_W V - [V, W].$$

We have

$$\mathrm{II}(V, W) - \mathrm{II}(W, V) = T^N(V, W) - T^M(V, W),$$

so II is symmetric if ∇^M, ∇^N are torsion-free. If $\mathrm{II} \equiv 0$, then M is called totally geodesic.

Proposition 2.6.2. *If M is totally geodesic, then any ∇^M-geodesic in M is also a ∇^N-geodesic in N. When ∇^M, ∇^N are torsion-free, the converse is also true.*

Lemma 2.6.3. *Let $M \subset \mathbb{R}^N$ be a submanifold and $f\colon M \to \mathbb{R}^N$ be the coordinate function on M with lift $\widetilde{f} = f \circ \pi$ to the frame bundle $F(M)$. Then*

$$H_i H_j \widetilde{f}(u) = \mathrm{II}(ue_i, ue_j)$$

Proof. Let u_t be the horizontal lift starting from $u_0 = u$ of a curve x_t with $x_0 = \pi(u)$ and $\dot{x}_0 = ue_i$, i.e., $\dot{u}(0) = H_i(u)$, and $V(x_t) = u_t e_j$ is a ∇^M-parallel along x_t. So

$$\nabla^M_{\dot{x}_t} V = 0 \;\Rightarrow\; \nabla^M_{ue_i} V = 0.$$

It implies that

$$\mathrm{II}(ue_i, ue_j) = \nabla^{\mathbb{R}^N}_{ue_i} V.$$

On the other hand, recall when viewing $T_x M \subset \mathbb{R}^N$,

$$H_j \widetilde{f}(u_t) = u_t e_j = V(x_t).$$

Therefore,

$$H_i H_j \widetilde{f}(u) = \left.\frac{d}{dt}\right|_{t=0} V(x_t) = \nabla^{\mathbb{R}^N}_{ue_i} V = \mathrm{II}(ue_i, ue_j)$$

for viewing $V(x_t) = (V^1(x_t), \cdots, V^N(x_t)) \in \mathbb{R}^N$. □

Theorem 2.6.4. *Using notations as above and assume* $\dim M = l$ *and* $\dim N = d$. *Suppose that X is a semimartingale on $M \subset N$. Let U^M be a horizontal lift of X in $F(M)$ with the corresponding $\mathbb{R}^d$-valued anti-development W^M, and U^N be a horizontal lift of X in $F(N)$ with the corresponding $\mathbb{R}^l$-valued anti-development W^N. Then*

$$W_t^N = \int_0^t (U_s^N)^{-1} U_s^M dW_s^M + \frac{1}{2}\int_0^t (U_s^N)^{-1}\mathrm{II}(dX_s, dX_s).$$

Proof. First,

$$\mathrm{II}(dX_s, dX_s) = \mathrm{II}(U_s^M e_i, U_s^M e_j) d\left\langle (W^M)^i, (W^M)^j \right\rangle_s.$$

So we need to show

$$W_t^N = \int_0^t (U_s^N)^{-1} U_s^M e_i d(W^M)_s + \int_0^t (U_s^N)^{-1}\mathrm{II}(U_s^M e_i, U_s^M e_j) d\left\langle (W^M)^i, (W^M)^j \right\rangle_s.$$

First, assume $N = \mathbb{R}^l$. WTLG, we can choose $U_0^N = I$. In such case, we have

$$W^N = X - X_0, \quad U^N \equiv I.$$

For simplicity, denote $U = U^M$ and $W = W^M$. Let $f\colon M \to \mathbb{R}^l$ be the coordinate map and $\widetilde{f} = f \circ \pi$. Then we have been shown

$$X_t = X_0 + \int_0^t H_i \widetilde{f}(U_s) dW_s^i + \frac{1}{2}\int_0^t H_i H_j \widetilde{f}(U_s) d\left\langle W^i, W^j \right\rangle_s.$$

Note that

$$H_i \widetilde{f}(u) = ue_i, \quad H_i H_j \widetilde{f}(u) = \mathrm{II}(ue_i, ue_j).$$

Therefore,

$$W_t^N = X_t - X_0 = \int_0^t U_s dW_s + \frac{1}{2}\int_0^t \mathrm{II}\left(dX_s, dX_s\right).$$

Next, for general N, assume $N \subset \mathbb{R}^N$ embedded manifold. Let II^M and II^N be the second fundamental forms of M and N in $\mathbb{R}^N$ respectively. Then

$$X_t = X_0 + \int_0^t U_s^M dW_s^M + \frac{1}{2}\int_0^t \mathrm{II}^M\left(dX_s, dX_s\right)$$

$$X_t = X_0 + \int_0^t U_s^N dW_s^N + \frac{1}{2}\int_0^t \mathrm{II}^N\left(dX_s, dX_s\right),$$

where the two equations are viewed in $\mathbb{R}^N$, $T_xM = \mathbb{R}^l$ and $T_xN = \mathbb{R}^d$. Note that

$$\mathrm{II}(X, Y) = \mathrm{II}^M(X, Y) - \mathrm{II}^N(X, Y).$$

By above, $dX_t = U_s^N dW_s^N + \frac{1}{2}\mathrm{II}^N\left(dX_s, dX_s\right)$,

$$\begin{aligned} W_t^N &= \int_0^t (U_s^N)^{-1} dX_t - \frac{1}{2}\int_0^t (U_s^N)^{-1}\mathrm{II}^N\left(dX_s, dX_s\right) \\ &= \int_0^t (U_s^N)^{-1} U_s^M dW_s^M + \frac{1}{2}\int_0^t (U_s^N)^{-1}\mathrm{II}^M\left(dX_s, dX_s\right) \\ &\quad - \frac{1}{2}\int_0^t (U_s^N)^{-1}\mathrm{II}^N\left(dX_s, dX_s\right) \\ &= \int_0^t (U_s^N)^{-1} U_s^M dW_s^M + \frac{1}{2}\int_0^t (U_s^N)^{-1}\mathrm{II}(dX_s, dX_s). \end{aligned}$$

□

Corollary 2.6.5. *Suppose that $M \subset N$ is a totally geodesic submanifold of N. Then every ∇^M-martingale on M is also a ∇^N-martingale on N.*

For $x \in M$, define

$$\operatorname{ran}\mathrm{II}_x := \{\mathrm{II}(X, Y)(x) \colon X, Y \in \Gamma(TM)\} \subset T_xN.$$

II is called independent of TM if $\operatorname{ran}\mathrm{II}_x \cap T_xM = \{0\}$ for every $x \in M$. Note that for Riemannian submanifold, II is obviously independent of TM.

Corollary 2.6.6. *Suppose that $M \subset N$ such that* II *is independent of TM. Then a ∇^N-martingale on N which lives on M is also a ∇^M-martingale on M.*

Proof. First, W^M is clearly a continuous semi-martingale, so it can be uniquely decomposed into

$$W_t = M_t + A_t,$$

where M is a local martingale and A is a process of bounded variation with $A_0 = 0$. It suffices to prove $A = 0$. By above,

$$W_t^N = \int_0^t (U_s^N)^{-1} U_s^M dW_s^M + \int_0^t (U_s^N)^{-1}\left(U_s^M dA_s + \frac{1}{2}\mathrm{II}(dX_s, dX_s)\right).$$

Because W^N is a local martingale,

$$\int_0^t (U_s^N)^{-1}\left(U_s^M dA_s + \frac{1}{2}\mathrm{II}(dX_s, dX_s)\right) = 0$$

and so

$$U_s^M dA_s + \frac{1}{2}\mathrm{II}(dX_s, dX_s) = 0.$$

However,

$$U_s^M dA_s \in T_{\pi(U_s^M)}M, \quad \mathrm{II}(dX_s, dX_s) \in \operatorname{ran}\mathrm{II}_{\pi(U_s^M)}.$$

The independence implies that $dA_s = 0$, and so $A = 0$. □

Example 2.6.7. (1) Let $M \subset N$ be a Riemannian hypersurface. It is called (strictly) convex if the scalar fundamental form is (strictly) positive definite at each point. Assume M is strictly convex. Because II is independent of TM, any ∇^N-martingale on N which lies on M is also a ∇^M-martingale. It implies that

$$\mathrm{II}(dX_s, dX_s) = 0.$$

However, because of the strict positivity, $dX_s = 0$, and so X is constant.

(2) Let M be a Riemannian manifold of dimension $\mathbb{R}^d$. A semimartingale X on M is called a Brownian motion if its anti-development W is a d-dimensional standard Brownian motion. Let $M \subset \mathbb{R}^N$ be a minimal submanifold, i.e., the mean curvature field $H = \operatorname{tr}_g \mathrm{II} = 0$. Then a Brownian motion on M is a local martingale on $\mathbb{R}^N$, because

$$\mathrm{II}\left(U_t e_i, U_t e_j\right) d\left\langle W^i, W^j \right\rangle_t = \sum_i \mathrm{II}\left(U_t e_i, U_t e_i\right) dt = H\left(X_t\right) dt = 0.$$

Note that for Riemannian manifold, we consider $O(M)$, and so $\{U_t e_i\}$ is orthonormal in $T_{\pi(U_t)}M$.

Proposition 2.6.8. *Suppose that $M \subset N$ be totally geodesic. For any $x \in M$, there exists a neighborhood O of p in N such that if X is an ∇^N-martingale in O on $[0, T]$ with $X_T \in M$, then $X_t \in M$ for all $t \in [0, T]$.*

Theorem 2.6.9. *Let M be a manifold with a connection. For any $x \in M$, there exists a neighborhood O of x such that: if $(X_t)_{t\in[0,T]}$ and $(Y_t)_{t\in[0,T]}$ are two martingales on V such that $X_T = Y_T$, then $X_t = Y_t$ for all $t \in [0, T]$.*

Chapter 3

Brownian Motion on Manifolds

3.1 Laplace-Beltrami operator

Let (M, g) be a Riemannian manifold equipped with the Levi-Civita connection ∇ and the standard Riemannian volume measure m. For $f, g \in C_c(M)$,

$$(f, g) := \int_M f(x)g(x)m(dx)$$

and for $X, Y \in \Gamma_c(TM)$, i.e., vector fields with compact support,

$$(X, Y) = \int_M \langle X, Y \rangle_x m(dx),$$

and for $\theta, \psi \in \Gamma_c(T^*M)$, i,e, 1-forms with compact support,

$$(\theta, \psi) = \int_M \langle \theta, \psi \rangle_x m(dx).$$

Let Δ_M be the Laplace-Beltrami operator. Because locally

$$\Delta_M f = \mathrm{tr}_g(\nabla^2 f) = g^{ij} \frac{\partial^2}{\partial x^i \partial x^j} f - g^{k\ell} \Gamma^i_{k\ell} \frac{\partial}{\partial x^i} f,$$

Δ_M is a nondegenerate second order elliptic operator. Moreover, by the local expression, for $L = \frac{1}{2}\Delta_M$, the corresponding

$$\Gamma(f, g) = g^{ij} \frac{\partial f}{\partial x^i} \frac{\partial g}{\partial x^j} = \langle \nabla f, \nabla g \rangle .$$

Let $\pi \colon O(M) \to M$ be the orthogonal frame bundle. Define a second order differential operator on $O(M)$ as

$$\Delta_{O(M)} := \sum_{i=1}^d H_i^2,$$

called Bochner's horizontal Laplacian.

Proposition 3.1.1. *For any $f \in C^\infty(M)$, let $\widetilde{f} = f \circ \pi \in C^\infty(O(M))$. Then for any $u \in O(M)$,*

$$\Delta_M f(x) = \Delta_{O(M)} \widetilde{f}(u),$$

where $x = \pi(u)$.

Proof. Fix a $f \in C^\infty(M)$. Note that for any 1-form, when viewing $(\mathbb{R}^d)^* = \mathbb{R}^d$, the scalarization $\widetilde{\theta} \colon O(M) \to \mathbb{R}^d$ satisfies

$$\widetilde{\theta}(u) = (\theta(ue_1), \cdots, \theta(ue_d))^\top \in \mathbb{R}^d.$$

Therefore,

$$(\widetilde{df})_i = df(ue_i) = ue_i(f) = H_i\widetilde{f}(u),$$

and

$$\widetilde{df} = \left(H_1\widetilde{f}, H_2\widetilde{f}, \cdots, H_d\widetilde{f}\right)^\top.$$

Because (ue_i) is a normal basis in $T_{\pi(u)}M$, we have

$$\sharp(ue_i)^* = ue_i.$$

It follows that, first

$$\widetilde{\operatorname{grad} f} = \left(H_1\widetilde{f}, H_2\widetilde{f}, \cdots, H_d\widetilde{f}\right)^\top.$$

because $\operatorname{grad} f((ue_i)^*) = \langle \operatorname{grad} f, \sharp(ue_i)^* \rangle = \langle \operatorname{grad} f, ue_i \rangle = df(ue_i)$. Second, for any $X \in \Gamma(TM)$,

$$\operatorname{div} X(\pi(u)) = \sum_i \nabla X((ue_i)^*, ue_i) = \sum_i \langle \nabla_{ue_i} X, \sharp(ue_i)^* \rangle = \sum_i \langle \nabla_{ue_i} X, ue_i \rangle.$$

Moreover, since $u \in O(M)$, orthogonal map,

$$\begin{aligned}
\operatorname{div} X(\pi(u)) &= \sum_i \left\langle u^{-1}\nabla_{ue_i} X(\pi(u)), e_i \right\rangle \\
&= \sum_i \left\langle \widetilde{\nabla_{ue_i} X}(u), e_i \right\rangle \\
&= \sum_i \left\langle H_i\widetilde{X}(u), e_i \right\rangle = \sum_i \left(H_i\widetilde{X}\right)^i(u).
\end{aligned}$$

It follows that

$$\Delta_M f(\pi(u)) = \operatorname{div}(\operatorname{grad} f)(\pi(u)) = \sum_i \left(H_i(\widetilde{\operatorname{grad} f})\right)^i(u) = \sum_i H_i(H_i\widetilde{f}) = \Delta_{O(M)}\widetilde{f}(u). \quad \square$$

Theorem 3.1.2 (Nash's Embedding Theorem)**.** *Every Riemannian manifold can be isometrically embedded in some Euclidean space with the standard metric.*

Remark 3.1.3. Note that if the Riemannian manifold is complete, then we can find an Euclidean space such that the image is closed, i.e., a complete Riemannian manifold can be viewed as a Riemannian submanifold of an Euclidean space, which is also a closed subset.

In the following, we assume a $M \subset \mathbb{R}^N$ be a Riemannian submanifold that is closed. So all above theories can be applied to such M. Similarly as above, let $P(x) \colon \mathbb{R}^N \to T_xM$ be the orthogonal projection and $P_\alpha = Pe_\alpha \in \Gamma(TM)$. Note that if $X \in T_xM$, we have

$$\begin{aligned}
X = P(x)X = P(x)\sum_{\alpha=1}^N \langle X, e_\alpha\rangle_{\mathbb{R}^N} e_\alpha &= \sum_{\alpha=1}^N \langle X, e_\alpha\rangle_{\mathbb{R}^N} P_\alpha(x) \\
= \sum_{\alpha=1}^N \langle P(x)X, e_\alpha\rangle_{\mathbb{R}^N} P_\alpha(x) &= \sum_{\alpha=1}^N \langle X, P(x)e_\alpha\rangle_{\mathbb{R}^N} P_\alpha(x)
\end{aligned}$$

$$= \sum_{\alpha=1}^{N} \langle X, P_\alpha(x) \rangle_{T_xM} P_\alpha(x).$$

Let $\widetilde{\nabla}$ be the standard Euclidean connection. The induced Levi-Civita connection on M is

$$\nabla_X Y = (\widetilde{\nabla}_X Y)^\top = P(\widetilde{\nabla}_X Y), \quad X, Y \in \Gamma(TM).$$

Moreover, for any $X \in \Gamma(TM)$ and any $f \in C^\infty(M)$,

$$X(f) = \langle \operatorname{grad} f, X \rangle_{TM} = \left\langle \overline{\operatorname{grad} f}, X \right\rangle_{\mathbb{R}^N}, \;\Rightarrow\; \operatorname{grad} f = P\overline{\operatorname{grad} f},$$

where $\overline{\operatorname{grad} f}$ is the usual gradient on $\mathbb{R}^N$.

Theorem 3.1.4. *Using notations as above,*

$$\Delta_M = \sum_{\alpha=1}^{N} P_\alpha^2.$$

Proof. Let $f \in C^\infty(M)$.

$$\operatorname{grad} f = \sum_{\alpha=1}^{N} \langle \operatorname{grad} f, P_\alpha \rangle P_\alpha = \sum_{\alpha=1}^{N} P_\alpha(f) P_\alpha.$$

Note that for any $X \in \Gamma(TM)$ and $h \in C^\infty(M)$, it is not difficult such that

$$\operatorname{div}(hX) = X(h) + h \operatorname{div}(X).$$

It follows that

$$\Delta_M f = \sum_{\alpha=1}^{N} P_\alpha^2(f) + \sum_{\alpha=1}^{N} P_\alpha(f) \operatorname{div}(P_\alpha).$$

For the second term, fix $x \in M$. First,

$$\operatorname{div}(P_\alpha)(x) = \sum_{i=1}^{d} \left\langle \nabla_{\frac{\partial}{\partial x^i}} P_\alpha(x), \left.\frac{\partial}{\partial x^i}\right|_x \right\rangle,$$

where $(x^1, \cdots, x^d)$ is a normal coordinate at x, because $\left\{ \frac{\partial}{\partial x^i}\big|_x \right\}$ is an orthonormal basis of T_xM. Moreover, because $\Gamma_{ij}^k(x) = 0$,

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) = 0.$$

So

$$\begin{aligned}
\left\langle \nabla_{\frac{\partial}{\partial x^i}} P_\alpha(x), \left.\frac{\partial}{\partial x^i}\right|_x \right\rangle_{T_xM} &= \frac{\partial}{\partial x^i} \left\langle P_\alpha, \frac{\partial}{\partial x^i} \right\rangle_{TM} (x) = \frac{\partial}{\partial x^i} \left\langle P e_\alpha, \frac{\partial}{\partial x^i} \right\rangle_{TM} (x) \\
&= \frac{\partial}{\partial x^i} \left\langle e_\alpha, P\frac{\partial}{\partial x^i} \right\rangle_{\mathbb{R}^N} (x) = \frac{\partial}{\partial x^i} \left\langle e_\alpha, \frac{\partial}{\partial x^i} \right\rangle_{\mathbb{R}^N} (x) \\
&= \left\langle \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} e_\alpha(x), \frac{\partial}{\partial x^i} \right\rangle_{\mathbb{R}^N} + \left\langle e_\alpha(x), \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N} \\
&= \left\langle e_\alpha, \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N}.
\end{aligned}$$

It follows that

$$\begin{aligned}
\sum_{\alpha=1}^{N} \operatorname{div}(P_\alpha)(x) P_\alpha(x) &= \sum_{\alpha=1}^{N}\sum_{i=1}^{d} \left\langle e_\alpha, \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N} P_\alpha(x) \\
&= \sum_{\alpha=1}^{N}\sum_{i=1}^{d} \left\langle e_\alpha, \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N} P(x) e_\alpha \\
&= \sum_{i=1}^{d} P(x) \sum_{\alpha=1}^{N} \left\langle e_\alpha, \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N} e_\alpha \\
&= \sum_{i=1}^{d} P(x) \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) = \sum_{i=1}^{d} \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) = 0.
\end{aligned}$$

Therefore, $\sum_{\alpha=1}^{N} P_\alpha(f) \operatorname{div}(P_\alpha) = 0$ for any $f \in C^\infty(M)$. Then

$$\Delta_M f = \sum_\alpha P_\alpha^2(f). \qquad \square$$

Proposition 3.1.5. *Using same notations as above,*

$$\Delta_M f = \sum_\alpha \nabla^2 f(P_\alpha, P_\alpha).$$

Proof. Fix $x \in M$. Let $\{X_i\}$ be an orthonormal basis of $T_x M$ and consider decomposition

$$X_i = \sum_{\alpha=1}^{N} \langle X_i, P_\alpha(x) \rangle P_\alpha(x).$$

Then

$$\begin{aligned}
\Delta_M f(x) &= \sum_{i=1}^{d} \nabla^2 f(x)(X_i, X_i) \\
&= \sum_{i=1}^{d} \sum_{\alpha,\beta=1}^{N} \nabla^2 f(P_\alpha, P_\beta)(x) \langle X_i, P_\alpha(x) \rangle \langle X_i, P_\beta(x) \rangle \\
&= \sum_{\alpha=1}^{N} \nabla^2 f \left(P_\alpha, \sum_{i=1}^{d} \sum_{\beta=1}^{N} \langle X_i, P_\alpha \rangle \langle X_i, P_\beta \rangle P_\beta \right)(x).
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\sum_{i=1}^{d} \sum_{\beta=1}^{N} \langle X_i, P_\alpha \rangle \langle X_i, P_\beta \rangle P_\beta &= \sum_{\beta=1}^{N} \left\langle \sum_{i=1}^{d} \langle X_i, P_\beta \rangle X_i, P_\alpha \right\rangle P_\beta \\
&= \sum_{\beta=1}^{N} \langle P_\beta, P_\alpha \rangle P_\beta = P P_\alpha = P_\alpha. \qquad \square
\end{aligned}$$

3.2 Brownian Motion on Manifolds

Theorem 3.2.1. *Let $X \colon \Omega \to W(M)$ be a measurable map defined on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $\mu = (X_0)_{\#}\mathbb{P}$ be the initial distribution on M. TFAE.*

(1) X is a $\frac{1}{2}\Delta_M$-diffusion process w.s.t. $\mathbb{F}^X$, i.e., X is a $\mathbb{F}^X$-semimartingale, and

$$M^f(X)_t = f(X_t) - f(X_0) - \frac{1}{2}\int_0^t \Delta_M f(X_s)ds, \quad 0 \le t < e(X),$$

is an $\mathbb{F}^X$-local martingale for all $f \in C^\infty(M)$.

(2) X is a $\mathbb{F}^X$-semimartingale on M whose anti-development is a standard Euclidean $\mathbb{F}^X$-Brownian motion.

Remark 3.2.2. An M-valued process X satisfies above condition is called a Riemannian Brownian motion on M. Moreover, $\mathbb{P}_\mu = X_\# \mathbb{P}$ is called a Wiener measure on $W(M)$. Note that a Wiener measure is uniquely determined by $\Delta_M/2$ and an initial measure μ.

Proof. $\Leftarrow$: Let U be a horizontal lift of X and W is its anti-development. Then we have

$$dU_t = H_i(U_t) \circ dW_t.$$

For any $f \in C^\infty(M)$, let $\widetilde{f} = f \circ \pi$ be its lift on $O(M)$. Then by above mention,

$$f(X_t) = f(X_0) + \int_0^t H_i \widetilde{f}(U_s)\, dW_s^i + \frac{1}{2}\int_0^t H_i H_j \widetilde{f}(U_s)\, d\left\langle W^i, W^j \right\rangle_s.$$

Because W is a Brownian motion, $\langle W^i, W^j \rangle_t = \delta^{ij} t$. Then

$$\begin{aligned}\int_0^t H_i H_j \widetilde{f}(U_s)\, d\left\langle W^i, W^j \right\rangle_s &= \int_0^t \sum_i H_i^2 \widetilde{f}(U_s) ds \\ &= \int_0^t \Delta_{O(M)} \widetilde{f}(U_s) ds \\ &= \int_0^t \Delta_M f(X_s) ds.\end{aligned}$$

Therefore,

$$M^f(X)_t = \int_0^t H_i \widetilde{f}(U_s) dW_s^i,$$

which is clearly a $\mathbb{F}^X$-local martingale.
$\Rightarrow$: Let $f = (f^\alpha)\colon M \to \mathbb{R}^N$ be the coordinate function $f(x) = x$. Let $\widehat{f^\alpha} = f^\alpha \circ \pi$ be the lift on $O(M)$. By assumption, for $X_t^\alpha = f^\alpha(X)_t$,

$$X_t^\alpha = X_0^\alpha + M_t^\alpha + \frac{1}{2}\int_0^t \Delta_M f^\alpha(X_s) ds,$$

where M_t^α is a local martingale, $\alpha = 1, 2, \cdots, N$. On the other hand, let H be the horizontal lift of X and W be its anti-development. Then

$$X_t^\alpha = X_0^\alpha + \int_0^t H_i \widetilde{f^\alpha}(U_s) dW_s^i + \frac{1}{2}\int_0^t \nabla^2 f^\alpha(dX_s, dX_s).$$

Claim: For any $f \in C^\infty(M)$, we can have

$$\int_0^t \nabla^2 f(dX_s, dX_s) = \int_0^t \Delta_M f(X_s) ds.$$

First, we know $dX_t = P_\alpha(X_t) \circ dX_t^\alpha$. Then we get

$$\int_0^t \nabla^2 f\left(dX_s, dX_s\right) = \int_0^t \nabla^2 f\left(P_\alpha, P_\beta\right)(X_s) d\left\langle X^\alpha, X^\beta\right\rangle_s.$$

By the assumption, because X is a $\Delta_M/2$-diffusion process,

$$d\left\langle X^\alpha, X^\beta\right\rangle_s = d\left\langle M^\alpha, M^\beta\right\rangle_s = \Gamma(f^\alpha, f^\beta)(X_s)ds,$$

where

$$\Gamma(f^\alpha, f^\beta) = \left\langle \nabla f^\alpha, \nabla f^\beta\right\rangle.$$

Because

$$\operatorname{grad} f^\alpha = P(\overline{\operatorname{grad}} f^\alpha) = Pe_\alpha = P_\alpha,$$

we have

$$\Gamma(f^\alpha, f^\beta) = \left\langle \operatorname{grad} f^\alpha, \operatorname{grad} f^\beta\right\rangle = \left\langle P_\alpha, P_\beta\right\rangle.$$

Moreover,

$$\sum_{\alpha,\beta} \nabla^2 f(P_\alpha, P_\beta)\left\langle P_\alpha, P_\beta\right\rangle = \sum_\alpha \nabla^2 f(P_\alpha, \sum_\beta \left\langle P_\alpha, P_\beta\right\rangle P_\beta) = \sum_\alpha \nabla^2 f(P_\alpha, P_\alpha) = \Delta_M f$$

Then

$$\begin{aligned}\int_0^t \nabla^2 f\left(dX_s, dX_s\right) &= \int_0^t \nabla^2 f\left(P_\alpha, P_\beta\right)(X_s)\Gamma(f^\alpha, f^\beta)(X_s)ds \\ &= \int_0^t \sum_{\alpha,\beta} \nabla^2 f\left(P_\alpha, P_\beta\right)(X_s)\left\langle P_\alpha, P_\beta\right\rangle ds \\ &= \int_0^t \Delta_M f(X_s)ds.\end{aligned}$$

So the claim has been proved.
Then we have

$$M_t^\alpha = \int_0^t H_i \widetilde{f^\alpha}(U_s)dW_s^i.$$

Next, we will use the Lévy's theorem to check W is a Brownian motion. First, because $\widetilde{f^\alpha} = f^\alpha \circ \pi$,

$$\begin{aligned}H_i \widetilde{f^\alpha}(u) &= (ue_i)(f^\alpha) \\ &= \left\langle ue_i, \operatorname{grad} f^\alpha\right\rangle_{TM} \\ &= \left\langle ue_i, Pe_\alpha\right\rangle_{TM} = \left\langle ue_i, Pe_\alpha\right\rangle_{\mathbb{R}^N} \\ &= \left\langle Pue_i, e_\alpha\right\rangle_{\mathbb{R}^N} = \left\langle ue_i, e_\alpha\right\rangle_{\mathbb{R}^N}\end{aligned}$$

Moreover,

$$\begin{aligned}\sum_{\alpha=1}^N \left\langle ue_i, e_\alpha\right\rangle \left\langle ue_j, e_\alpha\right\rangle &= \left\langle ue_i, \sum_{\alpha=1}^N \left\langle ue_j, e_\alpha\right\rangle e_\alpha\right\rangle \\ &= \left\langle ue_i, ue_j\right\rangle = \left\langle e_i, e_j\right\rangle = \delta_{ij}.\end{aligned}$$

Therefore, we have

$$dW_t^j = \left\langle e_\alpha, U_t e_j\right\rangle dM_t^\alpha,$$

which implies that W is a continuous $\mathbb{F}^X$-local martingale since M^α are $\mathbb{F}^X$-local martingales. Again, because

$$d\left\langle M^\alpha, M^\beta\right\rangle_s = \langle P_\alpha, P_\beta\rangle\, ds,$$

we have

$$\begin{aligned}
\left\langle W_t^i, W_t^j\right\rangle &= \sum_{\alpha,\beta}\int_0^t \langle e_\alpha, U_s e_j\rangle \langle e_\beta, U_s e_j\rangle \langle P_\alpha, P_\beta\rangle\, ds \\
&= \sum_{\alpha}\int_0^t \langle e_\alpha, U_s e_j\rangle \left\langle \sum_\beta \langle P_\alpha, P_\beta\rangle e_\beta, U_s e_j\right\rangle ds \\
&= \sum_{\alpha}\int_0^t \langle e_\alpha, U_s e_j\rangle \left\langle \sum_\beta \langle P P_\alpha, e_\beta\rangle e_\beta, U_s e_j\right\rangle ds \\
&= \sum_{\alpha}\int_0^t \langle e_\alpha, U_s e_j\rangle \langle P_\alpha, U_s e_j\rangle\, ds \\
&= \int_0^t \sum_{\alpha} \langle e_\alpha, U_s e_j\rangle \langle e_\alpha, U_s e_j\rangle\, ds = \delta^{ij} t.
\end{aligned}$$

Therefore, by the Lévy's theorem, W is a $\mathbb{F}^X$-Brownian motion. □

Remark 3.2.3. We can make $\mathbb{F}^X$ larger. X is called a $\mathbb{F}$-Brownian motion if it is defined as above w.s.t. $\mathbb{F}$ and it is a strong Markov process w.s.t. $\mathbb{F}$.

If X be a Brownian motion on M, then its horizontal lift U is called a horizontal Brownian motion. Since

$$dU_t = H_i(U_t)\circ dW_t^i,$$

where W is a Brownian motion, then U is a diffusion process generated by the operator

$$L = \frac{1}{2}\sum_i H_i^2 = \frac{1}{2}\Delta_{O(M)}.$$

Proposition 3.2.4. *Let $U\colon \Omega \to W(O(M))$ be a measurable map defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. TFAE.*

(1) U is a horizontal $\mathbb{F}^U$-semimartingale whose projection $X = \pi(U)$ is a Brownian motion on M.

(2) U is a horizontal $\mathbb{F}^U$-semimartingale whose anti-development is a standard Euclidean Brownian motion.

(3) U is a $\Delta_{O(M)}/2$-diffusion w.s.t. $\mathbb{F}^U$.

Remark 3.2.5. An $O(M)$-valued process U satisfies any of above is called a horizontal Brownian motion on $O(M)$.

Proof. (1) $\Rightarrow$ (2) : Since X is a Brownian motion, by above its anti-development W is a standard Euclidean Brownian motion.

(2) $\Rightarrow$ (3) : Then by

$$dU_t = H_i(U_t)\circ dW_t^i,$$

it is a diffusion process of $\Delta_{O(M)}/2$.

(3) $\Rightarrow$ (1) : For any $f \in C^\infty(M)$, let $\widetilde{f} = f\circ\pi \in C^\infty(O(M))$. Because

$$\Delta_M f(\pi(u)) = \Delta_{O(M)}\widetilde{f}(u),$$

we have

$$f(X_t) - f(X_0) - \frac{1}{2}\int_0^t \Delta_M f(X_s)ds = \widetilde{f}(U_t) - \widetilde{f}(U_0) - \frac{1}{2}\int_0^t \Delta_{O(M)}\widetilde{f}(U_s)ds.$$

The RHS is a $\mathbb{F}^U$-local martingale, which means that the LHS is also a $\mathbb{F}^U$-local martingale. Moreover, $X = \pi(U)$ implies $\mathbb{F}^X \subset \mathbb{F}^U$, so the LHS is a $\mathbb{F}^X$-local martingale. □

Remark 3.2.6. Note that let $M \subset \mathbb{R}^N$ and B is a Brownian motion on $\mathbb{R}^N$. Then if

$$dX_t = P_\alpha(X_t) \circ dB_t^\alpha,$$

then X_t is a diffusion process generated by

$$L = \frac{1}{2}\sum_\alpha P_\alpha^2 = \frac{1}{2}\Delta_M.$$

Therefore, X is a Brownian motion on M.

3.3 Examples of Brownian Motion

Riemannian Structure of Sphere. First, consider the standard Riemannian structure on $\mathbb{S}^d$.

1. $d = 1$: Choosing a chart $(0, 2\pi) \to \mathbb{S}^1$ and consider the embedding

$$\iota \colon \mathbb{S}^1 \hookrightarrow \mathbb{R}^2$$

given by $\theta \mapsto (x, y)$, where $x = \cos\theta$ and $y = \sin\theta$. Note that any $V_{(x,y)} \in T_{(x,y)}\mathbb{R}^2$ is equivalent to $V_{(x,y)} \in \mathbb{R}^2$ by setting

$$V_{(x,y)} = a(x,y)\frac{\partial}{\partial x} + b(x,y)\frac{\partial}{\partial y} = (a(x,y), b(x,y))^\top.$$

and for any $f(x,y) \in C^\infty(\mathbb{R}^2)$,

$$V_{(x,y)}f = a(x,y)\frac{\partial f}{\partial x}(x,y) + b(x,y)\frac{\partial f}{\partial y}(x,y).$$

So for $\iota_* \colon T\mathbb{S}^1 \to \mathbb{R}^2$, at θ_0

$$\iota_{*,\theta_0}\left(\left.\frac{d}{d\theta}\right|_{\theta_0}\right) = -\sin\theta_0\frac{\partial}{\partial x} + \cos\theta_0\frac{\partial}{\partial y}.$$

which is because for any $f(x,y) \in C^\infty(M)$,

$$\iota_{*,\theta_0}\left(\left.\frac{d}{d\theta}\right|_{\theta_0}\right)f = \left.\frac{d}{d\theta}\right|_{\theta_0} f \circ \iota = \left.\frac{d}{d\theta}\right|_{\theta_0} f(\cos\theta, \sin\theta) = -\sin\theta_0\frac{\partial f}{\partial x}(\theta_0) + \cos\theta_0\frac{\partial f}{\partial y}(\theta_0).$$

And note that ι_* is $C^\infty(\mathbb{S}^1)$-linear. Then the induce metric on $\mathbb{S}^1$ is given by

$$g_{\theta\theta}(\theta_0) = \left\langle \left.\frac{d}{d\theta}\right|_{\theta_0}, \left.\frac{d}{d\theta}\right|_{\theta_0} \right\rangle = \left\langle \iota_{*,\theta_0}\left.\frac{d}{d\theta}\right|_{\theta_0}, \iota_{*,\theta_0}\left.\frac{d}{d\theta}\right|_{\theta_0} \right\rangle = \langle(-\sin\theta_0, \cos\theta_0), (-\sin\theta_0, \cos\theta_0)\rangle = 1$$

Therefore, $g = d\theta \otimes d\theta$, which implies that the corresponding Christoffel coefficients

$$\Gamma^\theta_{\theta\theta} = \frac{1}{2} g^{\theta\theta} \frac{d}{d\theta} g_{\theta\theta} = 0,$$

It follows that that for any $X = a(\theta)\frac{d}{d\theta}, Y = b(\theta)\frac{d}{d\theta}$, we have

$$\nabla_X Y = a(\theta)\dot{b}(\theta)\frac{d}{d\theta},$$

because $\nabla_{\frac{d}{d\theta}} \frac{d}{d\theta} = 0$. Moreover, we also have

$$\Delta_{\mathbb{S}^1} h(\theta) = \frac{d^2}{d\theta^2} h \ \Rightarrow\ \Delta_{\mathbb{S}^1} = \frac{d^2}{d\theta^2}.$$

On the other hand, consider $\mathbb{S}^1 \subset \mathbb{R}^2$ embedded, let $T = \iota_*(d/d\theta)$. Then

$$\widetilde{X} = \iota_*(X) = a(\theta)T, \quad \widetilde{Y} = \iota_*(Y) = b(\theta)T.$$

Moreover, they can be extended to vector fields on $\mathbb{R}^2$. Let $\widetilde{T} = (-y, x)$, i.e., $T = \widetilde{T}|_{\mathbb{S}^1}$. Then

$$\widetilde{X}(x,y) = \widetilde{a}(x,y)\widetilde{T}, \quad \widetilde{Y}(x,y) = \widetilde{b}(x,y)\widetilde{T},$$

where $\widetilde{a}(\cos\theta, \sin\theta) = a(\theta)$ and $\widetilde{b}(\cos\theta, \sin\theta) = b(\theta)$. Note that

$$\begin{aligned}
\widetilde{T}(\widetilde{a})|_{\mathbb{S}^1}(\theta) &= -y\frac{\partial \widetilde{a}}{\partial x} + x\frac{\partial \widetilde{a}}{\partial y} = \dot{a}(\theta) \\
\widetilde{T}(\widetilde{b})|_{\mathbb{S}^1}(\theta) &= -y\frac{\partial \widetilde{b}}{\partial x} + x\frac{\partial \widetilde{b}}{\partial y} = \dot{b}(\theta)
\end{aligned}$$

and so we have

$$\widetilde{\nabla}_{\widetilde{T}}\widetilde{T}|_{\mathbb{S}^1}(\theta) = -\widetilde{T}(y)\frac{\partial}{\partial x} + \widetilde{T}(x)\frac{\partial}{\partial y} = -\cos\theta\frac{\partial}{\partial x} - \sin\theta\frac{\partial}{\partial y},$$

which mean that $\widetilde{\nabla}_{\widetilde{T}}\widetilde{T}|_{\mathbb{S}^1} = -\vec{n} \in N_\theta\mathbb{S}^1$. So

$$\nabla_T T = (\widetilde{\nabla}_{\widetilde{T}}\widetilde{T})^\top = 0,$$

and by viewing $T = d/d\theta$, $\Gamma^\theta_{\theta\theta} = 0$ as same calculation as above. Moreover,

$$\widetilde{\nabla}_{\widetilde{X}}\widetilde{Y}|_{\mathbb{S}^1}(\theta) = a(\theta)\left(\widetilde{T}(\widetilde{b})\widetilde{T} + \widetilde{b}\widetilde{\nabla}_{\widetilde{T}}\widetilde{T}\right) = a(\theta)\dot{b}(\theta)T - a(\theta)b(\theta)\vec{n}(\theta).$$

It follows that

$$\nabla_X Y(\theta) = (\widetilde{\nabla}_{\widetilde{X}}\widetilde{Y})^\top = a(\theta)\dot{b}(\theta)\frac{d}{d\theta},$$

and

$$\mathrm{II}(X,Y) = (\widetilde{\nabla}_{\widetilde{X}}\widetilde{Y})^\perp = -a(\theta)b(\theta)\vec{n}(\theta) = -\langle X, Y\rangle\,\vec{n}.$$

2. In general case, by embedding $\mathbb{S}^d \subset \mathbb{R}^{d+1}$,

$$\nabla_X Y = \widetilde{\nabla}_X Y + \langle X, Y\rangle\,\vec{n}$$

where $\vec{n} \in \Gamma(N\mathbb{S}^d)$ is the outer unit normal vector. Then let $\{X_i\}_{i=1}^d$ be an orthonormal frame of $T\mathbb{S}^d$ and so $\{X_i, \vec{n}\}$ is an orthonormal frame of $\mathbb{R}^{d+1}$. For any $f \in C^\infty(\mathbb{S}^d)$, let $F \in C^\infty(\mathbb{R}^{d+1})$ be an extension of f. As we know $\operatorname{grad} f = (\overline{\operatorname{grad}} f)^\top$, $\widetilde{\nabla}_{X_i}\vec{n} = X_i$, and

$$\Delta_{\mathbb{R}^{d+1}} F = \sum_{i=1}^d \widetilde{\nabla}^2 F(X_i, X_i) + \widetilde{\nabla}^2 F(\vec{n}, \vec{n}),$$

we have

$$\begin{aligned}
\Delta_{\mathbb{S}^d} f &= \sum_{i=1}^d \nabla^2 f(X_i, X_i) = \sum_{i=1}^d \langle \nabla_{X_i} \operatorname{grad} f, X_i \rangle \\
&= \sum_{i=1}^d \left\langle \widetilde{\nabla}_{X_i} \operatorname{grad} F + \langle \operatorname{grad} F, X_i \rangle \vec{n}, X_i \right\rangle = \sum_{i=1}^d \left\langle \widetilde{\nabla}_{X_i} \operatorname{grad} F, X_i \right\rangle \\
&= \sum_{i=1}^d \left\langle \widetilde{\nabla}_{X_i} (\overline{\operatorname{grad}} F)^\top, X_i \right\rangle = \sum_{i=1}^d \left\langle \widetilde{\nabla}_{X_i} (\overline{\operatorname{grad}} F - \langle \overline{\operatorname{grad}} F, \vec{n} \rangle \vec{n}), X_i \right\rangle \\
&= \sum_{i=1}^d \left\langle \widetilde{\nabla}_{X_i} \overline{\operatorname{grad}} F, X_i \right\rangle - d \left\langle \overline{\operatorname{grad}} F, \vec{n} \right\rangle = \sum_{i=1}^d \widetilde{\nabla}^2 F(X_i, X_i) - d \widetilde{\nabla}_{\vec{n}} F \\
&= \Delta_{\mathbb{R}^{d+1}} F - \widetilde{\nabla}^2 F(\vec{n}, \vec{n}) - d \widetilde{\nabla}_{\vec{n}} F
\end{aligned}$$

On the other hand, choose a polar coordinate chart for $\mathbb{R}^{d+1} \backslash \{0\}$, $(r, \theta = (\theta^1, \cdots, \theta^d))$. Then for Euclidean coordinate $(x^1, \cdots, x^{d+1})$,

$$x^i = r\iota^i(\theta),$$

where $\iota \colon \mathbb{S}^d \hookrightarrow \mathbb{R}^{d+1}$ is the inclusion. Then we have

$$\frac{\partial}{\partial r} = \frac{\partial x^i}{\partial r} \frac{\partial}{\partial x^i} = \iota^i \frac{\partial}{\partial x^i}$$

and

$$\frac{\partial}{\partial \theta^\alpha} = \frac{\partial x^i}{\partial \theta^\alpha} \frac{\partial}{\partial x^i} = r \frac{\partial \iota^i}{\partial \theta^\alpha} \frac{\partial}{\partial x^i}$$

Then we have, first

$$g_{rr} = \left\langle \frac{\partial}{\partial r}, \frac{\partial}{\partial r} \right\rangle = \delta_{ij} \iota^i \iota^j = 1,$$

which also implies that

$$0 = \frac{\partial}{\partial \theta^\alpha} \delta_{ij} \iota^i \iota^j = 2\delta_{ij} \iota^i \frac{\partial \iota_j}{\partial \theta^\alpha} = 0.$$

So, second

$$g_{\alpha r} = \left\langle \frac{\partial}{\partial \theta^\alpha}, \frac{\partial}{\partial r} \right\rangle = r\delta_{ij} \iota^i \frac{\partial \iota^i}{\partial \theta^\alpha} = 0.$$

Finally,

$$g_{\alpha\beta} = \left\langle \frac{\partial}{\partial \theta^\alpha}, \frac{\partial}{\partial \theta^\beta} \right\rangle = r^2 \delta_{ij} \frac{\partial \iota^i}{\partial \theta^\alpha} \frac{\partial \iota^i}{\partial \theta^\beta}.$$

For $g_{\alpha\beta}$, note that ι is the isometric inclusion, so

$$g_{\mathbb{S}^d} \left(\left.\frac{\partial}{\partial \theta^\alpha}\right|_{\mathbb{S}^d}, \left.\frac{\partial}{\partial \theta^\beta}\right|_{\mathbb{S}^d} \right) = g \left(\iota_* \left.\frac{\partial}{\partial \theta^\alpha}\right|_{\mathbb{S}^d}, \iota_* \left.\frac{\partial}{\partial \theta^\beta}\right|_{\mathbb{S}^d} \right) = \delta_{ij} \frac{\partial \iota^i}{\partial \theta^\alpha} \frac{\partial \iota^i}{\partial \theta^\beta},$$

because

$$\iota_* \left.\frac{\partial}{\partial \theta^\alpha}\right|_{\mathbb{S}^d} = \frac{\partial \iota^i}{\partial \theta^\alpha}\frac{\partial}{\partial x^i}.$$

It follows that

$$g_{\alpha\beta} = r^2 h_{\alpha\beta}, \quad h_{\alpha\beta} := g_{\mathbb{S}^d}\left(\left.\frac{\partial}{\partial \theta^\alpha}\right|_{\mathbb{S}^d}, \left.\frac{\partial}{\partial \theta^\beta}\right|_{\mathbb{S}^d} \right).$$

So we have

$$g = dr \otimes dr + r^2 h_{\alpha\beta} d\theta^\alpha \otimes d\theta^\beta.$$

It also implies that (r, θ) is a semi-geodesic coordinate of $\mathbb{R}^{d+1}$ w.s.t. $\mathbb{S}^d$ and distance function r. Moreover, based on this coordinate system, we have

$$g = (g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & r^2 h_{\alpha\beta} \end{pmatrix}, \quad g^{-1} = (g^{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & r^{-2} h^{\alpha\beta} \end{pmatrix},$$

which implies that

$$\sqrt{|g|} = r^d \sqrt{|h|}.$$

Let $(y^1, \cdots y^{d+1}) = (r, \theta^1, \cdots, \theta^d)$. For any $F = F(r,\theta) \in C^\infty(\mathbb{R}^{d+1})$, note that

$$g^{ij}\frac{\partial F}{\partial y^j} = \begin{cases} \frac{\partial F}{\partial r}, & i = 1 \\ \frac{h^{i\beta}}{r^2}\frac{\partial F}{\partial \theta^\beta}, & i \neq 1 \end{cases}$$

Then by the definition of Laplace-Beltrami operator,

$$\begin{aligned} \Delta_{\mathbb{R}^{d+1}} F &= \frac{1}{|g|}\frac{\partial}{\partial y^i}\left(\sqrt{|g|} g^{ij} \frac{\partial F}{\partial y^j}\right) \\ &= \frac{1}{|g|}\frac{\partial}{\partial r}\left(\sqrt{|g|}\frac{\partial F}{\partial r}\right) + \frac{1}{|g|}\frac{\partial}{\partial \theta^\alpha}\left(\sqrt{|g|}\frac{h^{\alpha\beta}}{r^2}\frac{\partial F}{\partial \theta^\beta}\right) = \mathrm{I} + \mathrm{II}. \end{aligned}$$

For I,

$$\begin{aligned} \mathrm{I} &= \frac{1}{|g|}\frac{\partial}{\partial r}\left(\sqrt{|g|}\frac{\partial F}{\partial r}\right) = \frac{1}{r^d\sqrt{|h|}}\frac{\partial}{\partial r}\left(r^d\sqrt{|h|}\frac{\partial F}{\partial r}\right) \\ &= \frac{d}{r}\frac{\partial F}{\partial r} + \frac{\partial^2 F}{\partial r^2}. \end{aligned}$$

For II,

$$\begin{aligned} \mathrm{II} &= \frac{1}{|g|}\frac{\partial}{\partial \theta^\alpha}\left(\sqrt{|g|}\frac{h^{\alpha\beta}}{r^2}\frac{\partial F}{\partial \theta^\beta}\right) \\ &= \frac{1}{r^d\sqrt{|h|}}\frac{\partial}{\partial \theta^\alpha}\left(r^{d-2}\sqrt{|h|}h^{\alpha\beta}\frac{\partial F}{\partial \theta^\beta}\right) \\ &= \frac{1}{r^2\sqrt{|h|}}\frac{\partial}{\partial \theta^\alpha}\left(\sqrt{|h|}h^{\alpha\beta}\frac{\partial F}{\partial \theta^\beta}\right) \\ &= \frac{1}{r^2}\Delta_{\mathbb{S}^d}\hat{F}_r, \end{aligned}$$

where $\hat{F}_r(\theta) = F(r,\theta) \in C^\infty(\mathbb{S}^d)$. So we have

$$\Delta_{\mathbb{R}^{d+1}} F = \frac{d}{r}\frac{\partial F}{\partial r} + \frac{\partial^2 F}{\partial r^2} + \frac{1}{r^2}\Delta_{\mathbb{S}^d}\hat{F}_r.$$

Then on $\mathbb{S}^d$, i.e., $r = 1$, for any $f \in C^\infty(\mathbb{S}^d)$, choose any extension F, then

$$\Delta_{\mathbb{S}^d} f = \left(\Delta_{\mathbb{R}^{d+1}} - d\frac{\partial}{\partial r} - \frac{\partial^2}{\partial r^2} \right) F|_{r=1}.$$

In particular, if we choose $F(r, \theta) = f(\theta)$ for any r, then

$$\Delta_{\mathbb{S}^d} f = \Delta_{\mathbb{R}^{d+1}} F.$$

Remark 3.3.1. If we embedded $\iota \colon \mathbb{S}^d \hookrightarrow \mathbb{R}^{d+1}$ by

$$\begin{aligned}
x_1 &= \cos\theta_1 \\
x_2 &= \sin\theta_1 \cos\theta_2 \\
x_3 &= \sin\theta_1 \sin\theta_2 \cos\theta_3 \\
&\vdots \\
x_d &= \sin\theta_1 \cdots \sin\theta_{d-1} \cos\theta_d \\
x_{d+1} &= \sin\theta_1 \cdots \sin\theta_{d-1} \sin\theta_d
\end{aligned}$$

then the induced metric metric is given by

$$g_{\mathbb{S}^d} = d\theta_1 \otimes d\theta_1 + \sin^2\theta_1 d\theta_2 \otimes d\theta_2 + \sin^2\theta_1 \sin^2\theta_2 d\theta_3 \otimes d\theta_3 + \cdots + \left(\prod_{i=1}^{d-1} \sin^2\theta_i \right) d\theta_d \otimes d\theta_d.$$

Note that it implies that $\left\{ \frac{\partial}{\partial\theta^\alpha} \right\}$ is a orthogonal frame of $\mathbb{S}^d$ but not normal.

Radially Symmetric Space. For a Riemannian manifold (M, g) of dimension d, consider a normal coordinate system, i.e., $\exp_p \colon U \to M$, and using the polar coordinate $(r, \theta) \in (0, \infty) \times \mathbb{S}^{d-1}$, then we have

$$g = dr \otimes dr + g_{\alpha\beta}(r, \theta) d\theta^\alpha \otimes d\theta^\beta.$$

Definition 3.3.2. A Riemannian manifold (M, g) is called radially symmetric if on any normal polar chart, for

$$g = dr \otimes dr + g_{\alpha\beta}(r, \theta) d\theta^\alpha \otimes d\theta^\beta,$$

it has

$$g_{\alpha\beta}(r, \theta) = G(r)^2 h_{\alpha\beta}(\theta),$$

where

$$G(0) = 0, \quad G'(0) = 1,$$

and $h_{\alpha\beta} d\theta^\alpha \otimes d\theta^\beta$ is the canonical Riemannian metric on $\mathbb{S}^{d-1}$.

Proposition 3.3.3. *Let (M, g) be a Riemannian manifold with constant sectional curvature κ. Then M is radially symmetric, i.e.,*

$$g = dr \otimes dr + G(r)^2 h_{\alpha\beta}(\theta) d\theta^\alpha \otimes d\theta^\beta,$$

where $G(0) = 0, \quad G'(0) = 1$. Moreover,

$$G''(r) + \kappa G(r) = 0.$$

Proof. Fix $p \in M$, and on T_pM, choosing an orthonormal basis such that $T_pM = \mathbb{R}^d$. For the $\Phi \colon (0, \delta) \times \mathbb{S}^{d-1} \to M$, $\Phi(r, \theta) = \exp_p(r\theta)$ is a geodesic, which also provides the normal polar coordination (r, θ). Fix a $\theta_0 \in \mathbb{S}^{d-1}$. Consider a normal radial geodesic $\gamma(r)$ with $\gamma(0) = p$ and $\dot\gamma(0) = \theta_0 \in \mathbb{S}^{d-1}$. Let $\theta_\alpha \colon (-\varepsilon, \varepsilon) \to \mathbb{S}^{d-1}$ be a smooth map with $\theta(0) = \theta_0$ and $\dot\theta(0) = \frac{\partial}{\partial\theta^\alpha}|_{\mathbb{S}^{d-1}, \theta_0}$. Note that:

(i) $W_\alpha = \frac{\partial}{\partial\theta^\alpha}|_{\mathbb{S}^{d-1},\theta_0} = \iota_* \frac{\partial}{\partial\theta^\alpha}|_{\mathbb{S}^{d-1},\theta_0} \in \mathbb{R}^d$, i.e., an vector in T_pM,

(ii) because $|\theta(s)| \equiv 1$, $W_\alpha \perp \theta_0$.

Consider the radial variation

$$F_\alpha(s,r) = \exp_p(r\theta_\alpha(s))$$

and its two vector fields

$$V_\alpha(r) = \frac{\partial F_\alpha}{\partial s}(r,0) = (d\exp_p)_{r\theta_0}(rW_\alpha), \quad T(r) = \frac{\partial F_\alpha}{\partial r}(r,0) = \dot\gamma(r).$$

Then $V_\alpha(r)$ is a Jacobian field. Moreover, we have

$$V_\alpha(0) = 0, \quad \nabla_T V_\alpha(0) = W_\alpha.$$

Because $W_\alpha \perp T(0) = \theta_0$ and $V_\alpha(0) \perp T(0)$, $V_\alpha \perp T$. It implies that

$$R(V_\alpha, T)T = \kappa V_\alpha,$$

since M has constant curvature, i.e.

$$\langle R(V_\alpha,T)T, V_\alpha\rangle = \kappa\left(|V_\alpha|^2 |T|^2 - \langle V_\alpha, T\rangle^2\right) = \kappa |V_\alpha|^2.$$

So the Jacobian equation is

$$\frac{D^2}{dt^2}V_\alpha(t) + \kappa V_\alpha = 0.$$

So when choosing an orthogonal frame $\{E_i(r)\}$ with $E_1(r) = T(r)$ along $\gamma(r)$ and $E_\alpha(r)$ are obtained by parallel moving W_α along $\gamma(r)$, which are orthogonal because $\langle W_\alpha, W_\beta\rangle = 0$. Using similar result, because

$$\langle V_\alpha(0), E_\beta(0)\rangle = \langle \nabla_T V_\alpha(0), E_\beta(0)\rangle = \langle W_\alpha, W_\beta\rangle = 0, \ \forall\ \beta \neq \alpha,$$

$V_\alpha \perp E_\beta$. So

$$V_\alpha(r) = G_\alpha(r) E_\alpha(r)$$

and $G_\alpha(r)$ should satisfy

$$f''(r) + \kappa f(r) = 0, \quad f(0) = 0,\ f'(0) = 1$$

By the uniqueness of ODE, for all α

$$G(r) = G_\alpha(r) = \begin{cases} \frac{1}{\sqrt{\kappa}}\sin(\sqrt{\kappa} r), & \kappa > 0, \\ r, & \kappa = 0, \\ \frac{1}{\sqrt{-\kappa}}\sinh(\sqrt{-\kappa} r), & \kappa < 0, \end{cases}$$

Next, the the coordinate tangent vectors on M are given by

$$\left.\frac{\partial}{\partial\theta^\alpha}\right|_M = \Phi_*\left(\left.\frac{\partial}{\partial\theta^\alpha}\right|_{\mathbb{S}^{d-1}}\right)$$

So by definition

$$d\Phi_{(r,\theta_0)}\left.\frac{\partial}{\partial\theta^\alpha}\right|_{\mathbb{S}^{d-1},\theta_0} = \left.\frac{d}{ds}\right|_{s=0}\Phi(r,\theta_\alpha(s)) = \left.\frac{d}{ds}\right|_{s=0}\exp_p(r\theta_\alpha(s)) = V_\alpha(r),$$

i.e.,

$$\left.\frac{\partial}{\partial\theta^\alpha}\right|_{M,(r,\theta_0)} = V_\alpha(r).$$

Therefore,

$$g_{\alpha\beta}(r,\theta_0) = \langle V_\alpha(r), V_\beta(r)\rangle = G(r)^2\langle E_\alpha(r), E_\beta(r)\rangle = G(r)^2\langle E_\alpha(0), E_\beta(0)\rangle = G(r)^2\langle W_\alpha, W_\beta\rangle$$

because $E_\alpha(r)$ are obtained by parallel moving. Let

$$h_{\alpha\beta}(\theta_0) = \langle W_\alpha, W_\beta\rangle = \left\langle \left.\frac{\partial}{\partial\theta^\alpha}\right|_{\mathbb{S}^{d-1},\theta_0}, \left.\frac{\partial}{\partial\theta^\beta}\right|_{\mathbb{S}^{d-1},\theta_0}\right\rangle,$$

i.e., $h_{\alpha\beta}d\theta^\alpha\otimes d\theta^\beta$ is the canonical Riemannian metric on $\mathbb{S}^{d-1}$. So we have

$$g = dr\otimes dr + G(r)^2 h_{\alpha\beta}(\theta)d\theta^\alpha\otimes d\theta^\beta. \qquad \square$$

For a radially symmetric space (M,g), using similar calculation as $\mathbb{S}^d$, we have

$$\Delta_M = L_r + \frac{1}{G(r)^2}\Delta_{\mathbb{S}^{d-1}},$$

where

$$L_r = \frac{\partial^2}{\partial r^2} + (d-1)\frac{G'(r)}{G(r)}\frac{\partial}{\partial r},$$

is called the radial Laplacian.

Examples.

Example 3.3.4 (Circle)**.** Consider

$$\mathbb{S}^1 = \left\{e^{i\theta} \colon 0\le\theta<2\pi\right\}\subset\mathbb{R}^2.$$

Let B be a Brownian motion. Define

$$X_t = e^{iB_t} = (\cos B_t, \sin B_t).$$

Then X is a Brownian motion on $\mathbb{S}^1$.

Proof by Laplacian Operator. First, When viewing $\mathbb{S}^1\subset\mathbb{R}^2$, $X_t = (\cos B_t, \sin B_t)$, by Itô formula,

$$dX_t = (-\sin B_t, \cos B_t)\circ dB_t = \iota_*\frac{d}{d\theta}(B_t)\circ dB_t.$$

Therefore,

$$V = \iota_*\frac{d}{d\theta} = \frac{d}{d\theta}.$$

Then X_t is a diffusion process with

$$L = \frac{1}{2}\frac{d^2}{d\theta^2} = \frac{1}{2}\Delta_{\mathbb{S}^1}.$$

So X is a Brownian motion on $\mathbb{S}^1$. $\square$

Proof by Anti-development. First, note that at any $\theta \in \mathbb{S}^{\nVdash}$, $O(\mathbb{S}^1)_\theta = \{-1, 1\}$ and so the orthonormal frame bundle $O(M) = \bigcup_{\theta \in \mathbb{S}^1} \{-1, 1\}$. By $\Gamma^\theta_{\theta\theta} = 0$, any horizontal vector $\dot{u}(0)$ vanishes on the vertical part. Then by $dU_t = P^H_\alpha \circ dX^\alpha_t$, the horizontal lift of X starting from $U_0 = 1$, $U_t \equiv 1$.
On the other hand, consider $P(\theta)\colon \mathbb{R}^2 \to T_\theta \mathbb{S}^1$. Note that $n = (-\sin\theta, \cos\theta) \in N_\theta \mathbb{S}^1$. So

$$P(\theta) = \begin{pmatrix} \sin^2\theta & -\sin\theta\cos\theta \\ -\sin\theta\cos\theta & \cos\theta^2 \end{pmatrix}.$$

Therefore,

$$\begin{aligned} P_1(\theta) &= (\sin^2\theta, -\sin\theta\cos\theta)^\top = \iota_* \left(-\sin\theta \frac{d}{d\theta}\right) \\ P_2(\theta) &= (-\sin\theta\cos\theta, \cos^2\theta)^\top = \iota_* \left(\cos\theta \frac{d}{d\theta}\right). \end{aligned}$$

Then by $dW^t = U_t^{-1} P_\alpha(X_t) \circ dX^\alpha_t$,

$$\begin{aligned} dW_t &= -\sin B_t \circ dX^1_t + \cos B_t \circ dX^2_t \\ &= \sin^2 B_t \circ dB_t + \cos^2 B_t \circ dB_t \\ &= dB_t, \end{aligned}$$

and $W_0 = 0$, so $W_t = B_t$. □

Example 3.3.5 (Sphere)**.** Consider $\mathbb{S}^d \subset \mathbb{R}^{d+1}$. Let $\vec{n}$ be the outer unit normal vector. Then the orthogonal projection

$$P(\theta)\colon \mathbb{R}^{d+1} \to T_\theta \mathbb{S}^d$$

is given by $P(\theta)(x) = x - \langle x, \vec{n}(\theta)\rangle\, \vec{n}(\theta)$. The matrix expression of $P(\theta)$ is

$$P(\theta)_{ij} = \delta_{ij} - n(\theta)_i n(\theta)_j$$

Or when embedding $\mathbb{S}^d \hookrightarrow \mathbb{R}^{d+1}$, $P(x)_{ij} = \delta_{ij} - x^i x^j$. Therefore, by $\Delta_M = \sum_\alpha P^2_\alpha$, X is a Brownian motion on $\mathbb{S}^d$, when it satisfies

$$X^i_t = X^i_0 + \sum_{j=1}^{d+1} \int_0^t (\delta_{ij} - X^i_s X^j_s) \circ dW^j_t,$$

where W is a standard Brownian motion on $\mathbb{R}^{d+1}$.

Example 3.3.6 (Radially Symmetric Space)**.** Let (M, g) be a radially symmetric space and $X_t = (r_t, \theta_t)$ be a Brownian motion on M expressed as polar coordinate. Consider its radial process and angular process separately.
(1) Radial process: Choose $f(r, \theta) = r$ to the martingale problem,

$$\begin{aligned} \beta_t &= r_t - r_0 - \frac{1}{2}\int_0^t \Delta_M f(X_s) ds \\ &= r_t - r_0 - \frac{d-1}{2}\int_0^t \frac{G'(r_s)}{G(r_s)} ds \end{aligned}$$

is a local martingale. Moreover, the quadratic variation is given by

$$\langle \beta, \beta \rangle_t = \int_0^t \Gamma(r, r)(X_s) ds,$$

where we have

$$\Gamma(r,r) = \langle \nabla r, \nabla r\rangle = g^{ij}\frac{\partial r}{\partial x^i}\frac{\partial r}{\partial x^j} = |\nabla r|^2 = 1.$$

So

$$\langle \beta, \beta\rangle_t = t,$$

which implies that β is a 1-dimensional Brownian motion. On the other hand, we can see the radial process r_t is generated by the radial operator L_r.

(2) Angular process: Let $f \in C^\infty(\mathbb{S}^{d-1})$, which can be extended to $f(r,\theta) = f(\theta) \in C^\infty(M)$. Then

$$f(X_t) = f(\theta_t), \quad \Delta_M f(r,\theta) = \frac{1}{G(r)^2}\Delta_{\mathbb{S}^{d-1}}.$$

It implies that

$$M_t^f = f(\theta_t) - f(\theta_0) - \frac{1}{2}\int_0^t \frac{1}{G(r_s)^2}\Delta_{\mathbb{S}^{d-1}} f(\theta_s)ds$$

Let

$$T_{t'} = \int_0^{t'} \frac{1}{G(r_s)^2}ds,$$

which is a nondecreasing process. So we can consider

$$\tau_t = \inf\{t' \geq 0\colon T_{t'} \geq t\} = T^{-1}(t),$$

where the final equality is because T is strictly increasing in t. Then by considering the time-change, let $z_t = \theta_{\tau_t}$, we have

$$M_{\tau_t}^f = f(z_t) - f(z_0) - \frac{1}{2}\int_0^t \Delta_{\mathbb{S}^{d-1}} f(z_s)ds.$$

Since M_t is a local martingale, $t \mapsto M_{\tau_t}^f$ is also a local martingale w.s.t. $\mathcal{F}_{\tau_t}$. Therefore, $t \mapsto z_t = \theta_{\tau_t}$ is a Brownian motion on $\mathbb{S}^{d-1}$

Claim: r and z are independent.

Proof. I. First, we embed $\mathbb{S}^{d-1} \hookrightarrow \mathbb{R}^d$ with $\theta \mapsto y = (y^\alpha)$ and choose $f(\theta) = f(y) = f^\alpha(y) = y^\alpha$, the coordinate function on $\mathbb{R}^\alpha$. Let U_t be a horizontal lift of $\theta_t = Y_t \in \mathbb{R}^d$ and W_t be its anti-development. Then

$$W_t = \int_0^t U_s^{-1}P_\alpha(Y_s)\circ dY_s^\alpha.$$

Because τ_t is continuous, for any continuous semimartingales H, K,

$$\int_0^{\tau_t} H_s dK_s = \int_0^t H_{\tau_s}dK_{\tau_s}, \quad \langle H\circ\tau, K\circ\tau\rangle_t = \langle H, K\rangle_{\tau_t}.$$

So U_{τ_t} is a horizontal lift of θ_{τ_t} and W_{τ_t} is the anti-development of θ_{τ_t}. Because θ_{τ_t} is a Brownian motion on $\mathbb{S}^{d-1}$, $\{W_{\tau_t}\}_{t\geq 0}$ is an Euclidean Brownian motion.

II. Next, we need to check

$$\langle W\circ\tau, \beta\rangle = 0$$

For any $f \in C^\infty(\mathbb{S}^{d-1})$,

$$\langle M^f, \beta\rangle_t = \int_0^t \Gamma(f,r)(X_s)ds = \int_0^t \langle \nabla f, \nabla r\rangle (X_s)ds.$$

Note that f is viewing as $f(r,\theta)=f(\theta)\in C^\infty(M)$. By

$$\langle \nabla f^\alpha, \nabla r\rangle = g^{ij}\frac{\partial f^\alpha}{\partial x^i}\frac{\partial r}{\partial x^j}=0,$$

we have

$$\left\langle M^{f^\alpha}, \beta\right\rangle = \langle Y^\alpha,\beta\rangle \equiv 0, \quad \forall\ \alpha=1,2,\cdots,d,$$

which implies that

$$\langle W,\beta\rangle_t = \int_0^t U_s^{-1}P_\alpha(Y_s)d\left\langle Y^\alpha,\beta\right\rangle = 0$$

By the following lemma, we have

$$\langle W\circ\tau,\beta\rangle_t = 0.$$

III. Because $\{W_{\tau_t}\}_{t\geq 0}$ and $\{\beta_t\}_{t\geq 0}$ are two Brownian motions, they are independent. Moreover, since

$$d\theta_{\tau_t} = U_{\tau_t}^{-1}e_i\circ dW_{\tau_t}^i$$

and

$$dr_t = d\beta_t + \frac{d-1}{2}\frac{G'(r_t)}{G(r_t)}dt,$$

r_t and $z_t=\theta_{\tau_t}$ are independent. □

Lemma 3.3.7. *Let W and M be two continuous local martingale and τ be a continuous time-change. If $\langle W,M\rangle\equiv 0$, then $\langle W\circ\tau, M\rangle\equiv 0$.*

Example 3.3.8 (In Local Coordinates)**.** Let $(x^1,\cdots,x^d)$ be a local chart of M and $(x^i, i=1,\cdots,d; e_j^i, 1\leq i\leq j\leq d)$ be a chart of $O(M)$. The equation of a horizontal Brownian motion is

$$dU_t = H_i(U_t)\circ dW_t^i,$$

where W is a d-dimensional Brownian motion. We have already shown that

$$H_i(u) = e_j^i\frac{\partial}{\partial x^j} - e_i^j e_m^\ell \Gamma_{j\ell}^k(x)\frac{\partial}{\partial e_m^k}.$$

So for $U_t=(X_t^i, e_j^i(t))$,

$$\begin{aligned} dX_t^i &= e_j^i(t)\circ dW_t^j,\\ de_j^i(t) &= -\Gamma_{k\ell}^i(X_t)e_j^\ell(t)e_m^k(t)\circ dW_t^m. \end{aligned}$$

For the first equation,

$$dX_t^t = e_j^i(t)dW_t^j + \frac{1}{2}d\left\langle e_j^i, dW^j\right\rangle_t.$$

Let

$$dM_t^i = e_j^i(t)dW_t^j \ \Rightarrow\ d\left\langle M^i, M^j\right\rangle_t = \sum_{k=1}^d e_k^i(t)e_k^j(t)dt.$$

Note that because $u\in O(M)$ and $ue_\ell = e_\ell^i\frac{\partial}{\partial x^i}$,

$$\delta_{\ell m} = \langle ue_\ell, ue_m\rangle = e_\ell^i g_{ij}e_m^j,$$

i.e., $ege^\top = I$ for $e = (e^i_j)$. Moreover, because e is invertible,

$$e^\top = g^{-1}, \quad i.e. \sum_{k=1}^d e^i_k e^j_k = g^{ij}.$$

Therefore,

$$d\left\langle M^i, M^j \right\rangle_t = g^{ij}\left(X_t\right)dt.$$

Then let $\sigma = g^{-\frac{1}{2}}$ and define

$$B_t = \int_0^t \sigma\left(X_s\right)^{-1} dM_s,$$

Because

$$d\left\langle B^i, B^j \right\rangle_t = \delta^{ij} dt,$$

B is an Euclidean Brownian motion. Also,

$$dM_t = \sigma(X_t)dB_t.$$

On the other hand, by the second equation,

$$d\left\langle e^i_j, dW^j \right\rangle_t = -\Gamma^i_{k\ell}\left(X_t\right) e^\ell_m(t) e^k_m(t) = -g^{\ell k}\left(X_t\right)\Gamma^i_{k\ell}\left(X_t\right).$$

Therefore, X_t satisfies

$$dX^i_t = \sigma^i_j\left(X_t\right) dB^j_t - \frac{1}{2} g^{\ell k}\left(X_t\right)\Gamma^i_{k\ell}\left(X_t\right)dt,$$

where B is a d-dimensional Brownian motion. Note that X_t is a diffusion process generated by

$$L = -\frac{1}{2} g^{\ell k}\Gamma^i_{k\ell}\frac{\partial f}{\partial x^i} + \frac{1}{2} g^{ij}\frac{\partial^2 f}{\partial x^i \partial x^j}.$$

Also, because

$$\Delta_M f = g^{ij}\frac{\partial^2 f}{\partial x^i \partial x^j} - g^{k\ell}\Gamma^i_{k\ell}\frac{\partial f}{\partial x^i} = 2L,$$

X is a $\Delta_M/2$-diffusion process, and so X is a Brownian motion on M.

3.4 Radial Process

Let (M, g) be a complete Riemannian manifold of dimension d. Fix $o \in M$ and let C_o be the cut locus and $E_o = M \backslash C_o$. Let

$$r(x) = d(x, o).$$

Then r is smooth on $M \backslash C_o \cup \{o\}$ and $|\nabla r| \equiv 1$.
Let X be a Brownian motion on M starting within E_o. Before it hitting C_o, i.e., $t < T_{C_o}$, where T_{C_o} is the hitting time of C_o,

$$\beta_t = r(X_t) - r(X_0) - \frac{1}{2}\int_0^t \Delta_M r(X_s) ds.$$

is a local martingale. Moreover, because

$$\langle r, r \rangle_t = \langle \beta, \beta \rangle_t = \int_0^t \Gamma\left(r, r\right)(X_s) ds = \int_0^t \left|\nabla r(X_s)\right|^2 ds = t,$$

$r(X)$ is a Brownian motion on $\mathbb{R}$.

Model Manifolds. Choose the polar coordinate (r,θ) of $\mathbb{R}^n$, then equip $\mathbb{R}^n$ with Riemannian metric

$$g = dr \otimes dr + G(r)^2 g_{\mathbb{S}^{n-1}}.$$

where $G\colon [0,\infty) \to [0,\infty)$ is smooth with $G(0)=0$, $G'(0)=1$, and $G(r)>0$ for $r \neq 0$. Then any Riemannian manifold $(\mathbb{M}, g)$ isometric to $(\mathbb{R}^n, g)$ is called a model manifold, and denote $0 \in \mathbb{M}$ be a fixed point corresponding to $0 \in \mathbb{R}^n$. For the Levi-Civita connection ∇, we have the Koszul formula

$$\begin{aligned} 2\left\langle \nabla_U V, W\right\rangle &= U\left\langle V, W\right\rangle + V\left\langle W, U\right\rangle - W\left\langle U, V\right\rangle \\ &\quad + \left\langle [U,V], W\right\rangle - \left\langle [V,W], U\right\rangle - \left\langle [U,W], V\right\rangle. \end{aligned}$$

So, let $X_r = \frac{\partial}{\partial r}$ and $Y, Z \in \Gamma(T\mathbb{S}^{n-1})$, because

$$\left\langle X_r, X_r\right\rangle = 1, \quad \left\langle Y, Z\right\rangle = G(r)^2 \left\langle Y, Z\right\rangle_{\mathbb{S}^{n-1}}, \quad [X_r, Y] = [X_r, Z] = 0,$$

we have the following calculations.

(1) First, because $\left\langle X_r, X_r\right\rangle = 1$,

$$0 = X_r\left\langle X_r, X_r\right\rangle = 2\left\langle \nabla_{X_r} X_r, X_r\right\rangle = 0.$$

So $\nabla_{X_r} X_r \in \Gamma(T\mathbb{S}^{n-1})$. On the other hand, because $Y \perp X_r$, by Koszul formula,

$$2\left\langle \nabla_{X_r} X_r, Y\right\rangle = -Y\left\langle X_r, X_r\right\rangle = 0.$$

Therefore,

$$\nabla_{X_r} X_r = 0.$$

(2) By Koszul formula, $\left\langle \nabla_{X_r} Y, X_r\right\rangle = 0$, which implies that $\nabla_{X_r} Y \in \Gamma(T\mathbb{S}^{n-1})$. Similarly,

$$2\left\langle \nabla_{X_r} Y, Z\right\rangle = X_r\left(\left\langle Y, Z\right\rangle\right) = 2G(r)G'(r)\left\langle Y, Z\right\rangle_{\mathbb{S}^{n-1}} = 2\frac{G'(r)}{G(r)}\left(\left\langle Y, Z\right\rangle\right).$$

Therefore,

$$\nabla_{X_r} Y = \frac{G'(r)}{G(r)} Y.$$

By torsion-free,

$$\nabla_Y X_r = [X_r, Y] + \nabla_{X_r} Y = \nabla_{X_r} Y = \frac{G'(r)}{G(r)} Y.$$

(3) Also, by Koszul formula,

$$\left\langle \nabla_Y Z, X_r\right\rangle = -\frac{G'(r)}{G(r)}\left\langle Y, Z\right\rangle.$$

Therefore,

$$\nabla_Y Z = \nabla_Y^{\mathbb{S}^{n-1}} Z - \frac{G'(r)}{G(r)}\left\langle Y, Z\right\rangle X_r,$$

where $\nabla^{\mathbb{S}^{n-1}}$ is the induced Levi-Civita connection.

The radical curvature κ_{rad} is defined as the sectional curvature restricted to radical section, that is a $\Pi_x \subset T_x M$, where $\Pi_x = \operatorname{span}\{X_r, Y\}$ for some unit $Y \in \Gamma(T\mathbb{S}^{n-1})$. To calculate the radical curvature, by above, we have

$$\nabla_Y Y = \nabla_Y^{\mathbb{S}^{n-1}} Y - \frac{G'(r)}{G(r)} X_r,$$

which implies that

$$\nabla_{X_r}\nabla_Y Y = \frac{G'}{G}\nabla_Y^{\mathbb{S}^{n-1}} Y - \frac{G''G-(G')^2}{G^2}X_r,$$

and

$$\begin{aligned}\nabla_Y\nabla_{X_r} Y &= \nabla_Y\left(\frac{G'}{G}Y\right) = \frac{G'}{G}\nabla_Y Y\\ &= \frac{G'}{G}\nabla_Y^{\mathbb{S}^{n-1}} Y - \left(\frac{G'}{G}\right)^2 X_r.\end{aligned}$$

Therefore,

$$\begin{aligned}R(X_r,Y)Y &= \nabla_{X_r}\nabla_Y Y - \nabla_Y\nabla_{X_r}Y - \nabla_{[X_r,Y]}Y\\ &= \nabla_{X_r}\nabla_Y Y - \nabla_Y\nabla_{X_r}Y\\ &= -\frac{G''}{G}X_r.\end{aligned}$$

It follows that

$$K(\Pi_x) = \frac{\langle R(X_r,Y)Y, X_r\rangle}{|X_r|^2|Y|^2 - \langle X_r, Y\rangle^2} = -\frac{G''(r)}{G(r)},$$

where $r = d(0,x)$. Note that the radial curvature is independent of the choice of Y, i.e.

$$\kappa_{\text{rad}}(r) = -\frac{G''(r)}{G(r)}.$$

i.e., G is the unique solution of ODE

$$G''(r) + \kappa_{\text{rad}}(r)G(r) = 0, \quad G(0) = 0,\ G'(0) = 1.$$

Moreover,

$$\mathrm{Ric}(X_r, X_r) = -(n-1)\frac{G''(r)}{G(r)}$$

Then let $\gamma(t) = (t,\theta_0)$ be a normal radial geodesic with $\dot\gamma(t) = X_r(t)$. Let $E_1 = X_r, E_2, \cdots, E_n$ be a orthonormal frame along γ. If

$$J(t) = f(t)E(t)$$

is an orthogonal Jacobian field, then f satisfies

$$f''(t) + \kappa_{\text{rad}}(t)f(t) = 0.$$

Therefore, for any $X \in T_{\gamma(t_0)}\mathbb{S}^{n-1}$, let $J(t)$ be the orthogonal Jacobian field with $J(0) = 0$ and $J(t_0) = X$. Note that for such $J(t) = f(t)E(t)$,

$$f''(t) + \kappa_{\text{rad}}(t)f(t) = 0, \quad f(0) = 0,$$

which implies that $f(t) = cG(t)$ for some constant c. Then

$$\nabla^2 r(X,X) = \langle \nabla_{X_r}J(t_0), J(t_0)\rangle = \frac{f'(t_0)}{f(t_0)}|X|^2 = \frac{G'(t_0)}{G(t_0)}|X|^2.$$

Therefore,

$$\Delta r(t) = \sum_{i=2}^{\infty}\nabla^2 r(E_i,E_i) = (n-1)\frac{G'(t_0)}{G(t_0)}.$$

Generalize Comparison Theorem. Let

$$\begin{aligned} K_M(x) &= \{K(\Pi)\colon \Pi \subset T_xM \text{ 2-dimensional section}\}, \\ \mathrm{Ric}_M(x) &= \{\mathrm{Ric}(X,X)\colon X \in T_xM,\ |X| = 1\}. \end{aligned}$$

Assume there are two model manifolds with radial curvature κ_1 and κ_2 such that

$$\begin{aligned} &\kappa_1(r) \geq \sup\{K_M(x)\colon r(x) = r\} \\ &\kappa_2(r) \leq \inf\{\mathrm{Ric}_M(x)\colon r(x) = r\}\,(d-1)^{-1}. \end{aligned}$$

Then by replacing the space form with the model manifold $(\mathbb{M}, g)$ in the proof of Hessian comparison theorem and Laplacian comparison theorem, we have the following comparison,

$$(d-1)\frac{G_1'(r)}{G_1(r)} \leq \Delta_M r \leq (d-1)\frac{G_2'(r)}{G_2(r)},$$

where

$$G''(r) + \kappa(r)G(r) = 0, \quad G(0) = 0, \quad G'(0) = 1.$$

Chapter 4

Riemannian Langevin Dynamics

Let (M, g) be a complete Riemannian manifold with Levi-Civita connection ∇ and the canonical Riemannian volume measure dm. In the following, we denote $\operatorname{grad} f = \nabla f$ for convenience. Let B^M is a Brownian motion on M. Let $U \in C^\infty(M)$. Consider the following SDE

$$dX_t = -\nabla U(X_t)dt + \sqrt{2}dB_t^M, \tag{4.1}$$

which is called a Riemannian Langevin dynamics.

4.1 Diffusion Operator

Because B^M is a Brownian motion on M, its anti-development W is a standard Euclidean Brownian motion. So

$$dB_t^M = V_\alpha \circ dW_t^\alpha,$$

with

$$\Delta_M = \sum_\alpha V_\alpha^2.$$

Therefore,

$$dX_t = -\nabla U(X_t)dt + \sqrt{2}V_\alpha \circ dW_t^\alpha,$$

which implies that the diffusion operator of X is

$$\begin{aligned} Lf &= \frac{1}{2}\sum_\alpha (\sqrt{2}V_\alpha)^2 f - \nabla U(f) \\ &= \Delta_M f - \langle \nabla U, \nabla f \rangle, \end{aligned}$$

i.e., $L = \Delta_M - \langle \nabla U, \nabla \cdot \rangle$. For such L, in note *Stochastic Analysis*, we have shown that

$$L^* p = \Delta_M p + \operatorname{div}(p\nabla U)$$

in the sense of on $L^2(dm)$. Also,

$$d\mu = e^{-U}dm,$$

is the invariant measure of such diffusion process, which is also symmetric for L. Note that L is essentially self-adjoint on $C_c^\infty(M)$ w.s.t. μ. Therefore, it has a unique self-adjoint extension L on $L^2(\mu, M)$. Let

$$P_t = e^{-tL}$$

be the functional calculus of L, defined on $L^2(\mu, M)$. On the other hand, define

$$Q_t f(x) = \mathbb{E}\left[f(X_t) \mid X_0 = x\right], \quad \forall\, f \in C_c^\infty(M),$$

By note *Markov Operator*, Q_t can be extended to $L^2(\mu)$ and has the generator A that is self-adjoint on $L^2(\mu)$. By Dynkin's formula,

$$Af = Lf, \quad \forall\ f \in C_c^\infty(M).$$

Therefore, $L \subset A$. By the essential self-adjointness of L, $L = A$, which implies that

$$P_t f(x) = Q_t f(x) = \mathbb{E}\left[f(X_t) \mid X_0 = x\right], \quad \forall\ f \in L^2(\mu).$$

Remark 4.1.1. Note that if ∇U is globally Lipschitz continuous (in particular, M is compact), i.e., $|\nabla^2 U| \le L$, then it can show that $(Q_t)_{t\ge 0}$ is a Feller-Dynkin semigroup so that it can be defined on $C_0(M)$.

Remark 4.1.2. For P_t, note that

$$\frac{\partial}{\partial t} P_t = LP_t = P_t L,$$

called the backward Kolmogorov equation. Moreover, it can define the Markov transition kernel

$$p_t(x, A) = P_t \mathbb{I}_A(x) = \mathbb{E}\left[\mathbb{I}_A(X_t) \mid X_0 = x\right] = \mathbb{P}_x(X_t \in A).$$

Then

$$P_t f(x) = \int_M f(y) p_t(x, dy).$$

It follows that we can define

$$\int_M f d(P_t^* \nu) = \int_M P_t f d\nu.$$

Then we have

$$\frac{\partial}{\partial t} P_t^* = L^* P_t^* = P_t^* L^*,$$

called the forward Kolmogorov equation/Fokker-Planck equation, where L^* acts on ν by

$$\int_M f d(L^*\nu) = \int_M Lf d\nu, \quad \forall\ f \in \mathcal{D}(L).$$

Note that $L^*\nu$ may not be a measure. Note that μ is an invariant measure if and only if $L^*\mu = 0$, i.e.,

$$\int_M Lf d\mu = 0, \quad \forall\ f \in C_c^\infty(M).$$

4.2 Deterministic Flow

Let $V(t,x)$ be a time-dependent C^∞-vector field. Consider ODE on M

$$\begin{cases} \dfrac{dx_t}{dt} = V(t, x_t) \\ x_0 = x \end{cases} \tag{4.2}$$

and denote $\varphi_t(x)$ be its flow.

Theorem 4.2.1. *Let $\{\varphi_t\}_{t\in[0,T)}$ be a locally Lipschitz family of diffeomorphisms of M with $\varphi_0 = id$. Let $V(t,x)$ be the time-dependent vector fields associated with $\{\varphi_t\}_{t\in[0,T)}$. Let ν be a probability measure on M and $\nu_t = (\varphi_t)_\# \nu$. Then ν_t is the unique solution of the following PDE*

$$\begin{cases} \dfrac{\partial \nu_t}{\partial t} + \operatorname{div}(\nu_t V_t) = 0, \quad t \in [0,T) \\ \nu_0 = \nu \end{cases} \tag{4.3}$$

in $C([0,T), \mathcal{P}(M))$, where $\mathcal{P}(M)$ is equipped with the weak topology.

Remark 4.2.2. For the measure PDE, it means that for any $f \in C_c^\infty(M)$,

$$\langle \partial_t \nu_t, f \rangle + \langle \operatorname{div}(\nu_t V_t), f \rangle = 0,$$

where

$$\langle \partial_t \nu_t, f \rangle = \frac{\partial}{\partial t} \langle \nu_t, f \rangle = \frac{\partial}{\partial t} \int_M f d\nu_t,$$

$$\langle \operatorname{div}(\nu_t V_t), f \rangle = -\int_M f d(\operatorname{div}(\nu_t V_t)) = -\int_M \operatorname{div}(f V_t) d\nu_t = \int_M \langle \nabla f, V_t \rangle \, d\nu_t.$$

Moreover, if $d\nu_0 = pdm$ for $p \in C^\infty$, which implies there are $p_t \in C^\infty$ such that $d\nu_t = p_t dm$, then the PDE is equivalent to

$$\begin{cases} \dfrac{\partial p_t}{\partial t} + \operatorname{div}(p_t V_t) = 0, \quad t \in [0, T) \\ p_0 = p. \end{cases}$$

The theorem implies that if there is a family of probability measures $\{\nu_t\}_{t \in [0,T)}$ satisfies above PDE (4.3), then for a random process $\widetilde{X}_t$ on M satisfying the ODE (4.2) with $\widetilde{X}_0 \sim \nu$, we have $\widetilde{X}_t \sim \nu_t$.

Consider the Riemannian Langevin dynamics (4.1) with $X_0 \sim \nu$, where $d\nu = pdm$ for a smooth p. Then $X_t \sim \nu_t$, for $d\nu_t = d(P_t^* \nu) = p_t dm$. By the Fokker-Planck equation, we have

$$\frac{\partial \nu_t}{\partial t} = L^* \nu_t,$$

which is equivalent to

$$\frac{\partial p_t}{\partial t} = \Delta_M p_t + \operatorname{div}(p_t \nabla U) = \operatorname{div}(p_t(\nabla U + \nabla \log p_t)).$$

Therefore, if $\widetilde{X}_t$ satisfies

$$\begin{cases} \dfrac{d\widetilde{X}_t}{dt} = -\nabla U(\widetilde{X}_t) - \nabla \log p_t(\widetilde{X}_t), \\ \widetilde{X}_0 \sim \nu, \end{cases}$$

then $\widetilde{X}_t \sim \nu_t$, i.e., $\widetilde{X}_t \overset{\mathrm{d}}{=} X_t$.

4.3 Convergence to Invariant Measure

KL Divergence. If we have

$$\operatorname{Ric}_U = \operatorname{Ric} + \nabla^2 U \geq \rho g,$$

for some $\rho > 0$, then above process satisfies the $\mathrm{CD}(\rho, \infty)$ and so $\mathrm{LSI}(1/\rho)$, which implies that

$$\operatorname{End}_\mu(P_t g) \leq e^{-2\rho t} \operatorname{Ent}_\mu(g), \quad \forall \, g \in L^1(\mu).$$

(see note *Stochastic Analysis*). If $X_0 \sim \nu_0$ with $d\nu_0 = f d\mu$ (assume $f \in L^1(\mu)$) and $\mathrm{KL}(\nu_0 \parallel \mu) < \infty$, then

$$X_t \sim \nu_t = P_t^* \nu_0,$$

where $d(P_t^* \nu_0) = P_t f d\mu$. Then we can get

$$\mathrm{KL}(\nu_t \parallel \mu) \leq e^{-2\rho t} \mathrm{KL}(\nu_0 \parallel \mu).$$

Therefore,

$$\mathrm{KL}(\nu_t \parallel \mu) \leq \mathcal{O}(e^{-t}).$$

On the other hand, by Pinsker inequality $d_{\mathrm{TV}}(\nu_t, \mu) \leq \frac{1}{2} \mathrm{KL}(\nu_t \parallel \mu)$,

$$d_{\mathrm{TV}}(\nu_t, \mu) \leq \mathcal{O}(e^{-t}).$$

Wasserstein Distance. Under the same condition,

$$\mathrm{Ric}_U = \mathrm{Ric} + \nabla^2 U \geq \rho g,$$

we can have

$$W_2(\nu_t, \mu) \leq e^{-\rho t} W_2(\nu_0, \mu).$$

Remark 4.3.1. It needs techniques from the optimal transport theory, like W_2-geodesic convexity, et. al., or it can be obtained by directly calculating from the SDE.

4.4 Time-reversal Process

This section is based on results in De Bortoli et al. 2022. Let (M, g) be a compact Riemannian manifold (without boundary) with Levi-Civita connection ∇ and canonical volume measure dm and the corresponding probability measure $dp_{\mathrm{ref}} = dm/m(M)$. In be following, we consider a more general equation.

$$dX_t = b(X_t)dt + dB_t^M, \quad t \in [0, T]$$

where $b \in \Gamma(TM)$, B^M is a Brownian motion on M. The main goal of this section is to consider its time-reversal process $Y_t = X_{T-t}$. It can show that

$$dY_t = (-b(Y_t) + \nabla \log p_{T-t}(Y_t))dt + dB_t^M,$$

where $X_t \sim \mu_t$ with $d\mu_t = p_t dp_{\mathrm{ref}}$. It is equivalent to prove that for any $f \in C^\infty(M)$,

$$M_t^{Y,f} := f(Y_t) - f(Y_0) - \int_0^t \langle -b(Y_s) + \nabla \log p_{T-s}(Y_s), \nabla f(Y_s) \rangle + \frac{1}{2}\Delta_M f(Y_s) ds$$

is a martingale, by the weak uniqueness of Martingale problem. Furthermore, it suffices to show that for any $g \in C^\infty(M)$ and any $t \geq s \in [0, T]$,

$$\mathbb{E}\left[g(Y_s)(f(Y_t) - f(Y_s))\right] = \mathbb{E}\left[g(Y_s)\int_s^t \tilde{\mathcal{A}}_s f(Y_s) ds\right],$$

where

$$\tilde{\mathcal{A}}_t f = \langle -b + \nabla \log p_{T-t}, \nabla f \rangle + \frac{1}{2}\Delta_M f.$$

Note that for $t \geq s \in [0, T]$

$$\begin{aligned}
\mathbb{E}\left[g(Y_s)(f(Y_t) - f(Y_s))\right] &= \mathbb{E}\left[g(X_{T-s})(f(X_{T-t}) - f(X_{T-s}))\right] \\
&= \mathbb{E}\left[\mathbb{E}[g(X_{T-s}) \mid X_{T-t}] f(X_{T-t})\right] - \mathbb{E}[g(X_{T-s}) f(X_{T-s})] \\
&= \mathbb{E}\left[v(T-t, X_{T-t}) f(X_{T-t})\right] - \mathbb{E}\left[v(T-s, X_{T-s}) f(X_{T-s})\right],
\end{aligned}$$

where $v \colon [0, T-s] \times M \to \mathbb{R}$ is given by

$$v(u, x) = \mathbb{E}[g(X_{T-s}) \mid X_u = x] = P_{T-s-u} g(x),$$

where P_t is the semigroup associated with X_t. Assume v is regular enough. So it satisfies the backward Kolmogorov equation

$$\partial_u v(u, x) = -\mathcal{A} v(u, x).$$

Let $h \colon [0, T-s] \times M \to \mathbb{R}$ be defined as

$$h(u, x) = v(u, x) f(x).$$

Note that by Itô's formula, because $\mathcal{A}$ is the generator for X,

$$M_t^{X,h} = h(t, X_t) - h(0, X_0) - \int_0^t (\partial_u + \mathcal{A})h(u, X_u)du$$

is martingale. It implies that

$$\mathbb{E}\left[v(T-t, X_{T-t})f(X_{T-t})\right] - \mathbb{E}\left[v(T-s, X_{T-s})f(X_{T-s})\right] = \int_{T-s}^{T-t} \mathbb{E}\left[(\partial_u + \mathcal{A})h(u, X_u)\right] du.$$

To calculate it, first, we have

$$\begin{aligned}\partial_u h(u,x) + \mathcal{A}h(u,x) &= f(x)\partial_u v(u,x) + f(x)\mathcal{A}v(u,x) + v(u,x)\mathcal{A}f(x) + \langle \nabla f(x), \nabla v(u,x)\rangle \\ &= v(u,x)\mathcal{A}f(x) + \langle \nabla f(x), \nabla v(u,x)\rangle,\end{aligned}$$

because $\Delta_M(fg) = f\Delta_M g + g\Delta_M f + 2\langle \nabla f, \nabla g\rangle$ by $\operatorname{div}(fX) = \langle \nabla f, X\rangle + f \operatorname{div} X$. By the divergence theorem,

$$\mathbb{E}\left[\langle \nabla f(X_u), \nabla v(u, X_u)\rangle\right] = -\mathbb{E}\left[v\left(u, X_u\right)\Delta_M f\left(X_u\right)\right] - \mathbb{E}\left[v\left(u, X_u\right)\langle \nabla f\left(X_u\right), \nabla \log p_u\left(X_u\right)\rangle\right].$$

Therefore, it implies that

$$\begin{aligned}\mathbb{E}\left[(\partial_u + \mathcal{A})h\left(u, X_u\right)\right] &= \mathbb{E}\left[v\left(u, X_u\right)\left(\langle b\left(X_u\right) - \nabla \log p_u\left(X_u\right), \nabla f\left(X_u\right)\rangle - \frac{1}{2}\Delta_M f\left(X_u\right)\right)\right] \\ &= -\mathbb{E}\left[v\left(u, X_u\right)\tilde{\mathcal{A}}_{T-u} f\left(X_u\right)\right].\end{aligned}$$

Therefore, we have

$$\begin{aligned}\mathbb{E}\left[g(Y_s)(f(Y_t) - f(Y_s))\right] &= \int_{T-t}^{T-s} \mathbb{E}\left[v\left(u, X_u\right)\tilde{\mathcal{A}}_{T-u} f\left(X_u\right)\right] du \\ &= \int_{T-t}^{T-s} \mathbb{E}\left[\mathbb{E}\left[g(X_{T-s}) \mid X_u\right]\tilde{\mathcal{A}}_{T-u} f\left(X_u\right)\right] du \\ &= \int_{T-t}^{T-s} \mathbb{E}\left[\mathbb{E}\left[g(X_{T-s})\tilde{\mathcal{A}}_{T-u} f\left(X_u\right) \mid X_u\right]\right] du \\ &= \int_{T-t}^{T-s} \mathbb{E}\left[g(X_{T-s})\tilde{\mathcal{A}}_{T-u} f\left(X_u\right)\right] du \\ &= \mathbb{E}\left[g(Y_s)\int_s^t \tilde{\mathcal{A}}_u f\left(Y_u\right) du\right].\end{aligned}$$

Integration by Parts. Let $(\mathsf{X}, \mathcal{X})$ be a polished space. Let $\mathbb{P}$ be a probability measure on $\Omega = C([0,T], \mathsf{X})$ equipping with canonical measurable sets. Let X be the canonical process valued on $\mathcal{X}$ defined on Ω, i.e., $X\colon C([0,T], \mathsf{X}) \to \mathsf{X}$,

$$X_t(\omega) := \omega_t,$$

and $\mathbb{F}^X = (\mathcal{F}_t)_{t\in[0,T]}$ with $\mathcal{F}_t = \sigma(X_s\colon s \in [0,t])$. For any $t \in [0,T]$, $\mathbb{P}_t := (X_t)_\#\mathbb{P}$ and more general, $\mathbb{P}_\mathcal{T} = (X_\mathcal{T})_\#\mathbb{P}$.
Let $M = (M_t)_{t\in[0,T]}$ be a X-valued and $\mathbb{F}^X$-adapted process on Ω, which is called a $\mathbb{P}$-local martingale if it is a local martingale w.s.t. $\mathbb{F}^X$.
For such $\mathbb{P}$, it can define the extended generator $\bar{\mathcal{A}}_\mathbb{P}$ (No Markov in advance). For a function,

$$u\colon [0,T] \times \mathsf{X} \to \mathbb{R}$$

if there exists a X-valued process $A = (A_t)_{t\in[0,T]}$ on Ω such that

(a) A is $\mathbb{F}^X$-adapted,

(b) $\int_0^T |A_s|\, ds < \infty$ $\mathbb{P}$-a.e.,

(c) $M = (M_t)_{t\in[0,T]}$ defined as

$$M_t = u(t, X_t) - u(0, X_0) - \int_0^t A_s ds,$$

is a $\mathbb{P}$-local martingale.

Note that if A exists, then it is unique, because

$$\int_0^t A_s - B_s ds \text{ a local martingale} \quad \Rightarrow \quad A_t = B_t$$

$dt\otimes\mathbb{P}$-a.e.. Moreover, because A is $\left(\sigma\left(X_{[0,t]}\right); 0 \le t \le T\right)$-adapted, i.e., A_t is $\sigma(Y)$-measurable, for $Y = X_{[0,t]}$, $A_t = A_t(Y) = A_t(X_{[0,t]})$. Then we say $u \in \mathcal{D}(\bar{\mathcal{A}}_{\mathbb{P}})$ and

$$A_t =: \bar{\mathcal{A}}_{\mathbb{P}} u(t, X_{[0,t]}).$$

Let (u, v) be $u, v\colon [0, T] \times \mathsf{X} \to \mathbb{R}$ such that $u, v, uv \in \mathcal{D}(\bar{\mathcal{A}}_{\mathbb{P}})$. For (u, v), define the carré du champ $\bar{\Upsilon}_{\mathbb{P}}$

$$\bar{\Upsilon}_{\mathbb{P}} = \bar{\mathcal{A}}_{\mathbb{P}}(uv) - u\bar{\mathcal{A}}_{\mathbb{P}}(v) - v\bar{\mathcal{A}}_{\mathbb{P}}(u).$$

Assume there exists an algebra $\mathcal{U}_{\mathbb{P}} \subset \mathcal{D}(\bar{\mathcal{A}}_{\mathbb{P}}) \cap C_b(X)$. Let

$$\mathcal{U}_{\mathbb{P},2} = \left\{u \in \mathcal{U}_{\mathbb{P}} : \bar{\mathcal{A}}_{\mathbb{P}} u \in \mathrm{L}^2(\mathbb{P}), \bar{\Upsilon}_{\mathbb{P}}(u, u) \in \mathrm{L}^1(\mathbb{P})\right\}.$$

If $\mathbb{P}$ is Markov, it has the usual operator $\mathcal{A}_{\mathbb{P}} u\colon [0, T] \times \mathsf{X} \to \mathbb{R}$. We can see

$$\bar{\mathcal{A}}_{\mathbb{P}} u(t, X_{[0,t]}) = \mathcal{A}_{\mathbb{P}} u(t, X_t)$$

For $\mathbb{P}$, let $R\mathbb{P}$ be the distribution of the inverse process of X, i.e., $t \mapsto X_{T-t}$. If $\mathbb{P}$ is Markov, then $R\mathbb{P}$ is also a Markov. In the following, we only consider the Markov case. For $u\colon \mathsf{X} \to \mathbb{R}$ and $u \in \mathcal{D}(\mathcal{A}_{\mathbb{P}})$, we denote

$$\mathcal{A}_{\mathbb{P}} u(t, x) =: \mathcal{A}_{\mathbb{P},t} u(x).$$

For Markov case, we use $\mathcal{U}_{\mathbb{P}}$, $\mathcal{U}_{\mathbb{P},2}$ denotes time-independent u such that the conditions are satisfied for all $t \in [0, T]$.

Theorem 4.4.1 (Theorem 3.17, Cattiaux et al. 2023)**.** *Let $u, v \in \mathcal{U}_{\mathbb{P},2}$.*

(a) If the following holds,

(i) $(t, x) \mapsto \Upsilon_{\mathbb{P},t}(u, v)(x)$ is continuous.

(ii) $\mathcal{U}_{\mathbb{P},2}$ determines the weak convergence of Borel measures on $\mathcal{X}$.

(iii) the linear map

$$v \in \overline{\mathcal{U}}_{2,\mathbb{P}} \to \mathbb{E}\left[\int_0^T \Upsilon_{\mathbb{P},t}(u, v_t)(X_t) dt\right]$$

determines a finite measure on $[0, T] \times \mathsf{X}$, where

$$\overline{\mathcal{U}}_{2,\mathbb{P}} = \{v \in C([0, T] \times \mathsf{X}, \mathbb{R})\colon v(t, \cdot) \in \mathcal{U}_{\mathbb{P},2},\ \forall\, t \in [0, T]\}.$$

then $u \in \mathcal{D}(\mathcal{A}_{R(\mathbb{P})})$ and $\mathcal{A}_{R(\mathbb{P})} u \in \mathrm{L}^1(\mathbb{P})$.

(b) If $u \in \mathcal{D}(\mathcal{A}_{R(\mathbb{P})})$ and $\mathcal{A}_{R(\mathbb{P})} u \in \mathrm{L}^1(\mathbb{P})$, then for a.e. $t \in [0, T]$,

$$\mathbb{E}\left[\left(\mathcal{A}_{\mathbb{P},t} u(X_t) + \mathcal{A}_{R(\mathbb{P}),T-t} u(X_t)\right) v(X_t) + \Upsilon_{\mathbb{P},t}(u, u)(X_t)\right] = 0.$$

Girsanov Theory on Compact Riemannian Manifolds. Let (M, g) be a compact Riemannian manifold and isometrically and properly embedding in $\mathbb{R}^p$.

Theorem 4.4.2. *Let $\mathbb{Q}$ be the path measure of a Brownian motion on M. Let $\mathbb{P}$ be a Markov path measure on $C([0,T], M)$ such that $\mathrm{KL}(\mathbb{P} \parallel \mathbb{Q}) < \infty$. Then there exists a time-dependent vector field β such that $\mathbb{P}$ satisfies the martingale problem for generator $\mathcal{A}$, where for $t \in [0,T]$, $u \in C^2(M)$, and $x \in M$,*

$$\mathcal{A}(t,u)(x) = \langle \beta(t,x), \nabla u(x) \rangle + \frac{1}{2}\Delta_M u(x),$$

In addition, we have

$$\mathrm{KL}(\mathbb{P} \parallel \mathbb{Q}) = \mathrm{KL}(\mathbb{P}_0 \parallel \mathbb{Q}_0) + \frac{1}{2}\int_0^T \mathbb{E}\left[\|\beta(t, X_t)\|^2\right] dt.$$

Remark 4.4.3. For Euclidean case, if $\mathbb{P} \ll \mathbb{Q}$, then for $X \sim \mathbb{P}$, we can find an Euclidean Brownian motion W such that

$$dX_t = v(t, X_t)dt + dW_t,$$

which implies that the generator

$$\mathcal{A}_t = \langle v, \nabla \cdot \rangle + \frac{1}{2}\Delta \cdot .$$

Moreover, by Girsanov's theorem,

$$\mathrm{KL}(\mathbb{P} \parallel \mathbb{Q}) = \mathrm{KL}(\mathbb{P}_0 \parallel \mathbb{Q}_0) + \frac{1}{2}\int_0^T \mathbb{E}\left[\|v(t, X_t)\|^2\right] dt.$$

Proof. By embedding, if let B_t^M be a Brownian motion on M, we have

$$dB_t^M = P_i(B_t^M) \circ dB_t^i = P(B_t^M) \circ dB_t,$$

for the orthogonal projection P and an Euclidean Brownian motion B. Note that

$$dB_t^M = \bar{b}(B_t^M)dt + P(B_t^M)dB_t,$$

where $\bar{b}$ satisfies $P(x)\bar{b}(x) = 0$ for all $x \in M$. It implies that

$$\frac{1}{2}\Delta_M u(x) = \left\langle \overline{\nabla}\bar{u}(x), \overline{\nabla}\bar{v}(x) \right\rangle + \frac{1}{2}\left\langle P(x), \overline{\nabla}^2 \bar{u}(x) \right\rangle,$$

for any $u, v \in C_c^2(M)$, where $\bar{u}, \bar{v} \in C_c^2(\mathbb{R}^p)$ be their extensions. Because $\mathbb{Q}$ is the distribution of B^M, which can be viewed on $\mathbb{R}^p$, by uniqueness condition and finite entropy, there exists $\bar{\beta} \colon [0,T] \times \mathbb{R}^p \to \mathbb{R}^p$ such that

$$\mathrm{KL}(\mathbb{P} \parallel \mathbb{Q}) = \mathrm{KL}\left(\mathbb{P}_0 \parallel \mathbb{Q}_0\right) + \frac{1}{2}\int_0^T \mathbb{E}\left[\left\|P\left(X_t\right)\bar{\beta}\left(t, X_t\right)\right\|^2\right] dt,$$

and $\mathbb{P}$ satisfies the martingale problem for $\bar{A} \colon [0,T] \times C_c^2(\mathbb{R}^p) \to \mathbb{R}$ with

$$\bar{\mathcal{A}}(t, \bar{u})(x) = \langle \bar{b}(x) + P(x)\bar{\beta}(t,x), \overline{\nabla}\bar{u}(x) \rangle + \frac{1}{2}\left\langle P(x), \overline{\nabla}^2 \bar{u}(x) \right\rangle.$$

Let $\beta = P\bar{\beta}$. Note that for any $u \in C_c^2(M)$, if $\bar{u}$ is a extension of u constant-along-normals, then for any $X \in \Gamma(TM)$,

$$\nabla^2 u(X,Y) = \overline{\nabla}^2 \bar{u}(X,Y).$$

By $\Delta_M u = \sum_\alpha \nabla^2 u(P_\alpha, P_\alpha)$,

$$\begin{aligned}\sum_\alpha \nabla^2 u(P_\alpha, P_\alpha) &= \sum_\alpha \overline{\nabla}^2 \bar{u}(P_\alpha, P_\alpha)\\ &= \sum_\alpha P_\alpha^i P_\alpha^j \frac{\partial^2 \bar{u}}{\partial x^i \partial x^j}\\ &= \sum_{i,j} P_i^j \frac{\partial^2 \bar{u}}{\partial x^i \partial x^j}\\ &= \left\langle P, \overline{\nabla}^2 \bar{u}\right\rangle\end{aligned}$$

because $P = P^2$. Moreover, $P(x)\bar{b}(x) = 0$ implies that $\bar{b} \in \Gamma(NM)$, which follows that $\left\langle \bar{b}(x), \overline{\nabla}\bar{u}\right\rangle = 0$. So

$$\begin{aligned}\bar{\mathcal{A}}(t,\bar{u})(x) &= \langle \bar{b}(x) + P(x)\bar{\beta}(t,x), \overline{\nabla}\bar{u}(x)\rangle + \frac{1}{2}\left\langle P(x), \overline{\nabla}^2 \bar{u}(x)\right\rangle\\ &= \langle \beta(t,x), \nabla u(x)\rangle + \frac{1}{2}\Delta_M u.\end{aligned}$$

□

In particular, when viewing $\mathbb{P}$ as a process on M, it satisfies the martingale problem with generator $\mathcal{A}\colon [0,T] \times C_c^2(M) \times M \to \mathbb{R}$ given by

$$\mathcal{A}(t,u)(x) = \langle \beta(t,x), \nabla u(x)\rangle + \frac{1}{2}\Delta_M u(x).$$

Corollary 4.4.4. *Let $\mathbb{P}^1, \mathbb{P}^2$ be two Markov path measures on $C([0,T],M)$ with $\mathbb{P}_0^1 = \mathbb{P}_0^2$, associating with SDE*

$$dX_t^i = b_i(t, X_t^i)dt + dB_t^M.$$

Then we have

$$\mathrm{KL}(\mathbb{P}^1 \parallel \mathbb{P}^2) = \frac{1}{2}\int_0^T \mathbb{E}\left[\left\|b_1(t,X_t^1) - b_2(t,X_t^2)\right\|^2\right] dt.$$

Proof. Let M be a compact manifold isometrically embedded in $\mathbb{R}^p$ such that $M \subset B(0,R)$. Let $\varphi \in C^\infty(\mathbb{R}^p)$ such that $\varphi|_{\overline{B(0,R)}} = 1$ and $\varphi(x) = 0$ for $|x| \geq R+1$. Let $\bar{b}_i \in C_c^2([0,T]\times\mathbb{R}^p, \mathbb{R}^b)$ be the extension of $b_i(t,x)$. Consider

$$\begin{aligned}d\bar{X}_t^i &= \varphi\left(\bar{X}_t^i\right)\left(\bar{b}^i\left(t,\bar{X}_t^i\right)dt + dB_t^M\right)\\ &= \varphi\left(\bar{X}_t^i\right)\left\{\mathrm{P}\left(\bar{X}_t^i\right)\bar{b}^i\left(t,\bar{X}_t^i\right) + \bar{b}\left(\bar{X}_t\right)\right\}dt + \varphi\left(\bar{X}_t^i\right)\mathrm{P}\left(\bar{X}_t^i\right)\mathrm{dB}_t.\end{aligned}$$

So for any $x \in M$, $f \in C_c^2(M)$ with extension $\bar{f} \in C_c^2(\mathbb{R}^p)$,

$$\begin{aligned}\bar{\mathcal{A}}_t^i(\bar{f})(x) &= \left\langle \bar{b}^i(t,x) + \bar{b}(x), \nabla\bar{f}(x)\right\rangle + \frac{1}{2}\left\langle \mathrm{P}(x), \nabla^2\bar{f}(x)\right\rangle\\ &= \left\langle b^i(t,x), \nabla f(x)\right\rangle + \frac{1}{2}\Delta_M f(x).\end{aligned}$$

It follows that $\bar{X}^i$ viewing as a process on M has the same generator as X^i. So $\bar{X}^i$ has the distribution $\mathbb{P}^i$. Moreover, denote $\mathbb{P}^i$ as $\bar{\mathbb{P}}^i$ when it is viewed as $\mathbb{R}^p$-process. Note that because the

$$\varphi(x) = 0, \ \forall \ \|x\| \geq R+1$$

the global Lipschitz condition of SDE $\bar{X}^i$ is satisfied. Moreover, by the compactness,

$$\begin{aligned}
\left(d\bar{\mathbb{P}}^1/d\bar{\mathbb{P}}^2\right)\left(\left(\bar{X}_t^1\right)_{t\in[0,T]}\right) &= \exp\left[\int_0^T \left\langle \bar{b}^1\left(t,\bar{X}_t^1\right)-\bar{b}^2\left(t,\bar{X}_t^1\right), \mathrm{P}\left(\bar{X}_t^1\right) d\bar{X}_t^1\right\rangle\right. \\
&\left.-\frac{1}{2}\int_0^T \left\langle \bar{b}^1\left(t,\bar{X}_t^1\right)-\bar{b}^2\left(t,\bar{X}_t^1\right), \mathrm{P}\left(\bar{X}_t^1\right)\left(\bar{b}^1\left(t,\bar{X}_t^1\right)+\bar{b}^2\left(t,\bar{X}_t^1\right)\right)\right\rangle dt\right] \\
&= \exp\left[\frac{1}{2}\int_0^T \left\|\bar{b}^1\left(t,\overline{X}_t^1\right)-\bar{b}^2\left(t,\overline{X}_t^1\right)\right\|^2 \mathrm{d}t + \int_0^T \left\langle \bar{b}^1\left(t,\overline{X}_t^1\right)-\bar{b}^2\left(t,\overline{X}_t^1\right), \mathrm{P}\left(\overline{X}_t^1\right)\mathrm{d}B_t\right\rangle\right].
\end{aligned}$$

It follows that

$$\mathrm{KL}\left(\overline{\mathbb{P}}^1 \parallel \overline{\mathbb{P}}^2\right) = \frac{1}{2}\int_0^T \mathbb{E}\left[\left\|b^1\left(t,X_t^1\right)-b^2\left(t,X_t^1\right)\right\|^2\right] dt. \qquad \square$$

Proof of Time-reversal Formula.

Corollary 4.4.5. *Let $\mathbb{Q}$ be the path measure of a Brownian motion on M with $\mathbb{Q}_0 = p_{\text{ref}}$ and $\mathbb{P}$ be a path measure on $C([0,T],M)$ with $\mathrm{KL}(\mathbb{P} \parallel \mathbb{Q}) < \infty$. Then, there exist two time-dependent vector fields $\beta_{\mathbb{P}}$ and $\beta_{R(\mathbb{P})}$ such that $\mathbb{P}$ and $R(\mathbb{P})$ satisfy martingale problem associated with $\mathcal{A}_{\mathbb{P}}$ and $\mathcal{A}_{R(\mathbb{P})}$,*

$$\begin{aligned}
\mathcal{A}_{\mathbb{P}}(t,u)(x) &= \langle \beta_{\mathbb{P}}(t,x), \nabla u(x)\rangle + \frac{1}{2}\Delta_M u(x), \\
\mathcal{A}_{R(\mathbb{P})}(t,u)(x) &= \left\langle \beta_{R(\mathbb{P})}(t,x), \nabla u(x)\right\rangle + \frac{1}{2}\Delta_M u(x).
\end{aligned}$$

Finally, we have that

$$\int_0^T \mathbb{E}\left[\|\beta_{\mathbb{P}}(t,X_t)\|^2\right]\mathrm{d}t + \int_0^T \mathbb{E}\left[\left\|\beta_{R(\mathbb{P})}(t,X_{T-t})\right\|^2\right]\mathrm{d}t < +\infty,$$

for $X \sim \mathbb{P}$.

Proof. Just applying above theorem with

$$\mathrm{KL}(\mathbb{P} \parallel \mathbb{Q}) = \mathrm{KL}(R(\mathbb{P}) \parallel R(\mathbb{Q})) = \mathrm{KL}(R(\mathbb{P}) \parallel \mathbb{Q}) < \infty. \qquad \square$$

Then by above the Integral by parts, for any $u, v \in C_c^\infty(M)$,

$$\mathbb{E}\left[v(X_t)\left(\left\langle \beta_{\mathbb{P}}(t,X_t) + \beta_{R(\mathbb{P})}(T-t,X_t), \nabla u(X_t)\right\rangle + \Delta_M u(X_t)\right) + \langle \nabla u(X_t), \nabla v(X_t)\rangle\right] = 0.$$

It follows that

$$\begin{aligned}
&\int_M \left\{\left\langle \beta_{R(\mathbb{P})}(T-t,x), \nabla u(x)\right\rangle + \langle b(x), \nabla u(x)\rangle\right\} v(x)p_t(x)dp_{\text{ref}}(x) \\
&\quad = \int_M \langle \nabla \log p_t(x), \nabla u(x)\rangle v(x)p_t(x)dp_{\text{ref}}(x).
\end{aligned}$$

Therefore,

$$\left\langle \beta_{R(\mathbb{P})}(T-t,x), \nabla u(x)\right\rangle = \langle \nabla \log p_t(x), \nabla u(x)\rangle.$$

4.5 Discretization

Let (M, g) be a complete Riemannian manifold and B^M be a Brownian motion on it. Consider the SDE,

$$dX_t = v(X_t)dt + dB_t^M,\ t \in [0, T], \quad X_0 = Z$$

We always assume B^M is independent of Z. Fix $h = T/N$ for some $N \in \mathbb{N}$. Let $t_k = kh$. Constructing process X_k^h by

$$X_{k+1}^h = \exp_{X_k^h}(hv(X_k^h) + \sqrt{h}\xi),$$

where exp is the exponential map, and ξ is a standard Gaussian standard Gaussian with respect to any orthonormal basis of T_xM. In the following, we study the convergence of $(X_k^h)_{k=0}^N$.

Weak convergence. Assume M has bounded geometry, that is,

$$\operatorname{inj}(M) \geq \iota > 0, \quad \sup_{x \in M} \left|\nabla^k R(x)\right| < \infty,\ \forall\ k,$$

where R is the Riemannian curvature. Let $(R_t)_{t\in[0,T]}$ be the semigroup of X_t and $(R_k^h)_{k=0}^N$ be the semigroup of X_k^h. Assume that $v \in C_b^3(TM)$. Then for any $f \in C_b^3(M)$, we have

$$\left\|R_T f - R_N^h f\right\|_\infty \leq Ch.$$

Proof. First, for X_t, the generator L is

$$Lf = \langle v, \nabla f\rangle + \frac{1}{2}\Delta_M f, \quad \forall\ f \in C_b^3(M),$$

and $R_T = (R_h)^N$. For X_k^h, let $R^h = R_1^h$. Then $R_N^h = (R^h)^N$.

Step 1. Check: For any $f \in C_b^3(M)$, we have

$$R^h f(x) = f(x) + hLf(x) + \mathcal{O}(h^{3/2}).$$

To prove that, first fix $X_0^h = x$ with $x \in M$ and $f \in C_c^\infty(M)$. We can choose $0 < \rho \leq \iota$ (note that ρ independent of x) such that

$$\exp_x \colon B(0, \rho) \subset T_xM \to B(x, \rho) \subset M$$

is a diffeomorphism, moreover, we choose the normal coordinate on T_xM. Let

$$Z_h(x) := hv(x) + \sqrt{h}\xi,$$

so that

$$X_1^h = \exp_x(Z_h(x)).$$

Define

$$\varphi_x(z) := f(\exp_x(z)), \quad \forall\ z \in B(0, \rho).$$

Because $f \in C_b^3(M)$ and M has bounded geometry,

$$\sup_{x\in M} \sup_{|z|\leq\rho} \sum_{|\alpha|\leq 3} |\partial_z^\alpha \varphi_x(z)| < \infty$$

Note that

$$R^h f(x) = \mathbb{E}\left[f(X_1^h) \mid X_0^h = x\right] = \mathbb{E}\left[\varphi_x(Z_h(x))\right]$$

I. Fix $h_0 > 0$ sufficiently small such that

$$h_0 \sup_x \|v(x)\| \le \frac{\rho}{4}.$$

Then for $0 < h < h_0$, if $|\xi| < \rho/(2\sqrt{h})$, then

$$|Z_h(x)| \le h\|v(x)\| + \sqrt{h}|\xi| \le \frac{\rho}{4} + \frac{\rho}{2} < \rho.$$

Therefore,

$$A_h := \left\{|\xi| \le \rho/(2\sqrt{h})\right\} \subset \{Z_h(x) \in B(0,\rho)\}.$$

Moreover, because ξ is Gaussian, then

$$\mathbb{P}(A_h^c) = \mathbb{P}\left(|\xi| > \rho/(2\sqrt{h})\right) \le c_1 e^{-c_2/h},$$

for some $c_1, c_2 > 0$. Writing

$$R^h f(x) = \mathbb{E}\left[\varphi_x\left(Z_h(x)\right)\mathbb{I}_{A_h}\right] + \mathbb{E}\left[f\left(\exp_x\left(Z_h(x)\right)\right)\mathbb{I}_{A_h^c}\right].$$

Then for the second term,

$$\left|\mathbb{E}\left[f\left(\exp_x\left(Z_h(x)\right)\right)\mathbb{I}_{A_h^c}\right]\right| \le \|f\|_\infty \mathbb{P}(A_h^c) \le \|f\|_\infty c_1 e^{-c_2/h} \le \mathcal{O}(h^2).$$

So the main goal is to bound the first term.

II. For $z \in B(0,\rho)$, by Taylor expansion

$$\varphi_x(z) = \varphi_x(0) + D\varphi_x(0)[z] + \frac{1}{2}D^2\varphi_x(0)[z,z] + R_3(x,z),$$

where

$$R_3(x,z) = \frac{1}{6}\sum_{i,j,k} \partial_{ijk}\varphi_x(\theta z) z_i z_j z_k,$$

for some $\theta = \theta(x,z) \in (0,1)$. For R_4, it has

$$|R_3(x,z)| \le C_f |z|^3, \quad \forall\ |z| \le \rho.$$

for some $C_f > 0$ (independent of x).

For $z = Z_h(x)$, writing

$$\begin{aligned}\mathbb{E}\left[\varphi_x\left(Z_h(x)\right)1_{A_h}\right] =& \varphi_x(0)\mathbb{P}\left(A_h\right) + \mathbb{E}\left[D\varphi_x(0)\left[Z_h\right]1_{A_h}\right] \\ &+ \frac{1}{2}\mathbb{E}\left[D^2\varphi_x(0)\left[Z_h, Z_h\right]1_{A_h}\right] \\ &+ \mathbb{E}\left[R_3(x, Z_h(x))1_{A_h}\right]\end{aligned}$$

Next, consider these terms respectively.

(a) First, because

$$\mathbb{P}(A_h) = 1 - \mathbb{P}(A_h^c) = 1 + \mathcal{O}\left(e^{-c_1/h}\right),$$

the zero-order term can be bounded by

$$\varphi_x(0) = \varphi_x(0)\mathbb{P}\left(A_h\right) = f(x) + \mathcal{O}\left(e^{-c_2/h}\right).$$

(b) For the first-order term, we first recall that $Z_h(x) = hv(x) + \sqrt{h}\xi$ and $\mathbb{E}[\xi] = 0$. Because

$$\left\|\mathbb{E}\left[Z_h(x)1_{A_h^c}\right]\right\| \leq \left(\mathbb{E}\left|Z_h(x)\right|^2\right)^{1/2} \left(\mathbb{P}\left(A_h^c\right)\right)^{1/2} \leq \mathcal{O}\left(e^{-c_3/h}\right),$$

we have

$$\begin{aligned}\mathbb{E}\left[Z_h(x)1_{A_h}\right] &= \mathbb{E}\left[Z_h(x)\right] - \mathbb{E}\left[Z_h(x)1_{A_h^c}\right] \\ &= \mathbb{E}\left[Z_h(x)\right] + \mathcal{O}\left(e^{-c_3/h}\right) \\ &= hv(x) + \mathcal{O}\left(e^{-c_3/h}\right).\end{aligned}$$

Then we get

$$\mathbb{E}\left[D\varphi_x(0)\left[Z_h\right]1_{A_h}\right] = D\varphi_x(0)[hv(x)] + \mathcal{O}\left(e^{-c_2/h}\right).$$

Because $\varphi_x(z) = f(\exp_x(z))$,

$$D\varphi_x(0)[hv(x)] = h\left\langle \nabla f(x), v(x)\right\rangle.$$

So

$$\mathbb{E}\left[D\varphi_x(0)\left[Z_h\right]1_{A_h}\right] = h\left\langle \nabla f(x), v(x)\right\rangle + \mathcal{O}\left(e^{-c_2/h}\right).$$

(c) For the second-order term, first, we have

$$D^2\varphi_x(0)[z, z] = \nabla^2 f(x)(z, z) = z^\top H_x z,$$

where $H_x = \nabla^2 f(x)$ in matrix expression under the normal coordinate. Moreover, because $\xi \in \mathcal{N}(0, I)$,

$$\begin{aligned}\mathbb{E}\left[Z_h Z_h^\top 1_{A_h}\right] &= \mathbb{E}\left[Z_h Z_h^\top\right] + \mathcal{O}\left(e^{-c_2/h}\right) \\ &= \mathbb{E}\left[(hv + \sqrt{h}\xi)(hv + \sqrt{h}\xi)^\top\right] + \mathcal{O}\left(e^{-c_2/h}\right) \\ &= h^2 vv^\top + h\mathbb{E}\left[\xi\xi^\top\right] + \mathcal{O}\left(e^{-c_2/h}\right) \\ &= h\,\mathrm{Id} + h^2 vv^\top + \mathcal{O}\left(e^{-c_2/h}\right).\end{aligned}$$

So it implies that

$$\begin{aligned}\frac{1}{2}\mathbb{E}\left[D^2\varphi_x(0)\left[Z_h, Z_h\right]1_{A_h}\right] &= \frac{1}{2}\mathbb{E}\left[\operatorname{tr}\left(H_x Z_h Z_h^\top\right)1_{A_h}\right] \\ &= \frac{1}{2}\operatorname{tr}\left(H_x \mathbb{E}\left[Z_h Z_h^\top 1_{A_h}\right]\right) \\ &= \frac{1}{2}h\operatorname{tr}\left(H_x\right) + \mathcal{O}\left(h^2\right) + \mathcal{O}\left(e^{-c_2/h}\right) \\ &= \frac{1}{2}h\Delta_M f(x) + \mathcal{O}\left(h^2\right).\end{aligned}$$

(d) For the remainder term, because $Z_h(x) = hv(x) + \sqrt{h}\xi$ with $\xi \sim \mathcal{N}(0, I)$,

$$\mathbb{E}\left[\left\|Z_h\right\|^3\right] = \mathcal{O}(h^{3/2}).$$

Therefore,

$$\left\|\mathbb{E}\left[R_3\left(x, Z_h(x)\right)1_{A_h}\right]\right\| \leq C_f \mathbb{E}\left[\left\|Z_h\right\|^3\right] = \mathcal{O}\left(h^{3/2}\right).$$

Combining these results, we have

$$R^h f(x) = f(x) + hLf(x) + \mathcal{O}(h^{3/2}),$$

which implies that

$$\left\| R^h f - f - hLf \right\|_\infty \le C_1 h^{3/2}.$$

Step 2. Check:

$$\| R_h f - f - hLf \|_\infty \le C_2 h^2.$$

Let $u(t,x) = R_t f(x)$. The by the backward Kolmogorov equation,

$$\partial_t u_t = L u_t, \quad u_0(x) = f(x).$$

By the smoothness and boundedness, one can show that $u_t \in C_b^3(M)$ and

$$\sup_{0\le t\le T} \|\nabla^\alpha u_t\|_\infty \le C_3, \quad \forall\ |\alpha| \le 3. \tag{4.4}$$

In particular, $\|L^2 u_t\|_\infty \le C_3$.By

$$u_h(x) - f(x) = \int_0^h \partial_t u_s(x) ds = \int_0^h L u_s(x) ds,$$

we get

$$L u_s(x) - Lf(x) = \int_0^s \partial_t L u_r(x) dr = \int_0^s L^2 u_r(x) dr.$$

Therefore,

$$\begin{aligned} u_h(x) &= f(x) + hLf(x) + \int_0^h \int_0^s L^2 u_r(x) dr ds \\ &= f(x) + hLf(x) + \beta_h(x) \end{aligned}$$

with

$$|\beta_h(x)| \le \int_0^h \int_0^s \left\| L^2 u_r \right\|_\infty dr ds \le C_3 \int_0^h \int_0^s dr ds = \frac{1}{2} C_3 h^2.$$

It follows that

$$\| R_h f - f - hLf \|_\infty \le C_2 h^2.$$

Step 3. By Step 1 and Step 2, for all $f \in C_c^\infty$,

$$\begin{aligned} \left\| R^h f - R_h f \right\|_\infty &\le \left\| R^h f - f - hLf \right\|_\infty + \| R_h f - f - hLf \|_\infty \\ &\le C_1 h^{3/2} + C_2 h^2 \le C_4 h^{3/2}. \end{aligned}$$

Globally, note that $R_T = (R_h)^N$ and $R_N^h = (R^h)^N$. Writing

$$\left(R^h\right)^N - (R_h)^N = \sum_{k=0}^{N-1} \left(R^h\right)^{N-1-k} \left(R^h - R_h\right) (R_h)^k .$$

Because R^h and R_h are Markov, they are contractions. So

$$\left\| \left(R^h\right)^N f - (R_h)^N f \right\|_\infty \le \sum_{k=0}^{N-1} \sup_x \left| \left(R^h - R_h\right) g_k(x) \right|,$$

where by (4.4)

$$g_k(x) = \left(R_h^k f\right)(x) = \left(R_{kh} f\right)(x) \in C_b^3(M).$$

Therefore,

$$\begin{aligned}
\left\| \left(R^h\right)^N f - \left(R_h\right)^N f \right\|_\infty &\le \sum_{k=0}^{N-1} \left\| R^h g_k - R_h g_k \right\|_\infty \\
&\le \sum_{k=0}^{N-1} C_4 h^{3/2} = C_4 N h^{3/2} = \frac{T}{h} C_4 h^{3/2} = C_T h^{1/2}. \qquad \square
\end{aligned}$$

Remark 4.5.1. Above proof can be extended to time-inhomogeneous case

$$dX_t = v(t, X_t)dt + g(t)dB_t^M.$$

Strong convergence. Let M be a complete Riemannian manifold and it is properly and isometrically embedded in $\mathbb{R}^n$. Assume that

(I) the second fundamental form and its derivative are uniformly bounded, i.e.,

$$\sup_{x\in M} \|\mathrm{II}_x\|_{\mathrm{op}} < \infty, \ \sup_{x\in M} \|\nabla \mathrm{II}_x\|_{\mathrm{op}} < \infty.$$

(II) there exists a tubular neighborhood $V \subset \mathbb{R}^n$ of M, together with the canonical projection $\pi \colon V \to M \subset \mathbb{R}^n$ such that

$$\sup_{y\in V} \|D\pi(y)\|_{\mathrm{op}} < \infty.$$

(III) $v \in \Gamma(TM)$ satisfies

$$\sup_{x\in M} \|v(x)\| < \infty, \quad \sup_{x\in M} \sup_{\|X\|=1} \|\nabla_X v(x)\| < \infty.$$

(IV) $X_0 \in L^2$.

Then we have the strong convergence of geometric EM algorithm, i.e.,

$$\mathbb{E}\left[\max_{0\le k\le N} \left\|X_{t_k} - X_k^h\right\|^2\right] \le Ch$$

Proof. The proof needs following steps.

Step 1. SDE on Euclidean: For $\pi \colon V \to M$, by Corollary 4.5.6, $\Pi(x) := D\pi(x) \colon \mathbb{R}^n \to \mathbb{R}^n$ for $x \in M$ is the orthogonal projection onto T_xM. Then we can choose

$$dB_t^M = \Pi \circ dW_t,$$

where W is a standard $\mathbb{R}^n$-Brownian motion that is also independent of X_0. Then

$$dX_t = v(X_t)dt + \Pi(X_t) \circ dW_t = \tilde{a}(X_t)dt + \Pi(X_t)dW_t, \tag{4.5}$$

as an Euclidean SDE, where

$$\tilde{a}(x) = v(x) + A(x), \quad A^j(x) = \frac{1}{2}\sum_{\alpha=1}^{n} \Pi_\alpha(x)\left(\Pi_{j\alpha}(x)\right),$$

for $\Pi(x) = (\Pi_{j\alpha}(x))$ when fixing the standard basis $\{e_1, \cdots, e_n\}$ in $\mathbb{R}^n$. Moreover, for

$$\Pi_\alpha(x) = \Pi(x)e_\alpha \in T_xM,$$

we have

$$\Pi_{j\alpha}(x) = \Pi_\alpha(x^j)(x),$$

where $x^j : M \to \mathbb{R}$ is the j-th component of the canonical embedding $x\colon M \hookrightarrow \mathbb{R}^n$.

Claim: If $\{E_i(x)\}_{i=1}^d$ is an orthonormal frame of $T_xM \subset \mathbb{R}^n$, then we have

$$A(x) = \frac{1}{2}\sum_{i=1}^d \mathrm{II}_x(E_i(x), E_i(x)).$$

To prove that, first, by

$$\Delta_M x^j = \sum_{\alpha=1}^N \Pi_\alpha(\Pi_\alpha(x^j)) = \sum_{\alpha=1}^N \Pi_\alpha(\Pi_{j\alpha}),$$

we have

$$A^j(x) = \frac{1}{2}\Delta_M x^j(x).$$

On the other hand, we know

$$\Delta_M x^j(x) = \sum_{i=1}^d \mathrm{II}_x^j(E_i(x), E_i(x)).$$

Step 2. Extension on $\mathbb{R}^n$: Because v, A and Π are only make sense on M, we cannot deal with (4.5) as the usual SDE on $\mathbb{R}^n$. So we need to extend it on $\mathbb{R}^n$ with well-posed properties. First, because M is closed, it can choose a smaller open U such that $M \subset U \subset V$. Then it can choose a bump function $\chi\colon \mathbb{R}^n \to [0,1]$ such that $\chi|_{\bar{U}} = 1$, $\chi_{\mathbb{R}^n\setminus V} = 0$, and $\|D\chi\| \le C_\chi < \infty$.

- Extending Π: Let

  $$\tilde{\Pi}(y) = \chi(y)\Pi(\pi(y)), \quad \forall\, y \in \mathbb{R}^n.$$

  Next, we need to guarantee the global Lipschitz continuity of $\tilde{\Pi}$. By Lemma 4.5.7 with Assumption (I),

  $$\sup_{x\in M} \|D\Pi(x)[v]\|_{\mathrm{op}} = \sup_{x\in M} \left\|\overline{\nabla}_v \Pi(x)\right\|_{\mathrm{op}} < C_\Pi \|v\|, \quad \forall\, x \in M,\ \forall v \in T_xM.$$

  Then, for any $y \in \mathbb{R}^n$ and $w \in \mathbb{R}^n$, we have

  $$\begin{aligned}\left\|D\tilde{\Pi}(y)[w]\right\| &= \|\Pi(\pi(y))D\chi(y)[w] + \chi(y)D\Pi(\pi(y))[D\pi(y)[w]]\| \\ &\le C_\chi \|w\| + C_\Pi C_\pi \|w\|,\end{aligned}$$

  because $D\pi(y)[w] \in T_{\pi(y)}M$ and by Assumption (II), $\|D\pi(y)\| \le C_\pi < \infty$. Therefore,

  $$\sup_{y\in\mathbb{R}^n} \left\|D\tilde{\Pi}(y)\right\|_{\mathrm{op}} \le L_\Pi := C_\chi + C_\Pi C_\pi,$$

  i.e., $\tilde{\Pi}$ is globally Lipschitz continuous,

  $$\left\|\tilde{\Pi}(y_1) - \tilde{\Pi}(y_2)\right\|_{\mathrm{op}} \le L_\Pi \|y_1 - y_2\|, \quad \forall\, y_1, y_2 \in \mathbb{R}^n.$$

  In the following, we denote $\tilde{\Pi}$ by Π for convenience.

- Extending v: For v satisfying Assumption (III), by Corollary 4.5.9,

$$\sup_{x\in M}\|Dv(x)[u]\| = \sup_{x\in M}\left\|\overline{\nabla}_u v(x)\right\| < C_v\|u\|, \quad \forall\, u\in T_xM.$$

Similarly, let

$$\tilde{v}(y)=\chi(y)v(\pi(y)), \quad \forall\, y\in\mathbb{R}^n.$$

Then for any $y\in\mathbb{R}^n$ and $w\in\mathbb{R}^n$,

$$\begin{aligned}\|D\tilde{v}(y)[w]\| &= \|v(\pi(y))D\chi(y)[w]+\chi(y)Dv(\pi(y))[D\pi(y)[w]]\|\\ &\le C_\chi C_v'\|w\|+C_vC_\pi\|w\|,\end{aligned}$$

because $\sup_{x\in M}\|v(x)\|\le C_v'$ of Assumption (III). It follows that $\tilde{v}$ is globally Lipschitz continuous,

$$\sup_{y\in\mathbb{R}^n}\|D\tilde{v}(y)\|_{\mathrm{op}}\le L_v := C_\chi C_v' + C_vC_\pi,$$

i.e.,

$$\|\tilde{v}(y_1)-\tilde{v}(y_2)\|\le L_v\|y_1-y_2\|, \quad \forall\, y_1,y_2\in\mathbb{R}^n.$$

We also do not distinguish v and $\tilde{v}$.

- Extending A: by above, we have known

$$A(x)=\sum_{i=1}^d \mathrm{II}_x(E_i(x),E_i(x))\in N_xM, \quad \forall\, x\in M.$$

Moreover, we let $\{E_1,\cdots,E_d\}$ be geodesic at x, i.e., $\nabla_{E_i}E_j(x)=0$. Then from the proof of Lemma 4.5.10 with Assumption (I), and by $\{E_1,\cdots,E_d\}$ geodesic, i.e., $\nabla_v E_i=0$, we have

$$\sup_x\|DA(x)[u]\| = \sup_{x\in M}\left\|\overline{\nabla}_u A(x)\right\|\le C_A\|u\|, \quad \forall\, u\in T_xM.$$

Let

$$\tilde{A}(y)=\chi(y)A(\pi(y)), \quad \forall\, y\in\mathbb{R}^n.$$

Then for any $y\in\mathbb{R}^n$ and $w\in\mathbb{R}^n$,

$$\begin{aligned}\left\|\tilde{A}(y)[w]\right\| &= \|A(\pi(y))D\chi(y)[w]+\chi(y)DA(\pi(y))[D\pi(y)[w]]\|\\ &\le C_{\mathrm{II}}C_\chi\|w\|+C_AC_\pi\|w\|\end{aligned}$$

Therefore, $\tilde{A}$ is globally Lipschitz continuous. In the following, we also denote $\tilde{A}$ by A.

Then for such extended v, A, and Π, we get SDE (4.5) defined on Euclidean space with globally Lipschitz continuous coefficients.

Step 3. Euclidean EM discretization: We can consider the Euclidean EM algorithm. Let $t_k=kh, Nh=T, \Delta W_k = W_{t_{k+1}}-W_{t_k}$, and define

$$Y_{k+1}^h = Y_k^h + \tilde{a}\left(Y_k^h\right)h + \Pi\left(Y_k^h\right)\Delta W_k, \quad Y_0^h = X_0.$$

By the global Lipschitz continuity of $\tilde{a}$, and Assumption (IV) of $X_0\in L^2$, we have

$$\mathbb{E}\left[\max_{0\le k\le N}\left\|X_{t_k}-Y_k^h\right\|^2\right]\le C_Th.$$

So the next goal is to bound

$$\mathbb{E}\left[\max_{0\le k\le N}\left\|X_k^h-Y_k^h\right\|^2\right].$$

Moreover, let

$$\Phi_h^E(x,z)=x+\tilde{a}(x)h+\Pi(x)z.$$

So

$$Y_{k+1}^h=\Phi_h^E(Y_k^h,\Delta W_k)$$

Step 4. Geometric EM discretization: For

$$X_{k+1}^h=\exp_{X_k^h}\left(hv\left(X_k^h\right)+\sqrt{h}\xi_{k+1}\right)$$

with ξ_{k+1} a standard Gaussian on T_xM. WTLG, assume

$$\xi_{k+1}:=\frac{1}{\sqrt{h}}\Pi\left(X_k^h\right)\Delta W_k,\quad \Delta W_k=W_{t_{k+1}}-W_{t_k},$$

where W is the Brownian motion in (4.5). So the geometric EM is

$$X_{k+1}^h=\exp_{X_k^h}\left(hv\left(X_k^h\right)+\Pi\left(X_k^h\right)\Delta W_k\right).$$

Let

$$\Phi_h^G(x,z)=\exp_x(hv(x)+\Pi(x)z),$$

i.e., we have

$$X_{k+1}^h=\Phi_h^G(X_k^h,\Delta W_k).$$

Moreover, by Lemma 4.5.10, for $u=hv(x)+\Pi(x)z$,

$$\Phi_h^G(x,z)=x+hv(x)+\Pi(x)z+\frac{1}{2}\mathrm{II}_x(u,u)+R_3(x,u),$$

where

$$\|R_3(x,u)\|\le C_1\|u\|^3,$$

where C_1 is independent of x.
Claim: For $z\sim\mathcal{N}(0,hI_n)$, we have

$$\mathbb{E}\left[\mathrm{II}_x(\Pi(x)z,\Pi(x)z)\right]=2hA(x).$$

For

$$\Pi(x)z=\sum_{i=1}^d\zeta_iE_i(x),\quad \zeta_i=\langle z,E_i(x)\rangle,$$

then $\zeta=(\zeta_1,\cdots,\zeta_d)$ is a Gaussian with

$$\mathbb{E}\left[\zeta_i\zeta_j\right]=\mathbb{E}\left[\langle z,E_i\rangle\langle z,E_j\rangle\right]=E_i^\top\mathbb{E}\left[zz^\top\right]E_j=E_i^\top\left(hI_N\right)E_j=h\delta_{ij},$$

i.e., $\zeta\sim\mathcal{N}(0,hI_d)$. Because II_x is bilinear,

$$\mathrm{II}_x(\Pi(x)z,\Pi(x)z)=\sum_{i,j=1}^d\zeta_i\zeta_j\mathrm{II}_x(E_i,E_j).\tag{4.6}$$

Therefore,

$$\begin{aligned}\mathbb{E}\left[\mathrm{II}_x(\Pi(x)z,\Pi(x)z)\right] &= \sum_{i,j=1}^{d}\mathbb{E}\left[\zeta_i\zeta_j\right]\mathrm{II}_x(E_i,E_j)\\ &= h\sum_{i=1}^{d}\mathrm{II}_x(E_i,E_i) = 2hA(x).\end{aligned}$$

Therefore, for $u = hv(x) + \Pi(x)z$ and $z = \Delta W_k \sim \mathcal{N}(0, hI_n)$,

$$\mathbb{E}\left[\mathrm{II}_x(u,u)\right] = \mathbb{E}\left[\mathrm{II}_x(\Pi(x)z,\Pi(x)z)\right] + h^2\mathrm{II}_x(v,v) = 2hA(x) + \mathcal{O}(h^2).$$

It follows that

$$\begin{aligned}\mathbb{E}\left[\Phi_h^G(x,\Delta W_k)\right] &= x + hv(x) + hA(x) + \mathcal{O}(h^2)\\ &= x + \tilde{a}(x)h + \mathcal{O}(h^2).\end{aligned}$$

Note that by (4.6) and Assumption (I), we also have

$$\mathbb{E}\left[\left\|\mathrm{II}_x(\Pi(x)z,\Pi(x)z)\right\|^2\right] = \mathcal{O}(h^2).$$

Step 5. Local difference: For Φ_h^E, Φ_h^G defined as above, we have

$$\Phi_h^G(x,z) - \Phi_h^E(x,z) = -hA(x) + \frac{1}{2}\Pi_x(u,u) + R_3(x,u).$$

Let

$$\Delta_h(x) = \Phi_h^G(x,\Delta W_k) - \Phi_h^E(x,\Delta W_k) = Q_h(x) + R_3(x,u),$$

where

$$Q_h(x) := \frac{1}{2}\Pi_x(u,u) - hA(x), \quad u = hv(x) + \Pi(x)\Delta W_k.$$

Writing

$$\begin{aligned}Q_h(x) &= \frac{1}{2}\mathrm{II}_x(u,u) - hA(x)\\ &= \left(\frac{1}{2}\mathrm{II}_x(\Pi\Delta W_k,\Pi\Delta W_k) - hA(x)\right) + \frac{1}{2}\mathrm{II}_x(hv,hv) + \mathrm{II}_x(hv,\Pi\Delta W_k)\\ &= T_1 + T_2 + T_3.\end{aligned}$$

Before continuing, let's recall that for $u = hv(x) + \Pi(x)\Delta W_k$, by the boundedness of v and $\mathbb{E}[\|\Delta W_k\|^p] = \mathcal{O}(h^{p/2})$, we have

$$\mathbb{E}\left[\|u\|^3\right] = \mathcal{O}(h^{3/2}), \quad \mathbb{E}\left[\|u\|^4\right] = \mathcal{O}\left(h^2\right), \quad \mathbb{E}\left[\|u\|^6\right] = \mathcal{O}\left(h^3\right).$$

- First momentum: For T_1, by above, we have $\mathbb{E}[T_1] = 0$. It is obvious that $\|\mathbb{E}[T_2]\| = \mathcal{O}(h^2)$ and $\mathbb{E}[T_3] = 0$. So

  $$\|\mathbb{E}\left[Q_h(x)\right]\| = \mathcal{O}(h^2).$$

  For R_3,

  $$\|R_3\| \le C_1\|u\|^3 \;\Rightarrow\; \|\mathbb{E}[R_3]\| \le \mathbb{E}\left[\|R_3\|\right] \le C_1\mathbb{E}\left[\|u\|^3\right] = \mathcal{O}(h^{3/2}).$$

  Therefore,

  $$\|\mathbb{E}\left[\Delta_h(x)\right]\| \le \|\mathbb{E}\left[Q_h(x)\right]\| + \|\mathbb{E}\left[R_3\right]\| \le \mathcal{O}(h^{3/2}).$$

- Second momentum: By above, because of the boundedness of second fundamental, we have
$$\mathbb{E}\left[\|T_1\|^2\right] = \mathcal{O}(h^2), \quad \mathbb{E}\left[\|T_2\|^2\right] = \mathcal{O}(h^4), \quad \mathbb{E}\left[\|T_3\|^2\right] = \mathcal{O}(h^3),$$
which implies that
$$\mathbb{E}\left[\|Q_h(x)\|^2\right] \leq \mathcal{O}(h^2).$$
And
$$\|R_3\| \leq C_1 \|u\|^3 \;\Rightarrow\; \mathbb{E}\left[\|R_3\|^2\right] \leq C_1^2 \mathbb{E}\left[\|u\|^6\right] = \mathcal{O}(h^3).$$
Therefore,
$$\mathbb{E}\left[\|\Delta_h(x)\|^2\right] = \mathcal{O}(h^2)$$

Step 6. Global difference: Let $e_k = X_k^h - Y_k^h$. Note that

$$X_{k+1}^h = \Phi_h^G\left(X_k^h, \Delta W_k\right), \quad Y_{k+1}^h = \Phi_h^E\left(Y_k^h, \Delta W_k\right),$$

where $\Delta W_k = W_{t_{k+1}} - W_{t_k}$. Therefore,

$$\begin{aligned}
e_{k+1} &= \Phi_h^G\left(X_k^h, \Delta W_k\right) - \Phi_h^E\left(Y_k^h, \Delta W_k\right) \\
&= \left(\Phi_h^E\left(X_k^h, \Delta W_k\right) - \Phi_h^E\left(Y_k^h, \Delta W_k\right)\right) + \left(\Phi_h^G\left(X_k^h, \Delta W_k\right) - \Phi_h^E\left(X_k^h, \Delta W_k\right)\right) \\
&= G_k + D_k,
\end{aligned}$$

where
$$G_k = \Phi_h^E\left(X_k^h, \Delta W_k\right) - \Phi_h^E\left(Y_k^h, \Delta W_k\right), \quad D_k = \Delta_h(X_k^h).$$
Let $\mathbb{F}_t = \sigma(X_0, W_s \colon s \leq t)$. Therefore, X_k^h, Y_k^h are $\mathcal{F}_{t_k} =: \mathcal{F}_k$-measurable. Moreover, because W is independent of X_0, W is also a Brownian motion w.s.t. $(\mathcal{F}_t)$. So

$$\mathbb{E}\left[\|e_{k+1}\|^2 \mid \mathcal{F}_k\right] = \mathbb{E}\left[\|G_k\|^2 \mid \mathcal{F}_k\right] + \mathbb{E}\left[\langle G_k, D_k\rangle \mid \mathcal{F}_k\right] + \mathbb{E}\left[\|D_k\|^2 \mid \mathcal{F}_k\right].$$

For these three terms, let's analyze them separably.

- For G_k, first note that

$$G_k = e_k + h(\tilde{a}(X_k^h) - \tilde{a}(Y_k^h)) + (\Pi(X_k^h) - \Pi(Y_k^h))\Delta W_k.$$

  So we have
$$\mathbb{E}\left[G_k \mid \mathcal{F}_k\right] = e_k + h(\tilde{a}(X_k^h) - \tilde{a}(Y_k^h)).$$
  Then by global Lipschitz continuity of $\tilde{a}$, we have

$$\|\mathbb{E}\left[G_k \mid \mathcal{F}_k\right]\| \leq (1 + L_1 h)\|e_k\|.$$

  On the other hand, by the global Lipschitz continuity of Π,

$$G_k - \mathbb{E}\left[G_k \mid \mathcal{F}_k\right] = (\Pi(X_k^h) - \Pi(Y_k^h))\Delta W_k \;\Rightarrow\; \mathbb{E}\left[\|G_k - \mathbb{E}\left[G_k \mid \mathcal{F}_k\right]\|^2 \mid \mathcal{F}_k\right] \leq L_2 h \|e_k\|^2.$$

  Therefore,

$$\begin{aligned}
\mathbb{E}\left[\|G_k\|^2 \mid \mathcal{F}_k\right] &= \|\mathbb{E}\left[G_k \mid \mathcal{F}_k\right]\|^2 + \mathbb{E}\left[\|G_k - \mathbb{E}\left[G_k \mid \mathcal{F}_k\right]\|^2 \mid \mathcal{F}_k\right] \\
&\leq (1 + C_2 h)\|e_k\|^2
\end{aligned}$$

- For D_k,
$$\mu_k := \mathbb{E}\left[D_k \mid \mathcal{F}_k\right] = \mathbb{E}\left[\Delta_h(x)\right]_{x=X_k^h}.$$
Therefore, by above step, we have
$$\|\mu_k\| \leq C_3 h^{3/2},$$
and
$$\mathbb{E}\left[\|D_k\|^2 \mid \mathcal{F}_k\right] = \mathbb{E}\left[\|\Delta_h(x)\|^2\right]_{x=X_k^h} \leq \mathcal{O}(h^2).$$
Moreover, let $\widetilde{D}_k := D_k - \mu_k$, so $\mathbb{E}\left[\widetilde{D}_k \mid \mathcal{F}_k\right] = 0$ and
$$\mathbb{E}\left[\|\widetilde{D}_k\|^2 \mid \mathcal{F}_k\right] \leq C_4 h^2$$

- For the cross term, by $D_k = \mu_k + \widetilde{D}_k$ and μ_k $\mathcal{F}_k$-measurable,
$$\mathbb{E}\left[\langle G_k, D_k\rangle \mid \mathcal{F}_k\right] = \langle \mathbb{E}\left[G_k \mid \mathcal{F}_k\right], \mu_k\rangle + \mathbb{E}\left[\left\langle G_k, \widetilde{D}_k\right\rangle \mid \mathcal{F}_k\right].$$
For the first term, by above, we have
$$\left|\langle \mathbb{E}\left[G_k \mid \mathcal{F}_k\right], \mu_k\rangle\right| \leq \left\|\mathbb{E}\left[G_k \mid \mathcal{F}_k\right]\right\| \|\mu_k\| \leq C_3(1 + L_1 h) h^{3/2} \|e_k\|.$$
Moreover, by $2ab \leq \varepsilon a^2 + (1/\varepsilon) b^2$ and setting $\varepsilon = h$,
$$h^{3/2} \|e_k\| \leq \frac{1}{2}\left(h \|e_k\|^2 + h^2\right).$$
So
$$\left|\langle \mathbb{E}\left[G_k \mid \mathcal{F}_k\right], \mu_k\rangle\right| \leq C_5 h \|e_k\|^2 + \mathcal{O}(h^2).$$
For the second term, first, because $\mathbb{E}[\widetilde{D}_k \mid \mathcal{F}_k] = 0$,
$$\mathbb{E}\left[\left\langle G_k, \widetilde{D}_k\right\rangle \mid \mathcal{F}_k\right] = \mathbb{E}\left[\left\langle G_k - \mathbb{E}\left[G_k \mid \mathcal{F}_k\right], \widetilde{D}_k\right\rangle \mid \mathcal{F}_k\right].$$
Then by the conditional CBS inequality,
$$\left|\mathbb{E}\left[\left\langle G_k, \widetilde{D}_k\right\rangle \mid \mathcal{F}_k\right]\right| \leq \sqrt{\mathbb{E}\left[\left\|G_k - \mathbb{E}\left[G_k \mid \mathcal{F}_k\right]\right\|^2 \mid \mathcal{F}_k\right]}\sqrt{\mathbb{E}[\|\widetilde{D}_k\|^2 \mid \mathcal{F}_k]}.$$
By above,
$$\mathbb{E}\left[\left\|G_k - \mathbb{E}\left[G_k \mid \mathcal{F}_k\right]\right\|^2 \mid \mathcal{F}_k\right] \leq L_2 h \|e_k\|^2, \quad \mathbb{E}[\|\widetilde{D}_k\|^2 \mid \mathcal{F}_k] \leq C_4 h^2.$$
So
$$\left|\mathbb{E}\left[\left\langle G_k, \widetilde{D}_k\right\rangle \mid \mathcal{F}_k\right]\right| \leq \sqrt{L_2 C_4} h^{3/2} \|e_k\| \leq C_6 h \|e_k\|^2 + \mathcal{O}(h^2).$$
Therefore,
$$\left|\mathbb{E}\left[\langle G_k, D_k\rangle \mid \mathcal{F}_k\right]\right| \leq (C_5 + C_6) h \|e_k\|^2 + C_7 h^2.$$

Therefore, we have
$$\mathbb{E}\left[\|e_{k+1}\|^2 \mid \mathcal{F}_k\right] \leq (1 + C_\alpha h) \|e_k\|^2 + C_\beta h^2,$$
which implies that
$$\mathbb{E}\left[\|e_{k+1}\|^2\right] \leq (1 + C_\alpha h)\mathbb{E}\left[\|e_k\|^2\right] + C_\beta h^2.$$

Then by the discrete Grönwall's inequality,

$$\mathbb{E}\left[\max_{0\le k\le N}\left\|X_k^h - Y_k^h\right\|^2\right] \le C_T' h.$$

Step 5. Final step: Combining all,

$$\begin{aligned}\mathbb{E}\left[\max_{0\le k\le N}\left\|X_{t_k} - Y_k^h\right\|^2\right] &\le 2\mathbb{E}\left[\max_{0\le k\le N}\left\|X_{t_k} - Y_k^h\right\|^2\right] + 2\mathbb{E}\left[\max_{0\le k\le N}\left\|X_k^h - Y_k^h\right\|^2\right] \\ &\le Ch.\end{aligned}$$

□

Remark 4.5.2. A direct result of $A(x) = \frac{1}{2}\sum_i \mathrm{II}_x(E_i(x), E_i(x)) \in N_x M$ is when viewing (4.5) as an Euclidean SDE with generator $\tilde{L}$,

$$\tilde{L}|_M = L.$$

For any $f \in C^\infty(M)$, let $\tilde{f} \in C^\infty(\mathbb{R}^n)$ be a normal extension, which means that

$$\overline{\nabla}\tilde{f}(x) = \nabla f(x), \quad \forall\ x \in M,$$

and for all $x \in M$ and $X, Y \in T_x M$,

$$\overline{\nabla}^2 \tilde{f}(x)(X,Y) = \nabla f(x)(X,Y).$$

First, we have

$$\tilde{L}\tilde{f}(x) = \left\langle \overline{\nabla}\tilde{f}(x), \tilde{a}\right\rangle + \frac{1}{2}\operatorname{tr}\left(\Pi(x)\Pi(x)^\top \overline{\nabla}^2\tilde{f}(x)\right).$$

For the first term, because $A(x) \in N_x M$,

$$\left\langle \overline{\nabla}\tilde{f}(x), \tilde{a}\right\rangle = \left\langle \nabla f(x), v\right\rangle.$$

For the second term, because $\Pi(x)$ is an orthogonal projection, we have

$$\Pi(x)\Pi(x)^\top = \Pi(x) = \sum_{i=1}^d E_i(x)\otimes E_i(x).$$

Therefore,

$$\begin{aligned}\operatorname{tr}\left(\Pi(x)\Pi(x)^\top \overline{\nabla}^2\tilde{f}(x)\right) &= \sum_{i=1}^d \overline{\nabla}^2\tilde{f}(x)(E_i(x), E_i(x)) \\ &= \sum_{i=1}^d \nabla^2 f(x)(E_i(x), E_i(x)) \\ &= \Delta_M f(x).\end{aligned}$$

So

$$\tilde{L}\tilde{f}(x) = \left\langle \nabla f(x), v\right\rangle + \frac{1}{2}\Delta_M f(x) = Lf(x).$$

Remark 4.5.3. If M is compact, then after embedding $M \subset \mathbb{R}^n$, it is a compact set. Then Assumptions (I)-(IV) are automatically satisfied:

- Because the second fundamental form II and vector field v are smooth on M, a compact Riemannian manifold, Assumptions (I) and (III) are obviously true.

- Since X_0 is valued on M a compact set, $\|X_0\| \leq \operatorname{diam}(M) < \infty$. So it is obvious that $X_0 \in L^2$.
- For compact M, it has the uniform tubular neighborhood, i.e., there exists a $r_0 > 0$ such that
$$U_{r_0} = \{y \in \mathbb{R}^n : d(y, M) < r_0\}$$
is a tubular neighborhood of M. Moreover, because M is compact, then for $M \subset V \subset U_{r_0}$, $\overline{V} \subset U_{r_0}$ is also compact. By the smoothness of π, if we choose V as the new tubular neighborhood, Assumption (II) obviously true on V.

Therefore, above result is automatically valid in compact case.

Remark 4.5.4. Above analysis can be easily extended to time-inhomogeneous case with $v = v(t, x)$ with an additional assumption of
$$\|v(t, x) - v(s, y)\| \leq C(1 + \|x\|) \, |t - s|^{1/2},$$
because this condition can be easily extended to Euclidean case such that the Euclidean EM converges, while the comparing of X_k^h and Y_k^h is not changed.

Lemma 4.5.5. *Let $M \subset \mathbb{R}^n$ be a Riemannian submanifold. Let $V \subset \mathbb{R}^n$ be a tubular neighborhood of M with the canonical projection $\pi \colon V \to M \subset \mathbb{R}^n$. Then for any $y \in V$, $D\pi(y) \colon \mathbb{R}^n \to \mathbb{R}^n$ with*
$$\ker D\pi(y) = N_{\pi(y)}M, \quad \operatorname{Im} D\pi(y) = T_{\pi(y)}M.$$

Proof. First, we know that for any $x \in M$,
$$\pi^{-1}(x) = \{x + v \colon \ \|v\| \ \text{ is small. } \} = (x + N_xM) \cap V.$$
For any $y \in V$, let $x = \pi(y)$, i.e., $y \in \pi^{-1}(x)$.

- $N_xM \subset \ker D\pi(y)$: Let $w \in N_xM$ and choose $\gamma(t) = y + tw$. Since $y \in \pi^{-1}(x) = (x + N_xM) \cap V$,
$$\gamma(t) = y + tw = x + (v + tw) \in x + N_xM,$$
which implies that $\pi(\gamma(t)) \equiv x$. Then
$$D\pi(y)[w] = \left.\frac{d}{dt}\right|_{t=0} \pi(\gamma(t)) = 0.$$
So $N_xM \subset \ker D\pi(y)$.
- $T_xM = \operatorname{Im} D\pi(y)$: First, because $\pi \colon V \to M$, it obviously $\operatorname{Im} D\pi(y) \subset T_xM$. Conversely, let $F \colon U \to V$ be the tubular neighborhood diffeomorphism, where
$$U = \{(x, v) \colon x \in M, \quad v \in N_xM \text{ with } \|v\| \leq \varepsilon(x)\} \subset NM,$$
i.e, $F(x, v) = x + v$. Let $pr \colon U \to M$ by $pr(x, v) = x$. Then $\pi = pr \circ F^{-1} \colon V \to M$. So
$$D\pi(y) = Dpr(x, v) \circ D(F^{-1})(y)$$
Note that $D(F^{-1})(y)$ is an isomorphism. So $\operatorname{rank} D\pi(y) = \operatorname{rank} Dpr(x, v)$. For pr, note that
$$Dpr(x, v) \colon T_{(x,v)}(NM) \cong T_xM \otimes N_xM \to T_xM$$
is a surjection. So
$$\operatorname{rank} D\pi(y) = \operatorname{rank} Dpr(x, v) = \dim(T_xM),$$
which implies that
$$T_xM = \operatorname{Im} D\pi(y).$$

- $N_xM = \ker D\pi(y)$: It is just by

$$\dim(\ker D\pi(y)) = n - \dim D\pi(y) = n - \dim(T_xM) = \dim(N_xM). \qquad \square$$

Corollary 4.5.6. *Let $M \subset \mathbb{R}^n$ be a Riemannian submanifold. Let $V \subset \mathbb{R}^n$ be a tubular neighborhood of M with the canonical projection $\pi\colon V \to M \subset \mathbb{R}^n$. Then for any $x \in M$, $D\pi(x)\colon \mathbb{R}^n \to \mathbb{R}^n$ is the orthogonal projection from $\mathbb{R}^n$ onto T_xM.*

Proof. Fix $x \in M$. For any $v \in T_xM \subset \mathbb{R}^n$, let $c\colon (-\varepsilon, \varepsilon) \to M$ be a curve with $c(0) = x$ and $\dot{c}(0) = v$. Because $\pi|_M = \mathrm{id}_M$, $\pi(c(t)) = c(t)$. It follows that

$$D\pi(x)[v] = \left.\frac{d}{dt}\right|_{t=0} \pi(c(t)) = \left.\frac{d}{dt}\right|_{t=0} c(t) = v.$$

So $D\pi(x)\colon \mathbb{R}^n = T_xM \otimes N_xM \to \mathbb{R}^n$ is identity on T_xM. By Lemma 4.5.5, $D\pi(x)$ is an orthogonal projection. $\square$

Lemma 4.5.7. *Assume $M \subset \mathbb{R}^n$ properly and isometrically embedded and assume the second fundamental form is uniformly bounded, i.e.,*

$$\sup_{x \in M} \|\mathrm{II}_x\|_{\mathrm{op}} < \infty.$$

Then we have

$$\sup_{x \in M} \|\nabla \Pi(x)\|_{\mathrm{op}} < \infty, \quad \sup_{x \in M} \sup_{v \in T_xM, \|v\|=1} \left\|\overline{\nabla}_v \Pi(x)\right\|_{\mathrm{op}} < \infty$$

Proof. We need to prove for the following separated steps.

- **Shape operator:** For any $\eta \in \Gamma(NM)$, consider the shape operator $s_\eta\colon \Gamma(TM) \to \Gamma(TM)$ as
$$\langle s_\eta(X), Y\rangle = \langle \mathrm{II}(X, Y), \eta\rangle, \quad X, Y \in \Gamma(TM).$$
Because the second fundamental form is uniformly bounded, i.e.,
$$\sup_{x \in M} \sup_{\|X\|, \|Y\|=1} \|\mathrm{II}_x(X, Y)\| \le C < \infty,$$
we have
$$\|s_\eta\|_{\mathrm{op}} \le C \|\eta\|,$$

- **Intrinsic Derivative:** For any $w \in \mathbb{R}^N$, $\Pi(x)w \in T_xM$, and by smoothness, we have $\Pi w \in \Gamma(TM)$, i.e, $\Pi \in \Gamma(M, \mathrm{Hom}(\mathbb{R}^N, TM))$. Therefore, we define $\nabla_X \Pi \in \Gamma(M, \mathrm{Hom}(\mathbb{R}^N, TM))$ as
$$(\nabla_X \Pi)(x)w := \nabla_X (\Pi(x)w), \quad \forall\, x \in M.$$
Note that for $w \in \mathbb{R}^N$,
$$w = w(x)^\top + w(x)^\perp,$$
where
$$w(x)^\top = \Pi(x)w \in T_xM, \quad w(x)^\perp = w - w(x)^\top \in N_xM.$$
Because w is constant on $\mathbb{R}^N$,
$$0 = \overline{\nabla}_X w = \overline{\nabla}_X w^\top + \overline{\nabla}_X w^\perp.$$

For $\overline{\nabla}_X w^\top$, we know

$$\overline{\nabla}_X w^\top = \nabla_X w^\top + \mathrm{II}(w^\top, X).$$

For $w^\perp \in \Gamma(NM)$, by Weingarten equation $(\overline{\nabla}_X w^\perp)^\top = -s_{w^\perp}(X)$, we have

$$\overline{\nabla}_X w^\perp = -s_{w^\perp}(X) + \nabla_X^\perp w^\perp.$$

Therefore,

$$\nabla_X w^\top - s_{w^\perp}(X) = -\left(\nabla_X^\perp w^\perp + \mathrm{II}(w^\top, X)\right).$$

Note that RHS $\in \Gamma(TM)$ and LHS $\in \Gamma(NM)$. So they are both 0, i.e.,

$$(\nabla_X \Pi)w = \nabla_X (\Pi w) = \nabla_X w^\top = s_{w^\perp}(X).$$

By the boundedness of s, we have

$$\|(\nabla_X \Pi)(x)w\| = \left\|s_{w^\perp(x)}(X)\right\| \le \left\|s_{w^\perp(x)}\right\|_{\mathrm{op}} \|X\| \le C \left\|w^\perp(x)\right\| \|X\|,$$

which implies that

$$\|\nabla_X \Pi(x)\|_{\mathrm{op}} = \sup_{\|w\|=1} \|(\nabla_X \Pi)(x)w\| \le C \|X\| \;\Rightarrow\; \|\nabla \Pi(x)\|_{\mathrm{op}} < C$$

Note that C is independent of $x \in M$.

- **Extrinsic Derivative:** For any $x \in M$, $v \in T_x M$ with $\|v\| = 1$, and $w \in \mathbb{R}^n$ with $\|w\| = 1$,

  $$\overline{\nabla}_v(\Pi(x)w) = \nabla_v(\Pi(x)w) + \mathrm{II}(\Pi(x)w, v).$$

  So

  $$\left\|\overline{\nabla}_v(\Pi(x)w)\right\| \le \|\nabla_v(\Pi(x)w)\| + \|\mathrm{II}(\Pi(x)w, v)\| \le 2C,$$

  which implies that

  $$\sup_{x\in M} \sup_{v\in T_x M, \|v\|=1} \left\|\overline{\nabla}_v \Pi(x)\right\|_{\mathrm{op}} < \infty. \qquad \square$$

Corollary 4.5.8. *Under same assumptions as Lemma 4.5.7, the orthogonal projection Π is globally Lipschitz continuous w.s.t. d_M on M, i.e., there exists some $L > 0$ such that*

$$\|\Pi(x) - \Pi(y)\|_{\mathrm{op}} \le L d_M(x, y), \quad \forall\ x, y \in M$$

Proof. For any $x, y \in M$, by the completeness, let $\gamma\colon [0, \ell] \to M$ be the minimizing unit geodesic from x to y, i.e.,

$$\gamma(0) = x, \quad \gamma(\ell) = y, \quad \|\dot\gamma(t)\| \equiv 1,$$

and $d_M(x, y) = \ell$. Let $F(t) := \Pi(\gamma(t))$. When viewing $\Pi\colon \mathbb{R}^N \to \mathbb{R}^N$ as a matrix and $\gamma(t) \in \mathbb{R}^N$

$$\frac{d}{dt} F(t) = \overline{\nabla}_{\dot\gamma(t)} \Pi(\gamma(t)).$$

By Lemma 4.5.7,

$$\left\|\frac{d}{dt} F(t)\right\|_{\mathrm{op}} = \left\|\overline{\nabla}_{\dot\gamma(t)} \Pi(\gamma(t))\right\|_{\mathrm{op}} \le 2C.$$

It implies that

$$\begin{aligned}\|\Pi(x) - \Pi(y)\|_{\mathrm{op}} &= \|F(\ell) - F(0)\|_{\mathrm{op}} \\ &= \left\|\int_0^\ell \frac{d}{dt} F(t) dt\right\|_{\mathrm{op}} \le \int_0^\ell \left\|\frac{d}{dt} F(t)\right\|_{\mathrm{op}} dt \le 2C d_M(x, y). \qquad \square\end{aligned}$$

Corollary 4.5.9. *Under same assumptions as Lemma 4.5.7, for $v \in \Gamma(TM)$ satisfying*

$$\sup_{x\in M} \|v(x)\| < \infty, \quad \sup_{x\in M} \sup_{\|X\|=1} \|\nabla_X v(x)\| < \infty$$

we have

$$\sup_{x\in M} \sup_{X\in T_xM, \|X\|=1} \left\|\overline{\nabla}_X v(x)\right\| < \infty.$$

Proof. For any $x \in M$, and $X \in T_xM$ with $\|X\| = 1$,

$$\overline{\nabla}_X v = \nabla_X v + \mathrm{II}(X, v).$$

So

$$\sup_{x\in M} \sup_{\|X\|=1} \left\|\overline{\nabla}_X v(x)\right\| \le \sup_{x\in M} \sup_{\|X\|=1} \|\nabla_X v(x)\| + C \sup_{x\in M} \|v(x)\| < \infty. \qquad \square$$

Lemma 4.5.10. *Let M be a Riemannian submanifold isometrically embedded in $\mathbb{R}^n$ such that the second fundamental form and its derivative are uniformly bounded, i.e.,*

$$\sup_{x\in M} \|\mathrm{II}_x\|_{\mathrm{op}} < \infty, \;\; \|\nabla \mathrm{II}_x\|_{\mathrm{op}} < \infty.$$

Then its exponential map has

$$\exp_x(u) = x + u + \frac{1}{2}\mathrm{II}_x(u,u) + R_3(x,u), \quad \forall\, x \in M, \;\; u \in T_xM$$

where

$$\|R_3(x,u)\| \le C \|u\|^3,$$

for some $C > 0$ independent of x.

Remark 4.5.11. Note that for any $X, Y, Z \in \Gamma(TM)$, $\mathrm{II}(X,Y) \in \Gamma(NM)$ and

$$(\nabla_X \mathrm{II})(Y,Z) := \nabla_X^{\perp}(\mathrm{II}(Y,Z)) - \mathrm{II}(\nabla_X Y, Z) - \mathrm{II}(Y, \nabla_X Z),$$

where $\nabla_X^{\perp} N = (\overline{\nabla}_X N)^{\perp}$ for $N \in \Gamma(TM)$.

Proof. Fix $x \in M$ and $u \in T_xM$. Let $\gamma\colon [0,1] \to M$ be the geodesic such that $\gamma(0) = x$ and $\dot\gamma(0) = u$. Note that $\gamma \subset \mathbb{R}^n$. Then

$$\gamma'(t) = \dot\gamma(t),$$

and

$$\gamma''(t) = \overline{\nabla}_{\gamma'(t)}\gamma'(t) = \nabla_{\dot\gamma(t)}\dot\gamma(t) + \mathrm{II}(\dot\gamma(t), \dot\gamma(t)) = \mathrm{II}(\dot\gamma(t), \dot\gamma(t)).$$

Then by the Euclidean Taylor formula for $\gamma(t)$,

$$\begin{aligned}\gamma(1) &= \gamma(0) + \gamma'(0) + \frac{1}{2}\gamma''(0) + \int_0^1 \frac{(1-t)^2}{2}\gamma^{(3)}(t)\, \mathrm{d}t \\ &= x + u + \frac{1}{2}\mathrm{II}(u,u) + R_3(x,u),\end{aligned}$$

where

$$R_3(x,u) = \int_0^1 \frac{(1-t)^2}{2}\gamma^{(3)}(t)\, \mathrm{d}t.$$

For $\gamma^{(3)}(t)$,

$$\gamma^{(3)}(t) = \overline{\nabla}_{\gamma'(t)}\gamma''(t) = \overline{\nabla}_{\dot\gamma(t)}\mathrm{II}(\dot\gamma(t), \dot\gamma(t)).$$

Let $X = \dot\gamma(t)$ and $S_N(X)$ be the Weingarten map. Then by the Weingarten formula,

$$\overline{\nabla}_X \mathrm{II}(X,X) = -S_{\mathrm{II(X,X)}}(X) + \nabla_X^{\perp}\mathrm{II}(X,X).$$

- For the first term, because $\sup_{x\in M} \|\mathrm{II}_x\|_{\mathrm{op}} \leq C_1 < \infty$, similarly as the proof in Lemma 4.5.7,
$$\left\|S_{\mathrm{II(X,X)}}(X)\right\| \leq C_1^2 \|X\|^3 = C_1^2 \|u\|^3 ,$$
because $\|\dot{\gamma}(t)\| \equiv \|\dot{\gamma}(0)\|$.
- For the second term, by the definition
$$\begin{aligned}\nabla_X^{\perp}\mathrm{II}(X,X) &= \left(\nabla_X \mathrm{II}\right)(X,X) + \mathrm{II}\left(\nabla_X X, X\right) + \mathrm{II}\left(X, \nabla_X X\right) \\ &= \left(\nabla_X \mathrm{II}\right)(X,X),\end{aligned}$$
because $\nabla_X X = 0$ for $X = \dot{\gamma}(t)$. Then by $\|\nabla \mathrm{II}_x\|_{\mathrm{op}} \leq C_2 < \infty$,
$$\left\|\nabla_X^{\perp}\mathrm{II}(X,X)\right\| \leq C_2 \|X\|^3 = C_2 \|u\|^3 .$$

Therefore,
$$\left\|\gamma^{(3)}(t)\right\| \leq (C_1^2 + C_2) \|u\|^3 .$$
It implies that
$$\|R_3(x,u)\| \leq \frac{C_1^2 + C_2}{6} \|u\|^3 . \qquad \square$$

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